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**PawTect: A Community-Based Wireless Dog Detection
and Alert System for Stray and Missing Dogs using
Computer Vision and Raspberry Pi**

A Thesis Presented To
The Faculty of the College of Engineering

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Requirements for the Degree Of
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I

APPROVAL SHEET

This research design entitled "**PawTect: A Community-Based Wireless Dog Detection and Alert System for Stray and Missing Dogs using Computer Vision and Raspberry Pi**", in partial fulfillment of the requirements for the subject **CpE Practice and Design 2**, has been examined and recommended for ORAL EXAMINATION.

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ABSTRACT

Persistent stray dog populations and rabies fatalities are critical public health challenges in the Philippines, exacerbated by manual, real-time-limited monitoring systems that hinder law enforcement and stray animal management. To address this, the study designed and developed PawTect, an intelligent, community-based dog detection, identification, and registration system utilizing decentralized edge intelligence and low-power, long-range wireless telemetry. PawTect employs a lightweight YOLOv8n-based computer vision model, trained on open-source datasets, for accurate dog detection on hardware-camouflaged Raspberry Pi Zero 2W field nodes. A secure SQL-based dog registration application stores canine profiles and owner traits. Detected cropped images are transmitted via LoRa wireless technology to a central master device for vector-based re-identification using a CNN feature extractor and Cosine Similarity matching against the database. Finally, a dual-channel notification module uses SMS and Telegram Bot APIs to instantly alert owners or administrators about detected missing or stray dogs. Field testing validated the system, which achieved a precision score of 86.73% for dog detection and successful differentiation of registered from unregistered dogs. LoRa telemetry demonstrated reliable data transmission with a



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III

stable relation between payload and time. The central processing script introduced negligible delay, taking 1.474 seconds for feature matching and dispatch. Crucially, the system achieved a 100% message delivery success rate with a mean time of 8.800 seconds for handset reception, proving its dependability. For areas with inconsistent internet connectivity, PawTect offers a cost-effective, real-time automated alternative to manual tracking, optimizing administrative response times for public safety and rabies control.

Keyword: *Computer Vision Model, Dog Registration Program, Raspberry Pi, LoRa Wireless Transmission Systems, SMS Notification.*



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V

support and understanding throughout the grueling research process. We are exceptionally grateful to her and her dog, **Berry**, for graciously allowing us to bring, handle, and utilize Berry as a critical subject for testing and validating our prototype on-site.

Finally, to our **families and friends**, thank you for your unwavering emotional and financial support, your endless encouragement, and for being our sanctuary during moments of stress.

This achievement belongs to all of us.

L.J.M.C.
O.H.J.P.
R.U.A.Q.
T.H.C.J.



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VI

DEDICATION

This work is wholeheartedly dedicated to:

*Our beloved parents, for their endless love, sacrifices,
and unwavering belief in our potential;*

*Our friends and colleagues, who provided constant
motivation, support, and laughter throughout this journey;*

*And above all, to the Almighty God, the source of all our
knowledge, strength, and inspiration.*

**L.J.M.C.
O.H.J.P.
R.U.A.Q.
T.H.C.J.**



TABLE OF CONTENT

APPROVAL SHEET	I
ABSTRACT	II
ACKNOWLEDGMENT	IV
DEDICATION	VI
CHAPTER I: PROBLEM AND ITS BACKGROUND	1
Introduction	1
Statement of the Problem	4
Objective of the Study	5
Theoretical Framework	7
Conceptual Framework	11
Significance of the Study	12
Scope and Limitation of the Study	14
Definitions of Terms	16
CHAPTER II: REVIEW OF RELATED LITERATURES	19
Related Readings	19
Foreign Related Literature	22
Local Related Literature	30
Foreign Related Studies	37
Local Related Studies	48
Synthesis of the Study	57
CHAPTER III: METHODOLOGY	60
IPO Model	61
Block Diagram	63
Conceptual Design	64
Phases of the System	68
Research Locale	98
Research Design	99
Population Sampling	99



EULOGIO “AMANG” RODRIGUEZ
INSTITUTE OF SCIENCE AND TECHNOLOGY
College of Engineering



VIII

Research Instrument	100
Statistical Treatment of Data	102
CHAPTER IV: RESULT, ANALYSIS AND INTERPRETATION OF DATA ..	106
Confusion Matrix for the On-Field Device	108
Cosine Similarity for the Identification Program	110
Frequency Distribution for the Dog Registration Application	112
Registration Success Rate	113
Transfer Speed Over Different Chunk Size in Seconds	115
SMS Notification Delay Over Time	117
CHAPTER V: SUMMARY, CONCLUSION, AND RECOMMENDATION	121
Summary of Findings	121
Conclusions	123
Recommendations	124
References	126
Appendices	147
Appendix A: Letter to the Adviser	148
Appendix B: Student – Faculty Research Agreement	150
Appendix C: Conceptual Framework	153
Appendix D: Theoretical Framework	156
Appendix E: Methodological Framework	158
Appendix F: Project Timeline	162
Appendix G: Actual Design of the Prototype	164
Appendix H: CAD Layout of Case with Components	166
Appendix I: Schematic Diagram	168
Appendix J: Project Costing	170
Appendix K: Dashboard and Telegram Message	172
Appendix L: Prototype Deployment	175
Appendix M: User Manual	178
Appendix N: Researchers Profile	180



Appendix O: Grammar Checker	186
Appendix P: Plagiarism Checker	190
Appendix Q: IEEE Format	192

LIST OF TABLE

Table 1. Matrix of Related Literature.....	7
Table 2. Matrix of Related Studies.....	10
Table 3. Summarization of Dataset Used for Detection Model.	107
Table 4. Training Setting.....	107
Table 5. Confusion Matrix of the On-Field Detection Model..	108
Table 6. Cosine Similarity of the Vector Program.....	110
Table 7. Frequency Distribution of Sample Subjects for the Dog Registration Application.....	112
Table 8. Rate of Successful Registration.....	113
Table 9. Average Completion Time Over Different Chunk Sizes	115
Table 10. Average Receiving Time on the SMS Notification...	117
Table 11. Project Costing.....	171

LIST OF FIGURES

<i>Figure 1. Conceptual Framework of the System.....</i>	<i>11</i>
<i>Figure 2. Deployment of the Prototype.....</i>	<i>60</i>
<i>Figure 2. Input-Process-Output Model.....</i>	<i>61</i>
<i>Figure 3. Block Diagram.....</i>	<i>63</i>
<i>Figure 4. Conceptual Design.....</i>	<i>64</i>
<i>Figure 5. Main Page of the Registration System.....</i>	<i>81</i>
<i>Figure 6. Exploded View of On-Field Device.....</i>	<i>85</i>
<i>Figure 7. Enclosure of the On-Field Device.....</i>	<i>88</i>
<i>Figure 8. Container.....</i>	<i>89</i>
<i>Figure 10. Integrated Dog Detection Model.....</i>	<i>109</i>
<i>Figure 11. Main Page and Registration Page of the System...</i>	<i>114</i>



CHAPTER I: PROBLEM AND ITS BACKGROUND

Introduction

In the Philippines, dogs are considered beloved companions in several households, acting as loyal pets, guardians, and family members. The ongoing problem of stray and free-roaming dogs is a significant threat to public health, community safety, and animal welfare.

Estimates from entities such as the Philippine Animal Welfare Society or also known as *PAWS*, suggest that there are roughly 12 million stray dogs across the nation, a statistic often referenced in recent reports, from 2019 to 2024, and acknowledged as a pressing issue in discussions on animal homelessness and overpopulation in 2025. The large stray population poses considerable public health risks, especially with rabies, a viral illness that is basically lethal once symptoms manifest.

The Department of Health in 2024 documented 426 confirmed rabies cases, all of which were fatal. Out of the confirmed rabies cases, 45% were associated with bites from domestic pets, while several others were stray animals or those with indeterminate vaccination status. Despite early 2025 data indicating a promising decline—such as 55 cases from January to



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College of Engineering



2

March, a 39% decrease compared to the same timeframe in 2024, rabies continues as endemic, resulting in hundreds of human fatalities each year, disproportionately impacting children.

Despite national mandates under Republic Act No. 9482 or known as, the Anti-Rabies Act of 2007, which requires annual dog registration, immunization, and responsible ownership—enforcement remains inconsistent across many regions. While Local Government Units are tasked with maintaining registries and managing stray populations, progress is often hindered by limited public awareness and resource constraints.

To address these gaps, cities like Manila have introduced Ordinance No. 8587 of 2020, which mandates microchipping and centralized computerized databases, backed by fines of thousands in Pesos. However, the transition from manual to digital tracking remains a significant logistical hurdle.

Previous studies and programs have addressed aspects of this issue via mass vaccination, community education, Catch-Neuter-Vaccinate-Return programs, and centralized databases to enhance traceability and pet reunion. Emerging technologies, such as computer vision for animal recognition, affordable embedded systems like Raspberry Pi, Long Range (LoRa) wireless communication, and verification SMS notifications, have exhibited significant potential in both international and



domestic research for automated monitoring and prompt response. Nevertheless, these solutions are often integrated into a unified, community-oriented system designed for Philippine circumstances, where connectivity is inconsistent and manual reporting frequently results in delays.

A distinct gap is evident such as the lack of an accessible, integrated technological platform that incorporates accurate dog detection and identification, owner registration databases, dependable wireless data transmission, and immediate community alerts. Existing methods rely significantly on manual procedures, leading to slow responses in dynamic environments where dogs roam freely.

The research aims to fill this gap through the development of a prototype titled, PawTect: A Community-Based Wireless Dog Detection and Alert System for Stray and Missing Dogs using Computer Vision and Raspberry Pi. It utilizes machine learning and a mini-computer for the automatic detection, vector comparison for identification, a dog registration application and centralized database for owner profiles and pet information, Long Range or LoRa for long-range wireless communication from field devices or child nodes to a central hub or the parent node, and a Telegram bot integration for verification notifications to owners and administrators.



By integrating these technologies, the system provides a low-cost, long-range alternative to traditional monitoring, enabling faster recovery of missing pets and more efficient management of stray populations within a community setting. Ultimately, this research seeks to demonstrate how localized IoT solutions can bridge the gap between municipal ordinances and practical, real-time enforcement.

Statement of the Problem

Traditional methods of monitoring stray animals and identifying missing pets often rely on manual reporting and physical registries, which are prone to delays and human error. Despite local ordinances requiring pet registration, the lack of a real-time, automated monitoring infrastructure makes it difficult for communities to enforce these laws and ensure public safety.



Objective of the Study

The main objective of the study is to design and develop an intelligent dog detection, identification and registration system using field video surveillance integrated with computer vision that can automatically detect. A separate device through the Long-Range wireless transmission would receive boxes of images and identify and manage dogs within a community premise to enhance community safety, promote responsible pet ownership, mitigate animal-related issues, and foster a well-regulated, pet-friendly environment.

To achieve the goal of a fully automated and community-based detection system, the following technical and functional objectives have been established:

1. Create and train a computer vision model that can accurately detect dogs from the environment and identify if it is missing or a stray dog.
2. Develop and integrate dog registration application and a community dog database for the computer vision that enables dog owners to log their pets and store related data for identification purposes such as owner information, dog specific descriptions with uploaded images and vaccination records.



3. Implement a LoRa wireless transmission system to send cropped images from a field Raspberry Pi device to a central hub.

4. Develop an automatic SMS Notification module via a Telegram Bot API to inform owners or administrators of their missing or stray dogs, respectively.

5. Implement the system to verify its accuracy and latency in identifying dogs and sending real-time alerts.

Keyword: Computer Vision Model, Dog Registration Program, Raspberry Pi, LoRa Wireless Transmission Systems, SMS Notification.



Theoretical Framework

AUTHOR	LORA	DOG REG. APP.	REAL-TIME SMS NOTIF.	COMP. VISION MODEL	RASPBERRY PI
Petdentity Philippines Inc. (2017)		Mobile & web registration system with RFID			
Ashley Whittaker (2021)				YOLO v3-based real-time dog	Uses Raspberry Pi with camera and megaphone for real-time dog alerts
Gordon Charles (2021).				Uses SSD Mobilnet V2 Single-Shot multibox Detection (SSD) on OpenCV, V3	Uses Raspberry Pi as the primary device for using the dog detection for his project
Llantino et al. (2026)	Uses LoRa Wireless Transmission system to send compressed detected images over the decision-making node from the detection node.	Collects various owner and dog data for the utilization of computer vision model and real-time SMS notification.	Notifies users or administration if a pet or stray dog is detected on the streets.	Uses YOLOv8n to create a computer vision model and CNN's Feature Extractor for comparing vectors of images	Raspberry Pi Zero 2W Raspberry Pi Camera Module for Detection Node

Table 1. Matrix of Related Literature

The synthesis matrix in Table 1 identifies a significant gap in current community-based pet monitoring. While previous researchers like Gordon Charles (2021) and Ashley Whittaker (2021) successfully utilized Raspberry Pi and computer vision



models (SSD and YOLOv3), their systems lacked a long-range wireless communication framework and a centralized registration database. The proposed study by the proponents, Llantino et al. (2026) fills this technological void by incorporating LoRa for data transmission and a comprehensive registration app, creating a holistic ecosystem for stray and missing dog management that previous iterations did not fully address.

AUTHOR	LORA	DOG REG. APP.	REAL-TIME SMS NOTIF.	COMP. VISION MODEL	RASPBERRY PI
Babayiğit & Yarlı Doğan (2025).	Implemented point-to-point LoRa communication using SX1262 LoRa HAT modules mounted on Raspberry Pi 4. Images were aggressively down sampled, compressed (JPEG/WebP), and base64-encoded to fit LoRa's strict bandwidth and duty-cycle constraints. (mdpi.com)			Used face-specific ESRGAN and Real-ESRGAN models on the receiver Raspberry Pi 4 to reconstruct transmitted low-resolution images. Evaluated using PSNR, SSIM, LPIPS, identity preservation, and MOS (Mean Opinion Score). (mdpi.com)	Both sender and receiver nodes were built with Raspberry Pi 4 boards, handling image processing, compression, transmission, and super-resolution reconstruction. (mdpi.com)
Cabacunga n, P. M., et al. (2023).	Uses LoRa alongside NearCloud system to gather and store data for the growth parameters of their hydrophnics farmed black rice			Deployed a contrastive Siamese network to detect similarity between frames or patches. Used PCA for compression and enhanced SR-GAN for reconstruction. (sciencedirect.com)	Each node comprised a Raspberry Pi 3 B+, a camera module, and a Pycom LoPy4 LoRa device, handling patch-based image processing, similarity detection, and



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College of Engineering



9

AUTHOR	LORA	DOG REG. APP.	REAL-TIME SMS NOTIF.	COMP. VISION MODEL	RASPBERRY PI
					transmission logic. (avesis.erciyes.edu.tr)
Islam et al. (2025).	Designed a LoRaWAN-based pipeline with LoRa end devices, gateways, and a ChirpStack server. Used HARQ-II for reliable transmission. Employed image patching, similarity assessment, compression, and reconstruction to reduce payload. (sciencedirect.com)			Deployed a contrastive Siamese network to detect similarity between frames or patches. Used PCA for compression and enhanced SR-GAN for reconstruction. (sciencedirect.com)	Each node comprised a Raspberry Pi 3B+, a camera module, and a Pycom LoPy4 LoRa device, handling patch-based image processing, similarity detection, and transmission logic. (avesis.erciyes.edu.tr)
Crasicas (2024)		Web-based barangay registration system			
Azizi, E. & Zaman, L. (2023)				"Creation of a model with seven sequential layers for species and animal identification: data preprocessing, species classification, landmark detection and face/body detection, alignment, identification, soft biometrics, and recommendation." "	
Chakravarty, S. et al. (2022)			Email notifications using Embedded ML	Machine learning models such as SSD, YOLOv3, and YOLOv4 are used to detect dogs.	Detection runs on Raspberry Pi on the University Campus



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College of Engineering



10

AUTHOR	LORA	DOG REG. APP.	REAL-TIME SMS NOTIF.	COMP. VISION MODEL	RASPBERRY PI
				YOLOv3 outperforms all other mentioned models.	
Dede, J. (2024)	Uses LoRa for deterrents and sending status message			Using Machine Learning for wolves' detection in the wild	Uses Raspberry Pi 4 with a camera, Axis M2025-LE connected
Nolasco et al. (2025)				Central feature. The study implements a CNN using InceptionV3 (via TensorFlow Keras) to classify dog activities (sitting, standing, lying) with ~88% accuracy (mdpi.com).	Essential platform. A Raspberry Pi 4 runs the model, interfaces with a camera and displays results via LCD; OpenCV supports the vision pipeline (mdpi.com).
Dhanaraju & Kumar (2025)			Sends immediate SMS alerts to farmers upon detecting animal intrusion	Used deep learning to detect specific animal species in real-time.	Used as the edge processing node for running the YOLO algorithm.
Llantino et al. (2025)	Uses LoRa Wireless Transmission system to send compressed detected images over the decision-making node from the detection node.	"Collects various owner and dog data for the utilization of computer vision model and real-time SMS notification.			

Table 2. Matrix of Related Studies

Table 2 provides a comprehensive matrix of related studies, categorizing the evolution of animal detection and wireless communication technologies. While recent works such as Babayiğit



& Yarlı Doğan (2025) and Islam et al. (2025) demonstrate sophisticated methods for image reconstruction and similarity detection over LoRa, they primarily focus on agricultural or general surveillance contexts. In contrast, this study by Llantino et al. (2025) bridges these technical advancements with a localized social application: a community-based dog registration and notification system. Unlike the general species classification explored by Azizi & Zaman (2023) or the deterrent-focused messaging of Dede (2024), the proposed system integrates real-time identification with a centralized owner database to provide a direct solution for stray and missing dog management.

Conceptual Framework

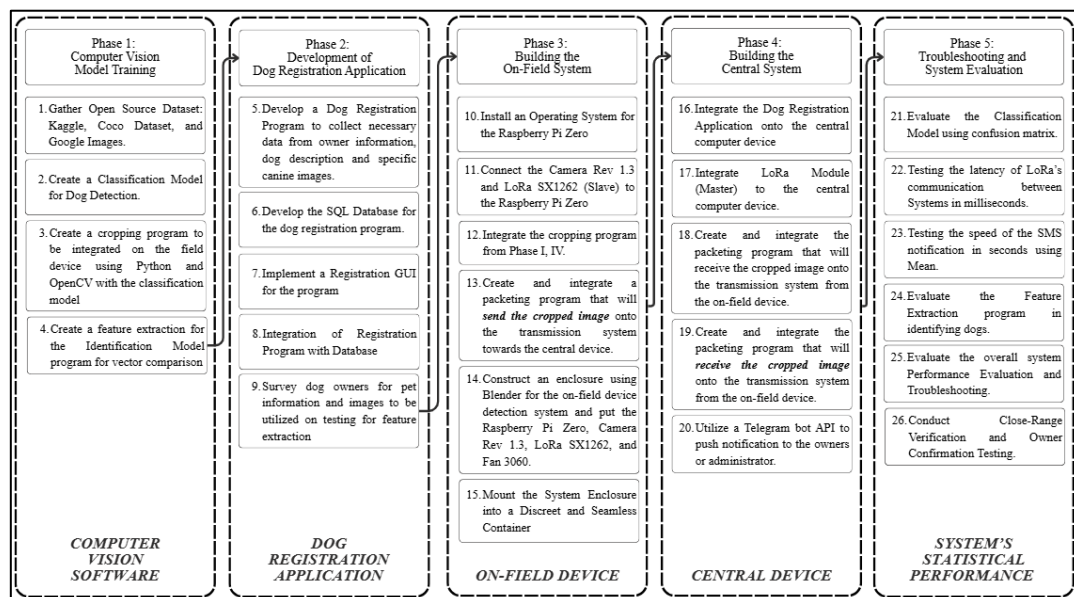


Figure 1. Conceptual Framework of the System



Figure 1 illustrates the five-phase developmental framework of the system, detailing the sequential progression from software engineering to field deployment. The methodology begins with Phase 1, focusing on the training of the YOLO-based computer vision model and feature extraction logic. This is followed by the creation of the dog registration application (Phase 2) and the hardware assembly of the On-Field Device (Phase 3) using the Raspberry Pi Zero 2W platform. Phase 4 details the integration of the central processing hub and the LoRa communication gateway. The process culminates in Phase 5, which outlines a rigorous statistical evaluation based on model accuracy, transmission latency, and notification speed.

Significance of the Study

The significance lies in its potential to improve community safety by making the identification of stray and missing dogs faster and easier, encouraging responsible pet ownership through simplified registration and notifications, supporting local government units in enforcing animal welfare and rabies control laws, and help in reducing rabies cases while enhancing animal welfare.



The study holds substantial significance for multiple stakeholders and contributes meaningfully to both technological innovation and societal welfare. To the respondents—including:

- dog owners,
- local government units (LGUs),
- barangay personnel, and
- community members.

The community dog detection offers a practical, accessible solution that simplifies dog registration, expedites the recovery of missing pets, and enhances public safety through automatic notification.

By automating the detection and identification of stray and missing dogs using computer vision and Raspberry Pi, the system directly addresses the problem of delayed response times inherent in manual reporting methods, thereby bridging a critical knowledge gap in community-based animal monitoring where existing solutions lack integration of automated detection, wireless long-range communication, and immediate SMS notification. Furthermore, this research contributes to improving social, economic, and health conditions by supporting the Philippines' rabies elimination goals under Republic Act No. 9482 and the National Rabies Prevention and Control Program.



- reducing stray dog populations through efficient identification and owner notification;
- minimizing public health risks associated with dog bites and rabies transmission;
- and lowering the economic burden on LGUs by streamlining animal control operations and reducing costs related to prolonged sheltering and manual search efforts

This study advances the field by enriching research instruments and methods through the integration of YOLOv8n-based computer vision models with edge computing on Raspberry Pi devices, the application of LoRa wireless technology for long-range, low-power data transmission in community IoT applications, and the validation of SMS API alerts as a notification mechanism in areas with inconsistent internet connectivity—thereby providing a replicable framework for future researchers developing intelligent monitoring systems in resource-constrained environments.

Scope and Limitation of the Study

The scope of the study focuses on the design and development of an intelligent dog detection, identification, and registration system, named PawTect, utilizing computer vision and Raspberry Pi technology. The system aims to enhance



community safety and promote responsible pet ownership by automatically detecting, identifying, and managing dogs within a village through independent camera handling. Key functionalities include a local application for dog registration with multiple images, a system integrated with computer vision model for detection capable of distinguishing dogs from the environment, two LoRas communicating to send the detected image for further inspection, and a feature extractor program that distinguishes the registered from unregistered dogs, a centralized database for storing owner and dog information, and an automated alert system that notifies administrators and owners via SMS API.

However, the limitations of this study are primarily related to its geographical focus on a single village. The system's effectiveness may also be influenced by environmental factors such as lighting conditions and weather, and the accuracy of the computer vision model is dependent on its input, Raspberry Pi Camera Rev 1.3. Furthermore, the system's ability to manage stray dogs is limited to detection and alerting of the authority, rather than direct intervention. Instead, manual obtaining of the detected stray is up-to the informed administrator.



Definitions of Terms

1. Central Device - the primary server or usually called, parent node. Typically, another Raspberry Pi or a computer- in this case, a personal computer was utilized by the proponents, it acts as the gateway for the system. It receives data from the field device, processes the identification comparison against the database, and triggers the notification module.
2. Computer Vision Model - a specialized artificial intelligence framework trained to interpret visual data. In the study, it refers to the YOLOv8n architecture and the subsequent vector-extraction layers used to perceive and analyze dog-related imagery.
3. Detection - the initial process in the computer vision pipeline where the system identifies the presence of a dog within a video frame and provides localized coordinates (bounding boxes) to isolate the subject from the background.
4. Embeddings - the high-dimensional numerical representation of a dog's unique visual characteristics per pixel. These are *compressed* summaries of features such as fur pattern, ear shape, and snout structure that allow the system to compare one dog against another mathematically.



5. Identification Program - the software logic responsible for *Re-Identification* (Re-ID). It compares the live embedding captured by the field device against the known vectors stored in the centralized database to find a match and identify a specific dog.
6. Field Device - usually called as the child node or edge-computing unit. In this case, Raspberry Pi Zero 2W deployed in the community. It is equipped with a camera to capture footage, run real-time detection, and transmit the relevant data back to central device.
7. Long Range Wireless Transmission System - a low-power, wide-area networking (LPWAN) protocol used to facilitate communication between the field device and the central device over long distances, up to 10 kilometers, based on optimal conditions, without the need for cellular data or high-bandwidth Wi-Fi.
8. Raspberry Pi - the single-board computer (SBC) that serves as the hardware backbone of the project. It provides the computational power necessary to run the detection model at the edge and manage the LoRa communication interface.
9. Feature Extraction - the process where the CNN layers of the model transform raw pixels from an image into meaningful mathematical descriptors. It extracts the most



important visual information while discarding irrelevant background data.

10. Vectors - the mathematical format used to represent embeddings. In this system, a vector is an array of numbers that serves as a *digital fingerprint* for each dog, allowing for fast and accurate similarity searches in the database.
11. YOLOv8n - the specific version of the YOLO architecture chosen for this study. The *Nano* variant is optimized for high-speed performance on resource-constrained devices like the Raspberry Pi Zero 2W, providing a balance between detection accuracy and processing efficiency.



CHAPTER II: REVIEW OF RELATED LITERATURES

This chapter contains reviews of related foreign and local literature connected to the research. The key concepts presented include computer vision model, a dog registration application, raspberry pi, LoRa wireless transmission methods, and real-time SMS notifications.

Related Readings

Recent improvements in computer vision, including the YOLO real-time object detection model (Redmon et al., 2016), have facilitated rapid and precise identification of animals via CCTV. The Anti-Rabies Act of 2007 requires all local government units (LGUs) in the Philippines to keep a register of dogs and issue identification tags for pets. Nonetheless, similar assessments indicate that compliance in addressing Local Government Units (LGUs) is inconsistent, often negligible, leading to challenges in managing stray dog numbers and preventing rabies. The municipal governments, including Bacolod City, have prioritized mass dog registration as a means to improve vaccination distribution and enhance public safety (Philippine News Agency, 2023). These changes suggest the need for an intelligent, automated monitoring and registration system



utilizing computer vision and CCTV to promote responsible pet ownership and enhance community safety.

Our proposed system aligns with three Sustainable Development Goals (SDGs): Sustainable Development Goals 3, 11, and 15. Initially, SDG 3 Good Health and Well-being is vital since it aids in illness prevention and guarantees public safety. The World Health Organization (2021) asserts that regulating dog populations and enhancing owner accountability can diminish rabies incidence globally. It also references SDG 11: Sustainable Cities and Communities, which aims to create inclusive, safe, resilient, and sustainable urban areas and human settlements. According to UN-Habitat (2020), the presence of stray dogs in communities affects safety, sanitation, and livability, which can be enhanced by data-driven animal management. This project corresponds with SDG 15: Life on Land, focusing on the ethical treatment of animals to safeguard biodiversity. The Food and Agriculture Organization (2019) asserts that monitoring mechanisms will prevent animal cruelty and mitigate the environmental consequences of uncontrolled dog populations. The SDG priorities indicate that responsible pet ownership, community safety, and animal welfare are significant global concerns that can be addressed through contemporary technological interventions.



EULOGIO “AMANG” RODRIGUEZ
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College of Engineering



21

The Philippine Development Plan (PDP) 2023–2028 at the national level substantiates the need for an intelligent dog monitoring system. The PDP emphasizes enhancing disease surveillance in Chapter 5 (Advance Health and Nutrition), underscoring the necessity of augmenting local monitoring initiatives and mitigating animal-related health hazards. Chapter 10 underscores the establishment of safer, more resilient barangays, highlighting that digital technologies and automated systems can improve community-based safety initiatives. Chapter 8 (Ensure Ecological Integrity and a Clean and Healthy Environment) addresses environmental and community sanitation issues associated with mismanaged dog populations, advocating for local governments to implement sustainable policies that safeguard both humans and animals. Chapter 6 (Promote Innovation) emphasizes the national imperative of integrating emerging technologies, including as artificial intelligence and automatic recognition systems, to enhance government services and local governance. The development of a computer vision-based dog monitoring and registration system enhances this national plan by improving animal management efficiency, data reliance, and responsiveness.



Foreign Related Literature

Computer Vision Model

Several studies have explored dog breed detection and recognition through the application of computer vision and deep learning techniques, revealing a growing trend in leveraging advanced neural networks for intelligent animal monitoring. A comprehensive review by Ambili et al. (2025) emphasized the potential of Convolutional Neural Networks (CNN) combined with transfer learning to improve the accuracy of dog breed classification, demonstrating that pre-trained models can significantly reduce training time while maintaining high performance across diverse datasets. Building upon these foundational insights, Sinnott, Wu, and Chen (2018) developed a mobile application for real-time dog breed detection, integrating deep learning algorithms with object detection and feature extraction techniques to enable accurate recognition even in dynamic or uncontrolled environments.

Expanding on practical detection systems, Bhosale et al. (2025) proposed a stray dog detection system utilizing YOLOv5, highlighting the efficiency of single-shot detectors and CNN architectures in rapidly identifying and localizing dogs across varying scenes. Similarly, Lin et al. (2024) presented a model that combines breed recognition with individual identity



verification, underscoring the critical importance of high-resolution imagery, precise feature extraction, and facial recognition methods for accurate and reliable classification. In a complementary approach, Mendoza-Montoya et al. (2023) applied computer vision and deep neural networks to the challenge of locating lost dogs, employing CNN-based breed sorting to enhance identification, tracking, and social networking coordination, demonstrating the social and practical implications of these technologies.

These studies illustrate the transformative impact of CNNs, YOLO, and other deep learning frameworks in automating and enhancing dog breed recognition, stray dog detection, and lost dog retrieval. They highlight not only the technical sophistication of modern computer vision models but also their potential for real-world applications, providing a strong foundation for future research aimed at developing intelligent, scalable, and socially impactful animal monitoring systems.

Dog Registration Program

Feral dog populations are closely linked to rabies transmission in many low-income countries, posing public health risks. Dog Population Management (DPM) aims to improve canine welfare, reduce conflicts, and stabilize populations. Central to DPM is dog identification and registration, which enables



accurate tracking of ownership, vaccination, and sterilization, supporting high vaccination coverage and traceability during disease outbreaks or biting incidents.

However, registration faces challenges such as poor data management, weak owner compliance, high fees, and the presence of unowned or loosely owned dogs. Many DPM programs lack comprehensive evaluations, limiting insights into the impact of registration alongside interventions like vaccination or sterilization. While surgical sterilization can reduce dog turnover, its sustainability is constrained by financial, technical, and governmental limitations. Scalable, cost-effective, data-driven strategies are needed to stabilize populations and support rabies elimination.

Studies highlight that centralized databases improve vaccination enforcement, prevent theft, reunite lost pets, and aid in identifying aggressive dogs. In low-income areas with many stray dogs, registration's effectiveness is limited. In Ireland, national databases reveal stable pet numbers but declining dogs in control centers, highlighting the need for integrated, real-time tracking systems.

The World Organization for Animal Health (WOAH, 2023) emphasizes registration as a core tool for monitoring dog populations, supporting vaccination and sterilization tracking,



reuniting lost dogs, and facilitating programs like Catch, Neuter, Vaccinate, Return (CNVR). Effective registration requires centralized, up-to-date records, cooperation with authorities, and compliance monitoring, despite challenges with free-roaming dogs and lost tags.

EU experiences show that microchipping linked to national registries is essential for responsible ownership, disease control, and cross-border monitoring, enabling authorities to enforce legislation and enhance public health and animal welfare outcomes.

Raspberry Pi

Comprehensive reviews by Ghael (2020), Mathe et al. (2024), and Umap et al. (2025) collectively highlight the broad application domains of Raspberry Pi, including IoT, automation, and computer vision. These papers underscore Raspberry Pi's impact on education, research, and industry, citing its low cost, adaptability, and strong community support.

Mathe et al. (2024) and Umap et al. (2025) specifically detail the advancements in Raspberry Pi hardware and software, enabling robust solutions in smart surveillance, animal tracking, and embedded AI. Directly aligning with the technical demands of the system.



Saari et al. (2017) focused on prototyping with Raspberry Pi, emphasizing its rapid development cycle and adaptability for custom applications, such as dog detection and alert systems.

LoRa Wireless Transmission Systems

In addition to applied research, several comprehensive surveys and reviews provide theoretical and practical insights into the deployment, challenges, and sensor integration of LoRa and LoRaWAN technologies.

Kamal (2023) provides an extensive survey on the requirements, deployments, and challenges of LoRa technology. The study emphasizes the technology's strengths in long-range, low-power communication ideal for IoT applications and highlights challenges such as interference, scalability, and security. These insights are particularly relevant for the study, as the system must ensure reliable detection and alert transmissions within a community setting, potentially dealing with interference from other wireless devices and the need for secure data handling to protect user privacy.

Kim et al. (2023) conducted a systematic literature review focusing on LoRaWAN sensors and applications. Their work underscores the importance of sensor fusion combining computer vision with environmental sensors to enhance system accuracy and functionality. This is critical for PawTect, where computer



vision must be supplemented by wireless sensor data to validate detection events and trigger timely alerts. The review also discusses various application domains, including smart cities and animal tracking, which align closely with PawTect’s objectives.

Fragkopoulos et al. (2023) presented an experimental assessment of factors affecting LoRaWAN performance in suburban and rural areas. Their findings on signal attenuation, packet loss, and environmental influences provide valuable design considerations for PawTect. Since stray dogs can be found in diverse community environments, understanding these performance factors helps optimize node, antenna design, and transmission settings to maximize detection coverage and reliability.

Syed et al. (2024) analyzed the design and feasibility of LoRa-based communication systems for disaster management. Their focus on real-time, reliable, and community-based alert systems parallels PawTect’s goal of providing immediate notifications to community members about missing and stray dogs. The study’s emphasis on system scalability and user engagement guides the development of PawTect’s network architecture and alert dissemination strategies.

These foreign literatures provide a comprehensive theoretical and practical framework to support the design,



deployment, and optimization of PawTect. They stress the importance of addressing technical challenges and leveraging sensor integration to ensure system effectiveness in real-world community scenarios.

Real-Time SMS Notification

The term "real-time" in IoT monitoring refers to the system's capacity to process data and trigger an output with negligible latency. Foreign literature establishes the theoretical framework for why this is non-negotiable in emergency and security applications.

MoldStud (2025) articulates that in emergency mobile applications, the psychological and practical impact of a notification is directly tied to its immediacy. Their analysis of emergency response systems suggests that delays exceeding a few seconds can render a warning useless, particularly when the target—in this case, a stray animal—is mobile. If a notification arrives minutes after detection, the subject has likely moved, negating the system's purpose. This is supported by Hossain and Muhammad (2021), who analyzed latency in IoT-based health monitoring frameworks. While their focus was on elderly care, their findings on cloud computing latency are highly relevant. They argue that relying solely on cloud-based app notifications can introduce delays due to server hops and bandwidth



fluctuation. This validates the PawTect approach of using edge computing (Raspberry Pi) combined with direct GSM communication to minimize the "decision-to-alert" timeframe.

Furthermore, Omnisend (2025) provides a comparative analysis between push notifications (app-based) and SMS. Their findings indicate that while push notifications allow for richer media, they suffer from lower deliverability rates when data connections are unstable. In contrast, SMS is described as a "store-and-forward" mechanism that ensures delivery once a signal is established. Horisen (2023) reinforces this by classifying SMS as an essential component of modern automation, arguing that despite the rise of 5G and fiber optics, the simplicity of the SMS protocol allows it to penetrate areas where high-speed internet fails. For a community-based system like PawTect, which may operate in the periphery of urban centers where WiFi coverage is spotty, this literature provides a foundational justification for choosing SMS over purely internet-based alerts.

Additionally, Vonage (2023) highlights that SMS provides reliable two-way communication for IoT devices, serving as a fail-safe when TCP/IP protocols (internet data) encounter packet loss. This underscores the reliability of the protocol for robust machine-to-machine (M2M) communication.



Local Related Literature

Computer Vision Model

Recent studies in the Philippines have increasingly leveraged computer vision and deep learning techniques to address challenges in animal monitoring, canine activity recognition, and visual surveillance. For instance, Nolasco, Aldea, and Villaverde (2025) explored dog activity recognition using Convolutional Neural Networks (CNN) with the InceptionV3 architecture, demonstrating that even resource-constrained devices such as Raspberry Pi can effectively classify and monitor canine behavior in real-time. This work underscores the potential of integrating lightweight yet powerful neural networks into practical, everyday applications.

Meanwhile, Figueroa et al. (2017) examined animal surveillance using thermal imaging mounted on quadcopters, highlighting innovative approaches to zoological monitoring that combine aerial robotics with advanced sensor technologies. Their research reflects the growing interest in applying computational and mechatronic systems to capture data in otherwise challenging or inaccessible environments.

Expanding the scope to canine-specific studies, Abesamis et al. (2024) developed “Pawsome Detect,” a system employing pattern recognition and data mining to determine both the age



and breed of dogs. This approach demonstrates the power of integrating data-driven algorithms with machine learning to extract nuanced biological features from visual inputs, which can support veterinary, shelter, and community animal management applications.

On a broader scale, Guillermo et al. (2020) explored the application of neural networks for public security, demonstrating how image-based recognition can identify potential threats. Although their study focused on human security, the underlying principles of detection, classification, and automated monitoring can be adapted to animal surveillance, highlighting the cross-disciplinary potential of deep learning methods.

Finally, Medel, Sevilla, and Perez (2018) addressed the challenges of color detection in robotic vision, offering insights into the interpretation of visual data in dynamic environments. Their work illustrates that even foundational visual perception tasks, such as color recognition, are crucial in enabling robots and monitoring systems to interact meaningfully with their surroundings.

Collectively, these local studies reveal a clear trend toward combining neural networks, sensor technologies, and intelligent data processing to enhance the understanding,



monitoring, and management of animals. They illustrate not only technical innovation but also the practical considerations of deploying these systems in real-world Philippine contexts, laying a solid groundwork for future research that bridges computational theory with applied animal and environmental monitoring.

Dog Registration Program

The Philippine Animal Welfare Society (PAWS) highlights Republic Act No. 9482 (Anti-Rabies Act of 2007), which mandates annual dog registration in every city and municipality. Registered dogs receive identity tags linked to vaccination records, and LGUs maintain centralized databases. Non-compliance can result in fines or confiscation. This program helps distinguish owned dogs from strays, supports rabies prevention, and promotes responsible pet ownership.

The National Rabies Prevention and Control Program Strategic Plan 2020-2025 (DOH & DA) reinforces the importance of LGU-led registration programs, requiring up-to-date records integrated into a national database. Effective registration enables mass vaccination, rapid stray identification, and community notifications for lost pets, supporting community-based interventions to reduce stray populations and rabies incidence.



Local ordinances illustrate implementation: Manila’s Ordinance No. 8587 (2020) requires microchipping, annual registration, and visible ID tags, with fines of ₱2,000-₱5,000 for violations. The City Veterinarian maintains a central electronic database, allowing rapid owner notification and real-time management of lost or stray dogs. Mandaue City’s ordinance (2020) combines affordable registration (₱50-₱100), microchipping, and digital tracking, with enforcement via impounding and community alerts, though compliance initially ranged 40-60%. Pasig City’s Ordinance No. 37 (2021) mandates annual registration, vaccination proof, microchips/RFID tags, and a centralized database, with fines up to ₱3,000 and 7-day holding periods for unclaimed strays. These programs demonstrate that combining microchipping, digital tracking, and LGU enforcement strengthens rabies prevention, responsible pet ownership, and stray dog management.

Raspberry Pi

Castillo et al. (2018) outlined the design of an automatic Wi-Fi controlled seeder, emphasizing wireless communication and automation using Raspberry Pi core concepts shared by PawTect.

Aldea et al. (2022) and Martinez et al. (2023) advanced the use of Raspberry Pi in ground fissure detection and biometric authentication, respectively, often integrating advanced



algorithms and wireless modules. Their success in real-world deployments supports the viability of deploying Raspberry Pi for intelligent animal detection and alerting.

Aranas et al. (2023) investigated web-based multimedia information displays using Raspberry Pi, illustrating its capacity for real-time notification and community information dissemination, crucial for the alert system in PawTect.

LoRa Wireless Transmission Systems

Further local literature focuses on performance analysis, system optimization, and specific IoT applications using LoRa, offering critical evaluations that can inform the refinement and deployment of PawTect.

Teves and Bañalacia (2021) conducted a performance analysis of LoRa-based networks in commercial environments, examining interference, data collection accuracy, and network reliability. Their findings highlight critical considerations for maintaining robust wireless communication in dense or interference-prone environments, which is essential for PawTect’s consistent performance in community settings with multiple wireless devices.

Alipio and Bures (2024) reviewed current testing and performance evaluation methodologies of LoRa and LoRaWAN in IoT



applications. Their classification of issues and future directives offers a roadmap for enhancing system designs to overcome challenges such as network scalability, power consumption, and latency. This literature supports PawTect’s need for continuous improvement to ensure that detection and alert systems remain responsive and energy-efficient.

Ramos and Segovia (2021) implemented a garbage collection notification system using LoRa, demonstrating the effectiveness of community-based alert systems leveraging long-range wireless communication. The practical application of community notifications and sensor-triggered alerts is directly applicable to PawTect’s model, which requires timely notification dissemination for stray dog detection.

Abella and Enriquez (2024) developed a community-based flood alert system using long-range technology, emphasizing the importance of local community involvement and real-time alerting for disaster preparedness. Their work stresses the social dimension of technology deployment, which is crucial for PawTect’s success as a community-driven initiative that requires active participation and trust from local residents.

Collectively, these local literatures offer critical evaluations and practical implementations that reinforce the feasibility and necessity of PawTect. They guide the system’s



design to ensure reliability, user engagement, and adaptability to local community dynamics.

Real-Time SMS Notification

A review of Local Literature reveals that SMS is the de facto standard for telemetry and remote monitoring in the Philippines due to the country's specific telecommunications landscape.

Batitis (2019) provides a critical sociological grounding for this technology. In investigating IoT for Barangay Labas, Santa Rosa, Laguna, Batitis observed that while smartphone penetration is high, data availability is inconsistent. Many constituents rely on prepaid load and may not have active data subscriptions at all times. However, the ability to receive SMS is universal across all mobile phone users. Therefore, Batitis concluded that SMS is the most inclusive platform for community notifications. This validates PawTect's "Community-Based" descriptor.

In terms of disaster risk reduction, Villanueva and Cruz (2024) assessed SMS rainfall warnings in Marikina, finding them highly effective for preparedness. Mamaril and Cruz (2016) found similar positive perceptions in Davao del Norte regarding SMS disaster warnings. Furthermore, Tallada (2020) discussed SMS information gateways for farmers in Laguna, centralizing the



concept of an "Information Gateway" between the physical world and the community.

Regarding security sensors, Gerpacio (2018) discussed security on doors and windows using magnetic sensors to trigger text messages. Consortia Academia (2022) explored Arduino-based fire alarm systems with GSM connectivity, highlighting the "life-safety" aspect of SMS independent of WiFi.

Several published local papers also utilize SMS for various triggers. Bondoc (2023) utilized it for attendance via QR codes. Natividad and Mendez (2018) used ultrasonic sensors for flood warnings. Tamayo (2022) and Cruz and Villaverde (2019) relied on PIR motion sensors for security systems. Tolentino et al. (2021) utilized image processing for flood detection, though focused on regression analysis.

Foreign Related Studies

Computer Vision Model

Recent foreign studies have increasingly demonstrated the capabilities of computer vision and deep learning in advancing animal detection, dog behavior modeling, and canine breed identification. For instance, Galeno, Sale, and Martin (2023) developed a system for SMS-based dog detection in residential areas using YOLOv5 integrated with ML.Net. Their work



illustrates the practical application of real-time object detection frameworks in community settings, enabling local authorities to monitor stray or roaming dogs efficiently through streaming cameras, thereby addressing public safety and urban management concerns.

Beyond applied detection, Ferrante et al. (2021) provided a comprehensive survey of animal detection and classification using computer vision technologies, analyzing the state-of-the-art approaches including support vector machines, deep learning, and big data architectures. Their study emphasizes how intelligent systems for animal monitoring are becoming central to smart city initiatives and ecological monitoring, reflecting a broader trend in integrating computational models with large-scale urban and environmental datasets.

Focusing on behavioral analysis, Ehsani et al. (2018) explored modeling dog behavior from visual data using CNNs and video-based detection techniques. This research highlights the potential of deep learning not only for recognizing static features such as breed or appearance but also for capturing dynamic behaviors and activity patterns, opening avenues for automated monitoring of animal welfare and behavior understanding.



In a methodological advancement, Zoph et al. (2018) proposed learning transferable neural architectures for scalable image recognition. While not limited to animals, their work has significant implications for designing highly efficient and generalizable models that can be adapted to animal detection, breed classification, and behavior recognition tasks, underscoring the importance of model architecture optimization in computer vision research.

Finally, Perez et al. (2025) addressed challenges in standardizing canine breed data within veterinary records, demonstrating that computer vision can serve as a robust alternative for breed assignment. Their findings highlight the intersection of AI and veterinary health, showing that visual-based breed recognition can complement traditional record-keeping and improve the accuracy of breed identification, which is critical for both clinical care and large-scale breed studies.

Collectively, these studies illustrate a converging trend in the application of advanced computer vision, deep learning, and scalable model architectures toward the automation of animal detection, behavior modeling, and breed identification. They demonstrate not only the technical sophistication of modern frameworks such as YOLO and CNNs but also their tangible



societal, ecological, and veterinary impacts, providing a solid foundation for further research in intelligent, data-driven animal monitoring systems.

Dog Registration Program

Choi and Koo (2020) found that in Korea, dog owners' emotional attachment influences awareness and participation in national registration programs. Owners with stronger bonds were more likely to register their dogs and preferred user-friendly registration methods. This suggests that registration systems are most effective when designed to consider both emotional engagement and practical ease, particularly for technology-based monitoring.

The One Health approach highlights the need for reliable pet population data to guide zoonosis prevention, welfare, and stray dog management. Surveys in central Italy found that dog populations are larger than previously thought, with only 75% properly registered. Registration was more common among neutered, purebred, urban dogs receiving regular veterinary care. Factors such as dog origin, feeding type, and household characteristics influenced ownership and care practices (Carvelli et al., 2020).

In Chile, Garde et al. (2022) reported low compliance with mandatory identification and registration, with only 25%



microchipped and 31% registered by 2021. They stressed the need for baseline surveys, measurable goals, public education, and enforcement to improve national dog management.

Ma et al. (2020) evaluated a multi-year Dog Population Management program across thirteen communities, finding that strengthened responsible ownership, microchipping, and sterilization reduced dog attacks and euthanasia, highlighting the importance of continuous monitoring and data-driven evaluation.

In Linzhi, Tibet, enhanced registration, monitoring, and systematic deworming from 2017-2019 increased dog registration from 75.9% to 98.6%, reduced stray populations, and cut infection rates from 7.3% to 2.8%, demonstrating the public health benefits of accurate identification and management (Wang et al., 2021).

Overall, these studies show that effective dog population control relies on comprehensive registration, identification, monitoring, and owner engagement, which support welfare, public health, and community safety.

Raspberry Pi

Several international studies have established the versatility and efficiency of the Raspberry Pi platform in computer vision,



wireless communication, and automation core components for systems like PawTect.

Ozkan et al. (2021) demonstrated the feasibility of using the Raspberry Pi for object detection and recognition in unmanned aerial vehicles, utilizing machine learning algorithms such as Faster-RCNN and SSD MobileNet V2. Their work emphasizes the capability of Raspberry Pi to deploy deep learning models for real-time object (including animal) recognition, which is directly relevant to dog detection tasks in PawTect.

Nagpal et al. (2018) and Mustakim et al. (2019) explored facial and object recognition using OpenCV on Raspberry Pi devices, confirming their effectiveness in image processing and real-time recognition, even on low-cost hardware. This underpins the computer vision aspect of PawTect, ensuring that missing or stray dogs can be detected efficiently in a community setting.

Pereira and Chaari (2025) evaluated the Raspberry Pi AI HAT+ for computer vision applications utilizing YOLO, a leading object detection algorithm. Their findings support the selection of advanced neural networks for accurate and fast detection essential for the responsiveness of an alert system.

Gragasin et al. (2017) and Deshmukh & Shinde (2016) investigated Raspberry Pi in wireless sensor networks, focusing



on throughput and integration with other platforms such as Arduino and Zigbee. These studies validate the use of Raspberry Pi in community-based wireless alert systems through mesh and sensor network configurations.

SaviÄ† & RadonjiÄ† (2015) and Nguyen Thi Huong et al. (2018) showcased the Raspberry Pi's role in environmental sensing and visual sensor networks, demonstrating its flexibility for real-time data acquisition, transmission, and monitoring.

Kam et al. (2023) provided comparative performance data between Raspberry Pi and similar boards, affirming its reliability and scalability for community-scale projects.

LoRa Wireless Transmission Systems

Wireless communication technologies, especially LoRa (Long Range), have been extensively studied and implemented for remote monitoring and alert systems. These studies provide foundational knowledge for the communication backbone of PawTect, which aims to detect and alert communities about missing and stray dogs.

Z. Qiang et al. (2025) developed a monitoring system for the stress state of power cables using LoRa wireless spread spectrum technology. This study demonstrated the effectiveness of LoRa in providing long-distance, low-power, and reliable data transmission for real-time monitoring in challenging



environments. The research highlighted the technology's robustness in maintaining communication with minimal packet loss, which is crucial for continuous monitoring systems like PawTect that require dependable transmission of detection alerts across community networks. The study's focus on stress monitoring through wireless sensors parallels the need for accurate, real-time detection in animal monitoring systems.

Similarly, A. Storozhko et al. (2025) investigated the use of LoRa for managing geospatial data transmission in wireless mapping systems. Their work on decentralized mapping and synchronization using LoRa communication showcased the technology's capacity to handle real-time data with optimization for autonomous aerial vehicles. This is significant for PawTect since it involves location-data transmission for detected dogs, which could benefit from geospatial accuracy and synchronization for community alerts. The use of ESP32 microcontrollers and UTM mapping in their study also aligns with the integration of microcontroller-based systems, like Raspberry Pi, in IoT applications.

In the domain of wildlife and environmental monitoring, Aditya et al. (2025) presented an efficient IoT system for wildlife monitoring using LoRa and Raspberry Pi integrated with EfficientDet, a computer vision algorithm for object detection.



This study is directly relevant to PawTect, as it combines wireless long-range communication with embedded computer vision for real-time animal detection and tracking. The use of Raspberry Pi as a local processing unit enables efficient image processing and decision-making at the edge, which reduces latency and reliance on constant internet connectivity. Their approach validated the feasibility of using low-cost, low-power devices for wildlife and stray animal monitoring, supporting the use of similar technologies for community-based dog detection systems.

D. Shen et al. (2024) explored real-time data wireless communication using LoRa technology in their research. They emphasized the importance of reliable transmission protocols and low-power consumption in wireless sensor networks. Their work reinforces the suitability of LoRa for applications like PawTect, where devices must operate for extended periods in potentially remote or distributed community environments without frequent maintenance or power replacement.

E.-H. Meftah et al. (2024) extended the application of LoRa by integrating it with BLE (Bluetooth Low Energy) and Wi-Fi for real-time ECG monitoring. Their innovative wireless solution demonstrated multi-technology interoperability for health monitoring systems, which can be extrapolated to animal health



and safety monitoring in PawTect. The integration of different communication protocols could influence future iterations of PawTect, enhancing device connectivity and alert dissemination through multiple wireless channels for better community reach.

Other studies such as those by Gao et al. (2023) and Ma et al. (2025) focused on wireless node devices and lightning monitoring systems using LoRa, respectively, showing the versatility of LoRa in various monitoring scenarios. Gao et al.'s implementation of ad hoc networking for wireless nodes could inspire mesh networking approaches in PawTect, allowing multiple detection nodes to communicate and relay information in a decentralized manner, improving system resilience.

These foreign studies collectively highlight the growing confidence in LoRa and embedded computing for long-range, low-power, real-time monitoring systems. They provide a strong technical foundation for PawTect's communication infrastructure and serve as a reference for integrating computer vision with wireless sensor networks to create an efficient stray dog detection and alert system.

Real-Time SMS Notification

Foreign studies provide high-tech precedents and empirical demonstrations of SMS integration in IoT systems. A. and Wibowo (2021) developed a smart aquarium monitoring system where the



Raspberry Pi acted as the central processing unit, similar to the proposed PawTect system. They utilized an SMS gateway not just for alerts, but for remote control (feeding). Their success proves that the Raspberry Pi architecture is perfectly capable of handling the serial communication required by GSM modules while simultaneously processing sensor data. This directly supports the technical feasibility of the current study's hardware selection.

Similarly, Zualkernan and Al-Rousan (2019) utilized GPRS and SMS for air pollution monitoring in mobile sensor arrays. Their study is significant because it addresses the "mobility" aspect of IoT assets. They concluded that SMS was the most cost-effective method for transmitting low-bandwidth alert data over long distances.

In the realm of security, Gao and Xu (2021) implemented an intelligent security system based on Raspberry Pi and GSM. Their methodology involved using the GSM module to send alerts when a specific security breach was detected. Crucially, they noted that SMS alerts were received faster than email alerts in 90% of their test cases. Jain et al. (2019) corroborated this in their Raspberry Pi surveillance system, noting that for motion detection, the "interrupt" nature of an SMS is vital for prompting immediate user action.



Specific to animal detection, Ahmad et al. (2024) and Dhanaraju and Kumar (2025) both utilized YOLOv8 algorithms. While their primary focus was computer vision, both studies integrated communication modules to alert farm owners of intrusions. Dhanaraju and Kumar explicitly mention the use of SMS alerts as a mechanism to notify farmers of "animal intrusion" in real-time. Anwar and Sultana (2020) and Maguluri et al. (2022) applied this to crop protection; their results highlighted a reduction in crop damage because farmers could respond immediately upon receiving the text.

Mwangi (2019) demonstrated that GSM technology is sufficient for geofencing-alerting authorities when an animal crosses a specific boundary. Finally, Vargas and Trovati (2022) utilized Raspberry Pi for low-cost intelligent camera traps, noting that GSM/SMS is the superior choice for "peri-urban" or rural areas where cellular towers are present but 4G data might be expensive or unreliable.

Local Related Studies

Computer Vision Model

Local research has made significant strides in applying computer vision, machine learning, and intelligent systems to animal monitoring, object recognition, and visual surveillance. Luta, Baldovino, and Bugtai (2017) investigated image



preprocessing techniques for color machine vision (CMV) systems, proposing a quick color averaging approach that enhances image quality and reduces noise. Their study underscores the importance of preprocessing in improving the accuracy and reliability of machine vision systems, particularly in applications requiring precise color analysis and inspection.

In a complementary approach, Reyes and Dadios (2004) explored dynamic color object recognition using fuzzy logic. By incorporating uncertainty handling into color-based recognition, their work demonstrated early advances in adaptive and intelligent image processing, laying the groundwork for more complex recognition systems capable of operating in variable environmental conditions.

Focusing specifically on animal identification, Estrella, Lucban, and Caya (2024) applied Convolutional Neural Networks (CNNs) for image recognition of different hamster breeds. Utilizing architectures such as VGG16 on devices like webcams and Raspberry Pi, their system achieved high classification accuracy and demonstrated the feasibility of deploying AI-driven animal recognition in resource-constrained environments, highlighting the potential for real-world monitoring in both research and domestic contexts.



Expanding beyond animals, Susa et al. (2022) addressed the challenge of person detection under low-illuminance conditions using high dynamic range imagery. Their study integrated deep learning and heuristic algorithms to enhance object recognition and tracking in visually challenging environments, illustrating how robust computer vision techniques can overcome real-world limitations such as variable lighting—a consideration that is equally relevant for animal monitoring applications in poorly lit or outdoor settings.

Finally, Pabico et al. (2015) explored automatic identification of animal breeds and species through bioacoustics and artificial neural networks, demonstrating that auditory data, when combined with computational intelligence, can serve as a complementary modality to visual recognition systems. Their work highlights the versatility of machine learning methods in monitoring animal populations, emphasizing the potential for multi-sensor integration in intelligent animal monitoring systems.

Collectively, these local studies showcase a continuum of innovation in image processing, deep learning, and intelligent recognition systems. They highlight not only the technical sophistication of contemporary approaches—ranging from fuzzy logic to CNNs—but also the practical challenges and



opportunities of implementing these systems in real-world Philippine contexts, providing a strong foundation for further research in intelligent, adaptive, and multi-modal animal monitoring and surveillance.

Dog Registration Program

Despite ongoing efforts, rabies persists in the Philippines, with dogs remaining the main source of transmission. Surveys in Central Luzon found that 54% of households owned dogs, with a dog-to-human ratio of 1:3.7. However, 35.9% of dogs were unvaccinated, only 3.5% were sterilized, and 31% roamed freely. Knowledge of rabies was higher among dog owners and lower among those with little education, highlighting the need for targeted awareness, monitoring, and registration programs (Dizon et al., 2022).

The Bohol Rabies Prevention and Elimination Project (BRPEP) combined mass vaccination, dog surveys, registration, school-based education, and community engagement. Registration fees supported program sustainability and facilitated tracking of owned and stray dogs, showing the effectiveness of integrated technical and educational interventions (Lapiz et al., 2012).

Studies of LGU implementation, such as in Davao del Sur, reveal variation in vaccination, registration, field control, and veterinary services, with only some areas maintaining dog



pounds. These findings underscore the need for centralized databases, standardization of practices, and enhanced veterinary capacity (Franco et al., 2024).

Economic barriers also limit registration. Birhane et al. (2016) found that while 86% of households were willing to pay for rabies vaccination, only 68% would pay for dog registration, and mean willingness-to-pay was low (₱29.13). Higher income, education, and rabies awareness increased willingness, indicating that subsidized or low-cost registration programs could improve compliance.

On Panglao Island, Tapdasan and Salces (2025) reported that only 44% of dog owners participated in registration, and 37% leashed their pets, with mongrels and free-roaming dogs more likely to contribute to rabies risks. Purebred dogs receiving regular care had higher vaccination and registration rates, highlighting socioeconomic and breed-specific disparities. The study emphasizes the need for intensified campaigns to improve responsible dog ownership, registration, and stray management to reduce rabies incidence.

Raspberry Pi

In the Philippine context, local studies have successfully applied Raspberry Pi in automation, monitoring, and intelligent systems.



Tolentino et al. (2022) developed an automated egg incubator using Raspberry Pi with camera-assisted candling and R-CNN-based maturity detection, showcasing the device's capability for real-time image analysis and automated decision-making closely paralleling the detection objectives of PawTect.

Romero et al. (2021) designed a smart roof mechanism for food preservation, and Medina et al. (2018) implemented IoT-based electricity monitoring, both reinforcing Raspberry Pi's suitability for IoT and community-oriented solutions.

Tabuada et al. (2018) and Alvarez et al. (2020) worked on home automation and controllable switches, demonstrating practical, user-friendly Raspberry Pi-based systems for the local setting.

LoRa Wireless Transmission Systems

Local research efforts have increasingly explored the application of LoRa and wireless sensor networks for various community monitoring systems, setting a precedent for PawTect's implementation in a local context.

Dumandan et al. (2024) developed a LoRa-based real-time train tracker with arrival time estimation, integrating GPS and mobile application alerts. Their system demonstrated how LoRa-enabled devices could provide timely and accurate location



tracking with community notifications through smartphones. This local study is relevant to PawTect as it highlights the integration of real-time location data and user-friendly mobile alerts, which can be adapted to notify communities about stray or missing dogs' positions.

Cabacungan et al. (2023) applied LoRa sensors for monitoring growth parameters in black rice hydroponics with near-cloud data storage. Their work shows the effective use of wireless sensor networks in precision agriculture, emphasizing real-time data collection and remote monitoring. The success of this system reinforces the feasibility of deploying LoRa-based sensor networks in local environments for continuous monitoring, a principle that can be applied to PawTect for persistent observation of stray dog movements.

Macaraeg et al. (2020) explored a LoRa-based mesh network designed for off-grid emergency communications. Their research demonstrated the potential of decentralized LoRa networks to operate independently of traditional infrastructure, ensuring communication continuity in remote or underserved areas. This is particularly useful for PawTect in rural or semi-urban communities where cellular coverage may be limited, and community members rely on mesh networking for timely alerts.



Canal et al. (2022) developed a wireless anti-theft and tracking device for wallets with alert notification via an Android application, integrating wireless communication and mobile alerts at the community level. The concept of community engagement and alert dissemination through mobile platforms in this study parallels PawTect’s approach to keeping communities informed about stray dog detections.

These local studies provide valuable insights into adapting LoRa technology and wireless sensor networks into community-based monitoring and alert systems. They demonstrate the adaptability and scalability of such technologies in diverse local applications, supporting the contextual relevance of PawTect.

Real-Time SMS Notification

Local studies (theses, capstones, and prototypes) reinforce the practical application of SMS in the Philippine setting.

In agriculture, Garcia and Ramos (2022) developed a poultry monitoring system in Batangas, while Alcantara (2024) developed a crab water monitoring system in Palawan. Both unpublished theses heavily relied on SMS notifications to alert farmers of critical levels. A ResearchGate (2025) publication on smart poultry for small-scale farmers in Ilocos Norte echoes this, prioritizing SMS because farmers are very comfortable reading



text messages. Abian et al. (2025) also used SMS for a prototype solar-powered water pump.

In the security and governance domain, the Lyceum of the Philippines University (2022) developed a GSM-based burglar alarm system. Delos Santos (2020) and Mendoza and Torres (2021) applied this to vehicle anti-theft systems. Their capstone projects demonstrated that SMS is fast enough to alert a motorcycle owner of theft in progress, allowing for rapid response. Corpuz and Magsino (2023) from Mapúa University explored systems for LPG leakage detection using SMS.

Regarding administrative applications, Brin (2021) studied Barangay helpdesk systems and text blasts, noting that current systems are manual. Hinunangan LGU (2025) developed an automated water billing system using SMS, and PUP (2021) created an automated attendance system.

Finally, regarding vision-based triggering, Ramos and Dizon (2023) implemented face recognition for smart door locks. However, the synthesis of these studies highlights a gap: while SMS is common (Abian et al., Hinunangan LGU), and computer vision is emerging (Ramos & Dizon), there is a scarcity of local systems combining high-level Computer Vision (like YOLO) specifically with SMS alerting for animal detection.



Synthesis of the Study

The survey of foreign and local literature reveals a strong convergence in the use of computer vision, wireless communication, microcontroller-based processing, and community-oriented notification systems for animal monitoring and public safety applications. Across numerous studies, computer vision—especially CNN-based architectures, YOLO object detectors, and transfer-learning models—has consistently demonstrated high accuracy and real-time performance in identifying dogs, animals, and moving objects in varied environments. These findings underscore the feasibility of applying deep learning to community-based dog detection, as illustrated by works that successfully integrate lightweight models with resource constrained devices such as the Raspberry Pi. At the same time, literature on dog registration, both local and foreign, emphasizes that effective dog management relies on accurate records, centralized identification systems, and community participation. Compliance with legal mandates such as the Philippine Anti-Rabies Act and international guidelines such as those of the WOAHP further highlight the need for tools that support systematic identification, rapid reporting, and active monitoring of stray and owned dogs. Complementing the computer vision evidence, studies on embedded systems consistently



validate the Raspberry Pi as a robust platform for real-time processing, rapid prototyping, and integration with communication modules including LoRa and GSM. Its widespread adoption in monitoring systems—from wildlife detection to security and agricultural applications—demonstrates that it can reliably execute detection algorithms while coordinating wireless data transmission. Likewise, both foreign and local research on LoRa communication confirms its superiority for long-range, low-power, and highly reliable transmission, even in signal-challenged or rural environments. These characteristics make LoRa ideal for decentralized community monitoring systems that require persistent communication across neighborhoods without dependence on costly cellular data. Finally, literature on SMS notification systems—particularly in the Philippine context—strongly affirms that SMS remains the most inclusive, reliable, and immediate communication channel, outperforming mobile push notifications in areas with unstable data connectivity. Local studies repeatedly demonstrate the success of SMS-based alerts in agriculture, disaster warning, security systems, and governance applications, reinforcing its practical value for real-time community alerts. Taken together, the related literature and studies provide clear justification for the development of PawTect. First, the consistent accuracy



of modern computer vision models establishes the technical viability of automated dog detection and breed classification in real-world environments. Second, evidence on national and local dog registration programs reveals persistent gaps in tracking, owner compliance, and stray management—gaps that technological integration can address. Third, hardware feasibility studies prove that Raspberry Pi-based systems can process visual data and communicate wirelessly with minimal latency, supporting the project’s goal of real-time operation. Fourth, decades of work on LoRa show that community-wide coverage and long-range communication are achievable at low cost and low power, which is essential for a proactive community alerting system. Lastly, SMS remains the most dependable communication method in the Philippines, ensuring that alerts reach owners and local authorities regardless of data availability or smartphone type. These converging strands of evidence justify the necessity and timeliness of an intelligent, community-based dog detection, registration, and alert system—one that enhances public safety, promotes responsible pet ownership, assists LGUs, and strengthens animal welfare initiatives.



CHAPTER III: METHODOLOGY

This chapter presents the methodology employed in this study, detailing the systematic processes involved in the development, training, and implementation of a computer vision-based dog identification system with an integrated canine database and LoRa wireless transmission system for recognizing missing and stray dogs. To accomplish these objectives, the study is structured into multiple phases, each encompassing specific tasks and procedures that are built upon one another.



Figure 2. Deployment of the Prototype



IPO Model

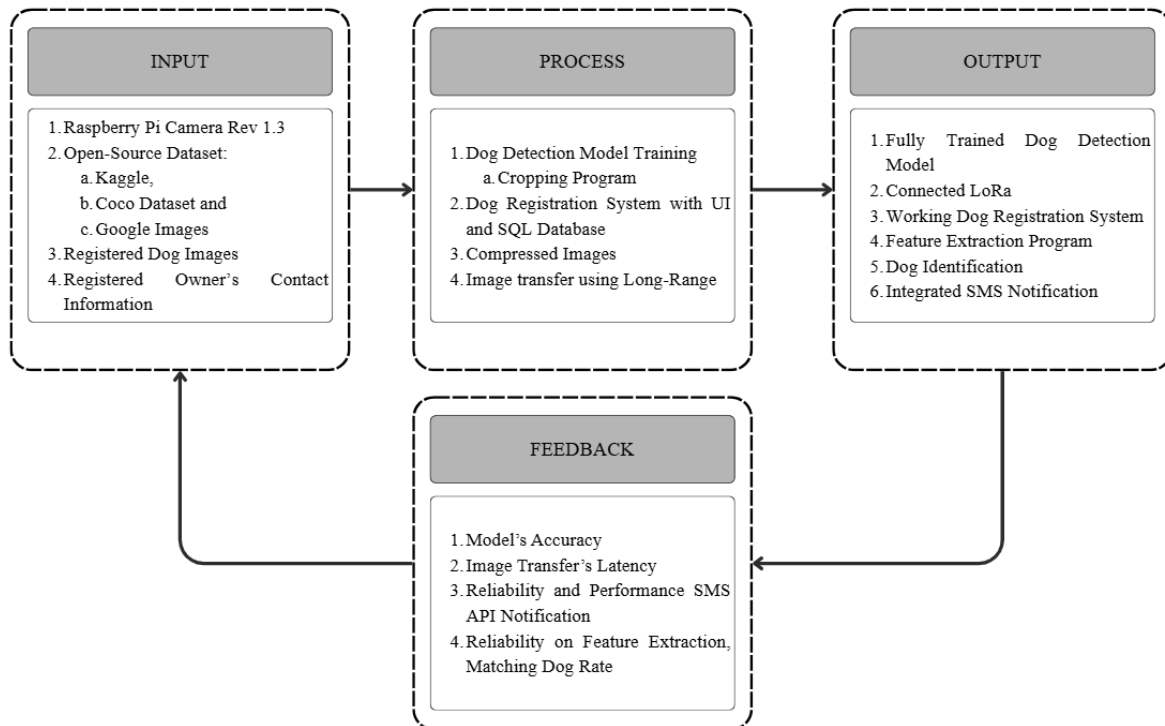


Figure 3. Input-Process-Output Model

Figure 2 illustrates the Conceptual Framework of the study, utilizing the Input-Process-Output (IPO) model with a Feedback Loop to delineate the functional flow of the study system. The diagram serves as a logical blueprint for the integration of hardware components and software algorithms, structured as follows:

Input-this stage identifies the foundational resources required, including hardware such as the Raspberry Pi Camera Rev 1.3, and software assets like the open-source datasets (Kaggle, COCO, and Google Images) used for model training. It also



encompasses the essential community data, specifically the registered dog images and owner contact information.

Process-this section represents the core "Edge Intelligence" of the system. It details the transformation of raw inputs through YOLOv8n model training, the implementation of the SQL-based registration system, and the technical handling of compressed image transmission over the LoRa (Long-Range) wireless network.

Output-the results of the system are presented here, characterized by a fully trained detection model and an integrated communication bridge. The primary outputs include the real-time identification of dogs, successful feature extraction, and the automated delivery of Telegram/SMS notifications to the registered owners.

Feedback-this iterative component ensures continuous system optimization. It focuses on the Statistical Performance of the project, measuring critical metrics such as Model Accuracy, Transmission Latency, and the Reliability of the API notifications to refine the inputs and processes for maximum efficiency.



Block Diagram

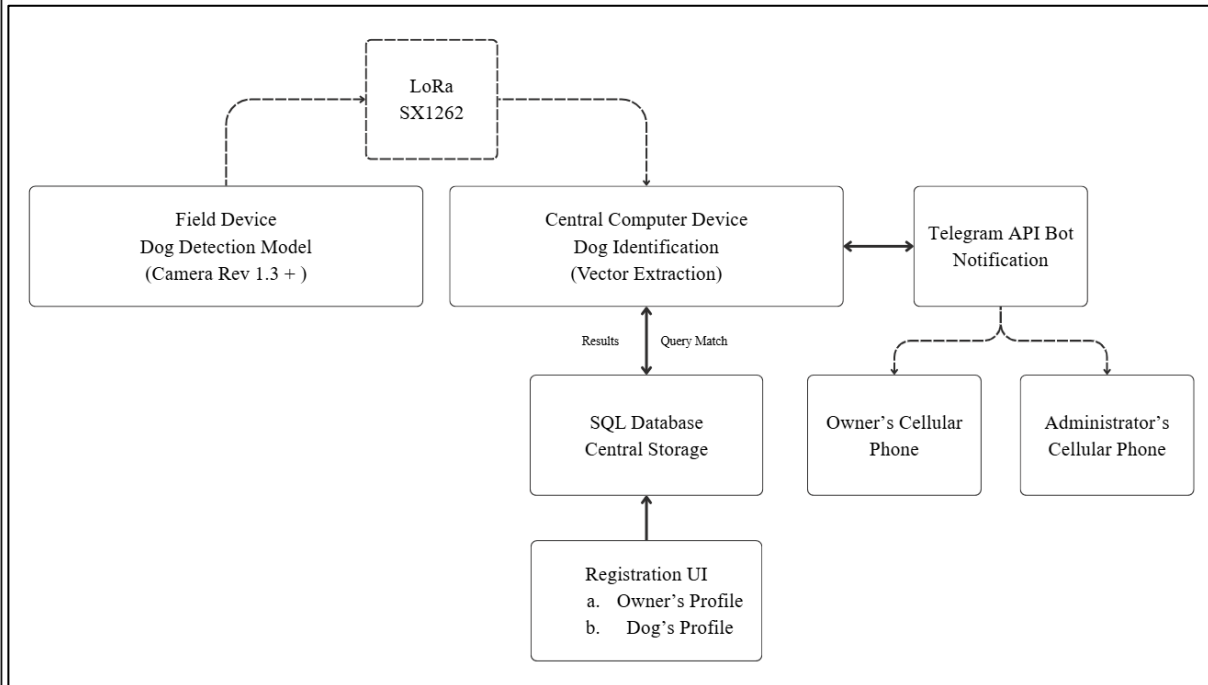


Figure 4. Block Diagram

The block diagram presented in Figure 3 illustrates the technical architecture and data communication flow of the system, highlighting the seamless integration between field-level hardware and central processing units. The workflow commences at the field device, where a Raspberry Pi Camera Rev 1.3 captures real-time footage to be processed by a localized dog detection model program. Upon successful detection, the data is encapsulated and transmitted via the LoRa SX1262 wireless module, which facilitates long-range communication to the central hub.



The receiving end consists of the central computer device, which is responsible for the computationally intensive task of dog identification through vector extraction. This unit performs a *Query Match* by interacting with the SQL Database and Central Storage, which houses the reference data for the system. This database is populated via a dedicated Registration UI, where owner and dog profiles are archived. Once the system confirms a match between the field data and the database records, the results are forwarded to the Telegram API bot. The bot triggers the final output of the system, delivering notifications to both the owner's or the administrator's cellular phone, completing the end-to-end identification and alert cycle.

Conceptual Design

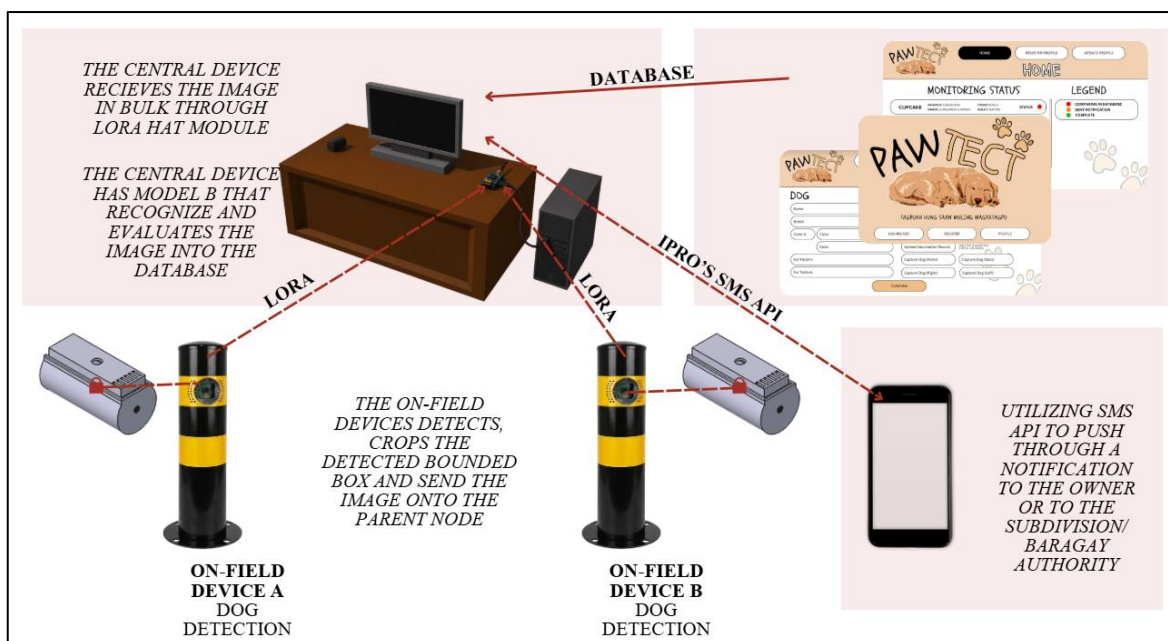


Figure 5. Conceptual Design



The system is engineered as an edge-to-centralized network topology optimized for open-set object detection, long-range remote telemetry, and real-time automated community alerting. The overall operational infrastructure is divided into two primary environments: the On-Field Detection Nodes, which are concealed within standard urban infrastructure components, and the Central Monitoring Base Station, which acts as the core processing hub. Together, these environments establish a continuous, closed-loop pipeline that moves data from raw camera acquisition to edge preprocessing, long-range wireless transmission, centralized high-dimensional vector matching, and localized database management. This dual-environment design minimizes field power and bandwidth consumption by delegating complex feature extraction and data routing to the central machine, while ensuring rapid localized responses at the edge.

The field deployment environment relies on structural camouflage and ruggedization by integrating the hardware sub-assemblies inside standard street bollards. Each bollard operates as an independent edge-computing node where an internal camera module captures real-time video through a low-profile cutout, maintaining an unobstructed line of sight across the designated monitoring zone. Frame by frame, an optimized object



detection model running locally on the microcomputer scans the visual input; once a target subject is localized with a confidence score exceeding a set minimum threshold, the system extracts the spatial coordinates of the bounding box and executes an immediate matrix crop. This isolated target region is retained while the environmental background is discarded, drastically reducing the payload size. To comply with radio band transmission limits, the cropped image array is converted to a compressed byte-stream and split into sequenced data packets wrapped with validation headers, which are then transmitted via a slave wireless transceiver using a high-gain antenna mounted through the bollard's top cap to avoid internal structural metal interference.

At the central device, a master transceiver remains in a continuous listening state on the network's exact radio frequency band to serve as the system's wireless uplink channel. As fragmented packets arrive from the field, a centralized reassembly script reads the serial communication buffer, validates incoming fragments using error-detection algorithms like Cyclic Redundancy Checks, and reorders the blocks in memory before decoding the byte sequence back into a structured image matrix. This reassembled target crop is immediately forwarded to a high-dimensional feature extraction engine that calculates



a 512-dimensional vector embedding representing the subject's unique physical traits. The system then evaluates this field vector using directional cosine similarity against a local database operating under an open-set paradigm: if the calculated similarity score breaks a designated acceptance threshold against an existing profile, a positive identity verification is flagged, whereas a persistent low similarity score across all known profiles triggers the dynamic registration of a new index entry for an unregistered stray target.

To ensure data privacy and high-speed execution, all user management profiles, raw reference imagery, system state metrics, and location tracking records are maintained within a standalone, local database engine stored directly on the central computer's native file system. Operators interface with this centralized environment via a local desktop web graphical user interface that consolidates live monitoring statuses, database registration tools, and administrative configuration options into a self-contained system. Closing the loop between target identification and immediate field action, the base station deploys an automated, dual-channel alerting mechanism that concurrently triggers two separate alert pipelines. For users with active data connections, the system executes an asynchronous HTTPS request to the Telegram Bot API to deliver a



rich push notification containing a structured text summary—including target location, timestamp, and owner metadata—paired with the exact high-resolution visual crop captured by the field node. Simultaneously, to bypass mobile internet dependencies and guarantee delivery under poor network conditions, the system maps the transaction details to the IPROG SMS API network gateway, structured as a concise, critical text summary transmitted securely over standard cellular infrastructure directly to the owner's registered mobile number.

Phases of the System

Phase I: Computer Vision Model Training

The primary goal of the first phase of the research study is to develop a robust computer vision model capable of accurately detecting and identifying dogs from diverse image datasets.

First, the model must differentiate dogs from other entities and capture their figure. The identified canine cropped image is analyzed for feature extraction.

Then the system extracts a set of biometric features from the image using a vector extraction program. The extracted feature vector is compared to a database of registered owners and their dogs. The system computes a similarity score using



distance metrics such as cosine similarity. If a match is found, the system fetches the owner’s contact information of the identified dog.

Step 1: Gather Open-Source Dataset: Kaggle, Coco Dataset, and Google Images.

The study utilizes open dataset sources that support each stage of the species detection. The datasets selected provide a diverse range of images, ensuring that the model learns to generalize across different environments, poses, and visual conditions.

1. COCO Open Database. This dataset serves as the foundation for species-level detection. It contains a wide variety of animal and non-animal classes, enabling the system to reliably distinguish dogs from other objects in complex scenes. Using this dataset, the model learns to detect the presence of a dog and capture its full figure for further analysis.
2. Kaggle. This hosts multiple open-source dog image datasets contributed by the community, covering species-level and breed-level tasks. These datasets are utilized primarily for testing and validation, as they provide diverse images and scenarios not seen during training. This helps evaluate the model’s generalization performance across different



environments, camera settings, and dog appearances, ensuring that the system performs reliably in real-world conditions.

3. Google Images (Randomized/Unlabeled Dataset): This data source serves as a repository for random imagery. Images gathered from this platform are completely unrelated to dogs. These may be street scenes, everyday objects including humans and animals, and varying environmental backgrounds. This portion of the dataset remains completely unlabeled and unannotated. By introducing these arbitrary, non-target backgrounds, the model is trained to effectively distinguish the specific features of the target subject from unrelated, real-world environments.

Step 2: Create a Classification Model for Dog Detection.

To develop a robust dog detection model, three primary datasets were utilized: the COCO Open Dataset, a Kaggle Dataset, and a randomized collection from Google Images. The COCO dataset provided a wide variety of annotated images, from which the 'dog' category was extracted to serve as the foundation for species-level detection. The Kaggle Dataset was subsequently incorporated to supplement the training data with localized and high-resolution variations of canine images, ensuring that the model could account for different dog breeds, orientations, and



lighting conditions. Lastly, Google Images was utilized to gather a randomized, non-target dataset. These images are completely unrelated to dogs—consisting of everyday objects, street scenes, and varying environmental backgrounds—and were left entirely unlabeled and unannotated. This serves as a negative control within the training pipeline, forcing the model to distinguish specific canine features from arbitrary real-world backgrounds, thereby minimizing false-positive detections and increasing overall model robustness. For the optimization of the dog detection model, the consolidated dataset was partitioned into training and validation sets. A total of 4,782 images were utilized, distributed into 4,559 images for the training phase and 223 images for the validation phase.

The specific architecture selected for this computer vision pipeline was YOLOv8n (Nano). This choice was dictated by the strict physical and computational limitations of the target deployment hardware, as the final trained model is designed to operate locally on a Raspberry Pi Zero 2 W.

The Raspberry Pi Zero 2 W operates under severe hardware constraints, featuring a quad-core 64-bit ARM Cortex-A53 processor clustered with only 512 MB of shared LPDDR2 SDRAM. Larger network architectures, such as YOLOv8s, YOLOv8m, or YOLOv8l, require significantly higher computational overhead,



larger memory allocations, and intense floating-point operations per second (FLOPS) that exceed the capacity of this single-board computer. Attempting to deploy a heavier model on this hardware would lead to severe latency penalties, thermal throttling, or system crashes due to Out-Of-Memory (OOM) errors. By choosing the YOLOv8n variant, the parameter size and computational complexity are drastically minimized, ensuring viable processing speeds (frames per second) during local edge inference.

The structural layout and data routing for the network were configured using a dedicated dataset configuration file, which explicitly defined the directories of the partitioned data. The paths mapped the training and validation inputs directly to their respective local directories within the project structure. As specified in the configuration, the model was established as a single-class detector, targeting index 0 assigned to the label dog. Unlabeled and unannotated negative samples from the Google Images dataset were processed through the pipeline without corresponding bounding box text files. When these images are loaded during training, the YOLO framework treats them as background imagery containing zero object instances. This enforces a mathematical penalty on false-positive localized predictions within non-canine environments, maximizing overall model specificity.



The training pipeline was implemented in Python using the Ultralytics YOLO framework. To ensure continuity and resource efficiency, the training configuration was designed to resume from a previously saved checkpoint, leveraging transfer learning and prior state parameters.

1. The model was scheduled to train for 100 iterations (epochs) to allow sufficient convergence of the bounding box and classification loss functions.
2. Input images were standardized and resized to a resolution of 640 x 640 pixels, balancing spatial detail with computational efficiency.
3. A batch size of 16 was selected to regulate the memory load during forward and backward propagation passes, while-
4. Data augmentation techniques were enabled to dynamically introduce variations in orientation, scaling, and exposure during training, thereby enhancing the model's ability to generalize to unseen environments. Because the training workflow was explicitly mapped to execute via the Central Processing Unit, all training outputs, weights, performance metrics, and validation curves were systematically directed to a designated tracking



folder inside the repository project root directory for structured analysis.

Validation was performed after each epoch using a reserved subset of the datasets, with performance measured in terms of mean Average Precision (mAP), precision, and recall.

$$\text{Precision} = \frac{TP}{TP + FP'} \quad \text{Recall} = \frac{TP}{TP + FN} \quad AP = \int_0^1 p(r) dr \quad mAP = \frac{1}{N} \sum_{i=1}^N AP_i$$

Step 3: Create a Cropping Program to be Integrated on the Field Device using Python And OpenCV with the Classification Model

To bridge the gap between object detection and biometric identification on the field device, a dedicated image cropping and optimization program was developed using Python and OpenCV. This subsystem acted as an automated data pre-processor that directly handled the raw bounding box outputs generated by the YOLOv8n classification model. Operating locally on the Raspberry Pi Zero 2 W, the program continuously listened for high-confidence inference signals. When the classification model detected a dog instance and outputted its corresponding localized bounding box coordinates—defined by the normalized spatial parameters—the OpenCV-based script intercepted these



coordinates and dynamically translated them into pixel-based matrix regions.

Once the precise spatial boundaries were established, the program executed a region-of-interest (ROI) slicing operation on the raw input frame matrix. This operation instantly isolated the canine's figure from the surrounding environment, effectively discarding the randomized background elements, non-target objects, and peripheral noise that were intentionally allowed by the negative control dataset. By slicing the frame down to the exact bounding box dimensions, the program ensured that subsequent computational tasks were completely focused on the target animal, preventing irrelevant environmental data from skewing the downstream biometric calculations.

After the targeted dog figure was isolated, the cropping program applied a series of matrix normalization techniques to prepare the image asset for the feature extraction pipeline. Because raw captures from the field device's camera module varied due to moving distances or environmental conditions, the cropped ROI was passed through an OpenCV resizing function to standardize its dimensions to a uniform pixel resolution. This geometric standardization was critical because deep learning vector extraction programs required fixed-size array inputs to perform consistent tensor operations. Along with dimensional



normalization, the script adjusted color channel alignments and pixel intensities neutralize sudden exposure shifts. The final optimized, cropped canine image was then temporarily cached in the edge device's volatile memory, ready to be ingested by the vector extraction program for distance metric comparison and owner verification.

*Step 4: Create a Feature Extraction for the Identification
Model Program for Vector Comparison*

In this phase of the identification framework, the system bridges the gap between raw visual data and statistical evaluation by transforming cropped target images into standardized mathematical signatures called feature vectors. When a target is detected in the field, its bounding box image is passed into a multi-step preprocessing pipeline designed to eliminate environmental noise. The image is first resized to a uniform resolution and converted to grayscale, a critical step that strips away inconsistent lighting and color variations to focus purely on structural textures and forms. These pixel intensities are then normalized to a standardized floating-point scale and flattened into a one-dimensional array. To satisfy a rigid, fixed-dimensional vector architecture, the top structural elements are sliced from this array, yielding a compact and



highly consistent embedding that represents the unique physical layout of the subject.

Once this mathematical embedding is extracted, the system executes vector comparison to determine if the detected target matches an identity in the registration database. This is achieved by calculating the Cosine Similarity between the newly captured field vector and known reference vectors from the database. Rather than measuring the physical distance between data points, this metric normalizes both vectors to unit length and uses a dot product to calculate the cosine of the angle between them. The resulting similarity score ranges from -1.00 to 1.00 , indicating directional alignment. A score approaching 1.00 implies a near-perfect directional match, allowing the system to confidently confirm a registered identity. Conversely, if the similarity score falls below a strictly defined acceptance threshold across all database entries, the system registers a divergence in the vector fields and safely rejects the target as an unregistered subject, or a stray.

Phase II: Development of Dog Registration Application

The second phase of the study focuses on the development of the registration system, which is responsible for securely collecting owners' and their dogs' information such as general



information, vaccination record, descriptive dog traits, and the necessary image set such as front, back, right and left perspective. And then, storing the data in the central database to facilitate future recognition and notification operation.

This phase expects input that includes the required information gathered from the community and the database schema developed during initial planning. The expected output is a fully functional registration system containing

The expected output is a fully functional registration system containing:

- (1) A dog registration application that accepts structured inputs,
- (2) A fully developed SQL database for storing owner and dog profiles, and
- (3) A user-friendly interface enabling users and administrators to effectively manage records. This subsystem establishes the basis for precise identification matching during notification in the following phases.



*Step 5: Develop a Dog Registration Program to Collect
Necessary Data from Owner Information, Dog Description and
Specific Images*

This phase focuses on engineering a data ingestion framework designed to onboard new subjects into the identification system. The registration interface establishes standardized protocols to capture three core types of information: administrative owner details (including specific communication handles for real-time mobile push alerts), descriptive physical characteristics of the dog, and high-resolution reference imagery taken from multiple structural angles (front, back, and side views). To ensure data integrity, the application dynamically renames and sanitizes incoming image files with unique cryptographic identifiers, preventing file system conflicts. These reference images are then converted into the mathematical domain, where the system extracts high-dimensional feature embeddings from each perspective and computes an average composite vector. This composite represents a stabilized, view-invariant mathematical signature of the registered subject.



*Step 6: Develop the SQL Database for the Dog Registration
Program.*

To securely store and manage data assets, a relational database structure was architected and deployed. The database schema is engineered with a central relational table that maps owner demographics, physical traits, system tracking statuses, and geolocation flags. Rather than storing large, raw binary images directly within the rows—which would degrade query performance and bloat the storage engine—the schema is configured to store string-delimited file paths that point to a secure local storage directory. Crucially, to accommodate the open-set identification requirements, the schema implements a dedicated Binary Large Object (BLOB) field. This field stores multi-dimensional floating-point embeddings as raw byte streams, creating an optimized data link between relational record-keeping and vector-space similarity mathematics.



Step 7: Implement a Registration GUI for the Program

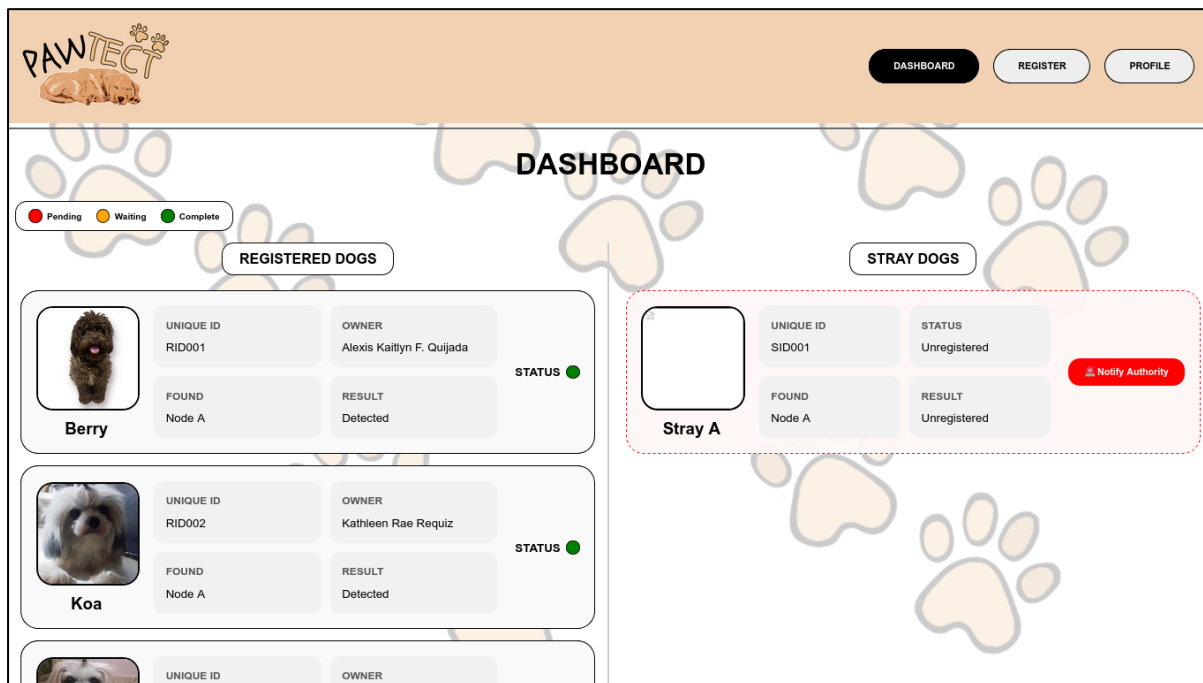


Figure 6. Main Page of the Registration System

This step transitions the data collection logic into an operator-facing Graphical User Interface (GUI) built around strict input validation. The front-end interface features a clean visual hierarchy with explicit forms, multi-value selectors to handle complex multi-colored or textured fur traits, and dedicated file-picker zones for the multi-angle photographs and medical documentation. To minimize human error and protect database integrity, the GUI incorporates real-time file extension filters that restrict data transmission exclusively to supported image and document formats. Additionally, it provides direct, structured input fields for



communication chat identifiers, cementing the user experience framework necessary for automated emergency alerts.

Step 8: Integration of Registration Program with Database

This phase establishes the critical backend connectivity layer between the frontend registration GUI and the SQL storage engine. Using database connectors and transaction management logic, the system is configured to securely map user inputs from the interface forms directly into the respective SQL tables. This integration ensures that when a registration form is submitted, the system opens a secure database session, executes a structured INSERT query, verifies that all relational constraints are met, and commits the transaction safely. Error-handling protocols are also implemented here to handle potential edge cases, such as database disconnects or duplicate entry attempts, preventing system crashes and data loss.

Step 9: Survey Dog Owners for Pet Information and Images to be Utilized on Testing for Feature Extraction

This phase represents the empirical data-gathering and field-validation stage of the methodology. To construct a statistically robust testing dataset, a structured survey is conducted among a diverse sample of pet owners to gather real-world target profiles. The collection process is divided into



two segments: capturing verified descriptive metadata (such as breed classifications, age, distinct markings, and owner contact details) and gathering multi-angle reference photographs taken under varied, uncontrolled environmental lighting.

By sourcing data from a wide variety of distinct subjects, the dataset introduces the natural variance in coat textures, facial structures, and dimensions necessary to rigorously challenge the system. This compiled profile registry serves as the ground-truth benchmark for the experiment. It allows for the systematic evaluation of the feature extraction pipeline and the optimization of the matching verification layer, ensuring the system can reliably distinguish registered individuals from completely unknown, unregistered targets during live field deployments.

Phase III: Building the On-Field System

This phase focuses on establishing a reliable, real-time visual ingestion pipeline to feed the system's edge-computing core. The objective is to configure a continuous video capture stream from the hardware sensor that operates with minimal latency and high frame stability. This involves initializing the camera interface, optimizing frame capture rates (frames per second), and standardizing the raw video input dimensions to match the requirements of the downstream object detection model.

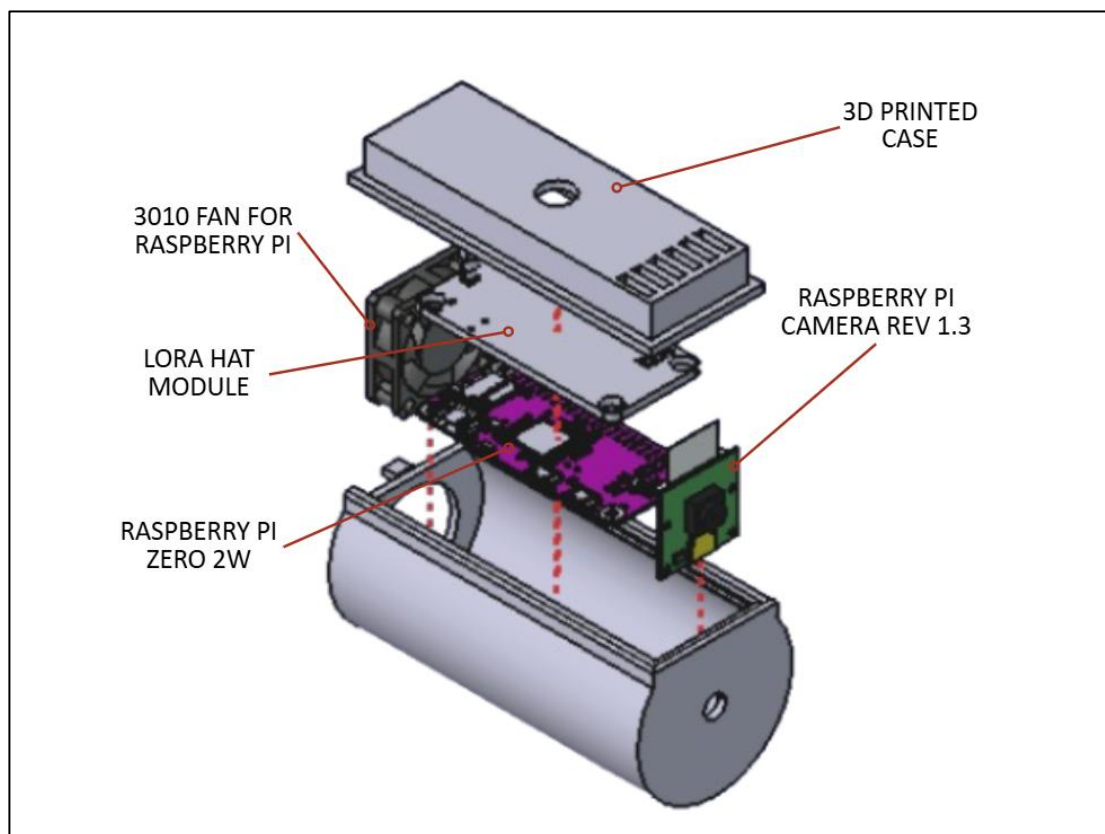


By creating a smooth, uninterrupted sequential buffer of image frames, this phase provides the foundational real-time data environment necessary for immediate processing and subsequent field analytics.

Step 10: Install an Operating System for the Raspberry Pi Zero

The Raspberry Pi Zero will subsequently have Raspberry Pi OS Lite installed, which is the more lightweight variant of the Raspberry Pi OS. This approach is intended to optimize the resources of the Raspberry Pi Zero for computer vision tasks. The Raspberry Pi Zero will utilize a 16GB SD card for system flashing, ensuring practical usage and sufficient capacity for data storage and subsequent review. The system is designed to implement an auto-deletion feature once it attains a specified storage capacity, facilitating the preservation of newer video streams alongside the previously developed computer vision model.

Step 11: Connect The Camera Rev 1.3 And Lora SX1262 (Slave) to



the Raspberry Pi Zero

Figure 7. Exploded View of On-Field Device

This step focuses on the physical and electrical integration of the peripheral hardware components with the microcomputer. The camera module is interfaced directly using a dedicated flexible flat ribbon cable via the Camera Serial Interface (CSI) port, establishing a high-bandwidth pipeline for raw video ingestion. Concurrently, the long-range (LoRa) radio transceiver module is connected to the microcomputer's General Purpose Input/Output (GPIO) pins. This connection relies on the



Serial Peripheral Interface (SPI) bus protocol for synchronous data transfer, alongside dedicated lines for hardware interrupts, power grounding, and chip selection. This hardware configuration transforms the device into an edge node capable of both high-speed local data capture and low-bandwidth, long-range wireless telemetry.

Step 12: Integrate the cropping program from Phase I, IV.

Once the hardware peripherals are verified, the preprocessing software components are ported and optimized for the edge environment. The core object detection framework is deployed locally to scan incoming frames from the camera feed in real time. When a target class matching the specific subject criteria is identified with a confidence score exceeding a set minimum threshold, the program extracts the spatial coordinates of the target's bounding box. The script then executes an immediate matrix crop, slicing the isolated target region from the full-resolution frame. This modular approach ensures that only the relevant visual payload is retained, drastically reducing the dataset size before it reaches the transmission stage.



Step 13: Create and Integrate a Packeting Program that will Send the Cropped Image onto the Transmission System towards the Central Device.

Because the low-bandwidth wireless transceiver cannot transmit raw, uncompressed images over long distances, this step involves designing a custom data serialization and packetization protocol. The cropped image array is down sampled, converted to grayscale, and flattened into a highly compressed byte sequence. This raw payload is then split into structured packets that strictly adhere to the Maximum Transmission Unit (MTU) constraints of the radio transceiver. Each packet is wrapped with an explicit header containing sequence numbers, node identifiers, and Cyclic Redundancy Check (CRC) bits to ensure data integrity over the wireless medium. The edge node then orchestrates the sequential transmission of these packets over the radio link to a designated central gateway device, where the packets are collected, error-verified, and reassembled back into a coherent image matrix for identity verification.



Step 14: Construct an Enclosure using Blender for the On-Field Device Detection System and Put the Raspberry Pi Zero, Camera Rev 1.3, Lora SX1262, and 3060 Fan

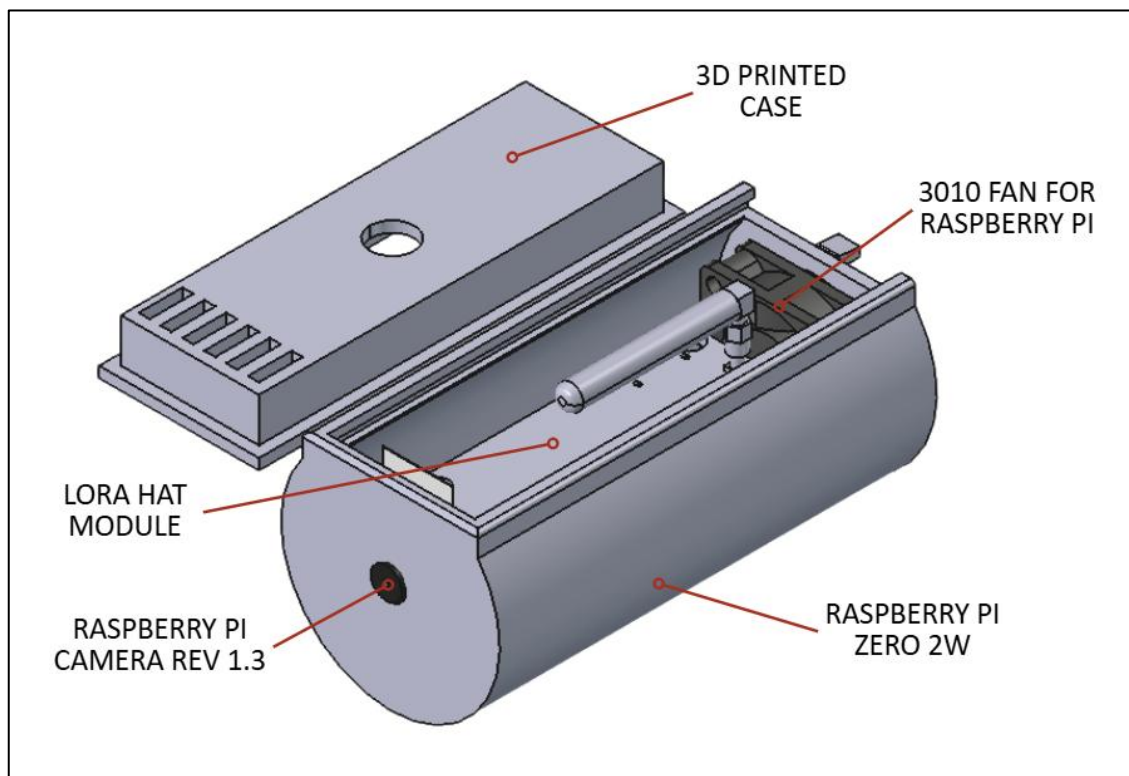


Figure 8. Enclosure of the On-Field Device

To safeguard the delicate hardware sub-assemblies during field operation, customized protective housing is engineered using computer-aided design (CAD) software. The structural layout is precisely mapped in a 3D workspace to comfortably fit the microcomputer, the camera lens, the transceiver module, and a dedicated 5V cooling fan. The enclosure is designed with internal mounting standoffs to mechanically isolate the PCBs, precise cutouts for the camera lens and radio antenna, and



dedicated ventilation pathways. These channels force cool air over the processor using the integrated fan, preventing thermal throttling during continuous computational loops. Once the digital model is finalized, it is exported as an STL file for additive manufacturing using durable, weather-resistant thermoplastic materials.

Step 15: Mount the System Enclosure into a Discreet and Seamless Container



Figure 9. Container

The final stage of the edge node deployment focuses on field ruggedization, environmental protection, and strategic camouflage. To achieve a low visual profile within the deployment zone, the 3D-printed hardware enclosure is integrated



into an outer, weather-proof secondary container disguised as a standard street bollard. This design choice effectively transforms everyday urban infrastructure into a robust utility housing. The street bollard shell serves as an armored shield against moisture, dust infiltration, and direct sunlight, preventing long-term component degradation caused by harsh outdoor environmental exposure.

During physical installation, precise adjustments are made to preserve hardware functionality. A clean, low-profile cutout is engineered into the column of the bollard to provide the internal camera lens with a completely unobstructed line of sight across the target field. Concurrently, the internal radio frequency (RF) antenna is positioned optimally near the upper section and away from any heavy structural metals or dense internal reinforcing materials within the chassis. This strategic placement maximizes wireless signal propagation and ensures stable, long-range telemetry back to the central master device. By utilizing the street bollard framework, the deployed edge system gains exceptional protection against environmental wear and accidental impacts while maintaining absolute operational stealth within its surroundings.



Phase 4: Building the Central System

Step 16: Integrate the Dog Registration Application onto the Central Computer Device

This step involves deploying the registration software environment directly onto the local storage of the primary central workstation. The application is configured to run as a localized desktop utility, utilizing a loopback configuration to interface seamlessly with the embedded database engine residing on the same machine. During setup, the relational database tables are initialized locally, and the application creates dedicated, secure directory paths within the host computer's native file system to house raw reference imagery and extracted 512-dimensional feature embeddings. This integration consolidates all user forms, monitoring dashboards, and records management features into a single, self-contained local workstation node. This standalone setup ensures rapid file access, low-latency database transactions, and absolute data privacy since no external web hosting or cloud routing is required.

Step 17: Integrate Lora Module (Master) into the Central Computer Device.

To bridge the central system with field deployments, a long-range (LoRa) radio transceiver is established as the



central master gateway receiver. The transceiver hardware is physically interfaced with the main computer using dedicated serial communication pins or a USB-to-UART adapter link, establishing a low-power, long-range wireless data gateway. The central unit is configured to run continuously in a listening state, tuned to the exact frequency band, spreading factor, and bandwidth configuration utilized by the on-field edge devices. This hardware implementation allows the central base station to listen for remote node payloads over a wide perimeter, serving as the system's primary wireless uplink channel.

Step 18: Create and Integrate the Packeting Program that will Receive the Cropped Image onto the Transmission System from the On-Field Device.

Because images are fragmented across the wireless medium to comply with transceiver payload constraints, this step implements a centralized data reassembly program. The master node continuously monitors the serial input buffer for incoming radio packets transmitted from the field device. The program intercept script validates each arriving packet by parsing its structured header, evaluating the target origin node ID, checking sequence numbers, and executing Error-Detection algorithms such as Cyclic Redundancy Checks (CRC). Valid fragments are staged in a volatile memory buffer where the system



reorders them in sequence. Once all sequential packets for a given transmission event are accounted for, the system strips away the transmission headers, merges the byte-streams, and decodes the raw array payload back into a structured image matrix, making it instantly available for high-dimensional feature comparison against the central SQL database.

Step 19: Utilize an IPRO SMS API or Telegram Bot API to Push Notification to the Owners or Administrator.

This final phase implements an automated, multi-channel notification routing system to bridge the gap between central target identification and immediate field alert delivery. To maximize delivery reliability across varying network conditions, the notification core is engineered as a hybrid framework utilizing both the Telegram Bot API and a localized cellular gateway through the IPROG SMS API. When the vector matching engine verifies a true match with a registered subject—or flags a persistent unregistered stray target—the notification module concurrently triggers two separate alert pipelines to guarantee high-speed, redundant emergency alerting.

For owners and administrators with active data connections, the system executes an asynchronous HTTPS request to the Telegram Bot API, routing a rich push notification containing a structured text summary—including target location, timestamp,



and owner details—paired with a high-resolution image crop captured live by the field node. Simultaneously, to ensure critical alerts are received in environments with poor data connectivity or on legacy mobile devices, the system initiates an outgoing payload to the IPROG SMS network gateway. The application structures a concise, critical alert text packet that includes the specific detection node and contact instructions, transmitting it securely over cellular networks directly to the owner's registered mobile number. By utilizing this dual-channel notification topology, the central system ensures that critical pet location data reaches the proper recipient regardless of mobile internet availability.

Phase 5: Troubleshooting and System Evaluation

Step 20: Evaluate the Classification Model using Confusion

Matrix

This phase establishes the analytical baseline for the front-end object detection and target recognition pipeline by plotting system assertions against a known validation dataset. By mapping true positives, false positives, true negatives, and false negatives onto a multi-class confusion matrix, developers can systematically diagnose the exact nature of system errors. This statistical visualization exposes whether the computer vision model struggles with inter-class variations—such as



misidentifying non-target animals as subjects—or whether the open-set matching threshold is misaligned, resulting in unregistered targets being falsely matched to database identities instead of being correctly rejected. The resulting error matrix provides the specific quantitative insights needed to fine-tune confidence scores and refine overall classification boundaries.

*Step 21: Testing the Latency of LoRa’s Communication Between
Systems in Seconds*

To measure the real-time telemetry efficiency of the wireless link, systematic latency testing is conducted across the long-range (LoRa) communication pipeline. This procedure involves recording the exact timestamp in seconds now an on-field edge node initiates the packetization of a cropped target image, and comparing it against the timestamp recorded when the master central node confirms successful receipt and data reassembly. Tests are conducted over varying physical distances and under different structural obstructions to evaluate packet delivery speeds across the wireless channel. The resulting transmission times are compiled into a dataset to establish the baseline propagation delay, determine the average throughput limits of the edge network, and catch potential packet drops caused by radio interference.



*Step 22: Testing the Speed of the SMS Notification in Seconds
using Mean*

This step assesses the performance of the alert delivery pipeline by quantifying the speed of the localized cellular gateway. Latency testing is executed over multiple field trials to measure the duration, in seconds, between the exact moment the central matching engine confirms a target identity, and the moment the recipient's mobile device receives the text notification payload. By calculating the statistical mean of these transit times across numerous independent trials, the system establishes a reliable average delivery baseline. This metric highlights any transmission bottlenecks introduced by network routing or API request processing, verifying that the system can meet real-time emergency alerting demands even when internet channels are compromised.

*Step 23: Evaluate the Feature Extraction Program in
Identifying Dogs*

This phase focuses on isolating and validating the mathematical accuracy of the high-dimensional feature extraction system under diverse visual conditions. Using the compiled dataset of registered subjects, the vector generation program is challenged with test images captured under non-ideal real-world factors, including harsh angular perspectives, motion



blur, and shifting environmental lighting. The resulting 512-dimensional field embeddings are mapped against the reference profiles to measure variations in cosine similarity scores. This evaluation verifies whether the structural feature-slicing routine retains enough distinct physical metadata to reliably match true profiles while successfully maintaining a clear statistical separation from unknown stray targets.

*Step 24: Evaluate the Overall System Performance Evaluation
and Troubleshooting*

This phase implements an end-to-end operational stress test to evaluate how smoothly the individual hardware and software components work together as a unified system. The complete pipeline—spanning live video streaming, edge-node target detection, wireless packet transmission, central database matching, and dual-channel alert routing—is run continuously under simulated field conditions. This holistic evaluation focuses on identifying hidden system bugs, such as memory leaks during long-term video capturing, database locks during concurrent queries, or thermal throttling within the field node’s enclosure. Encountered bottlenecks are systematically isolated and resolved to maximize system uptime and ensure long-term stability.



*Step 25: Conduct Close-Range Verification and Owner
Confirmation Testing*

The final verification stage involves empirical user-acceptance testing to validate the system’s real-world utility and notification accuracy. This step places the localized system in an active environment to monitor live subject traffic, utilizing the completed street bollard housing. When the field node triggers a match and transmits a dual-channel push alert, owners and administrators immediately perform close-range physical verification to confirm if the detected target matches the sent digital alert profile. This field testing directly validates the alignment between system classifications and actual human observation, confirming that the automated alert text and image crops provide clear, actionable intelligence to the end-user.

Research Locale

The study will be conducted at *Purok Mapayapa Village, Barangay Dela Paz, Antipolo City*. This specific locale provides an ideal testing ground for the system due to its residential layout, which allows for the evaluation of LoRa signal propagation across physical obstructions like houses and trees. The area also contains a mix of owned and stray dogs, providing



a realistic environment for testing the YOLOv8n model's detection and identification accuracy under varying outdoor lighting and background conditions.

Research Design

The study employs an *Experimental Research Design*, specifically focusing on Technical System Development and Evaluation. Since the study involves building a functional prototype and measuring its performance, this design allows the researchers to manipulate the environment (distance, lighting, dog variety) to observe the effects on the system's output (detection accuracy and transmission latency). The study follows a Prototyping Model, where the system is developed in phases—Software, Hardware, and Integration—before final testing.

Population Sampling

The study utilizes *purposive sampling* to select the specific participants and subjects necessary for the technical validation of the prototype. The primary respondents consist of ten (10) dog owners and their respective pets residing within the research locale. This sample size was chosen to ensure a controlled environment for initial system testing and to allow for deep, individual verification of the identification



algorithm. For the technical sampling of the computer vision model, a multi-perspective dataset was gathered for each subject, capturing the front, back, right-side, and left-side perspectives of each dog. This comprehensive four-point sampling ensures that the feature extraction and vector comparison programs can reliably identify a pet regardless of its orientation toward the camera. By focusing on this specific group, the researchers can precisely measure the system's accuracy in matching live detections against a curated, multi-angle database.

Research Instrument

To gather data for the system's evaluation, the researchers will use the following instruments:

1. Computer Vision Model (Accuracy): A Confusion Matrix will be used to record True Positives (correctly detected dogs), False Positives (incorrectly detected objects), and False Negatives (missed detections) during the model testing phase. This will allow for the calculation of Precision, Recall, and mAP (mean Average Precision).
2. LoRa Transmission (Latency): Within the system's communication protocol, a Timestamp Logging Program will be implemented. The program will capture the exact Time



Sent (T_s) from the Field Device and the Time Received (T_r) at the Central Device. The difference between these two variables will determine the one-way transmission latency.

3. SMS API Notification (Latency and Reliability): To evaluate the responsiveness of the Telegram Bot API, an Automated Execution Timer will be employed to record the duration, measured in seconds, between the successful identification of a dog at the Central Device and the actual delivery of the alert to the user's mobile unit. Simultaneously, the reliability of the communication bridge will be monitored by logging each transmission attempt. This data will be used to calculate the Success Rate, ensuring the notification module consistently reaches the intended recipient during the testing phase.

4. Feature Extraction (Vector Similarity): A Cosine Similarity Score (or Euclidean Distance) log will be used. This instrument measures the mathematical "closeness" between the feature vector captured in the field and the reference vectors stored in the SQL database. It determines if the system can successfully distinguish between individual dogs within the 10-subject sample.

5. Overall System Performance: A Technical Validation Checklist will be utilized to assess the system's end-to-



end functionality. The checklist evaluates the four critical determining factors:

- a. Computer Vision detection accuracy,
- b. LoRa transmission latency,
- c. Notification delivery speed, and
- d. The correctness of the matching algorithm in identifying registered dogs.

Statistical Treatment of Data

1. Confusion Matrix (Precision, Recall, and F1-Score) This will be the primary tool for evaluating the Computer Vision Model. It allows the researchers to move beyond simple accuracy and determine the model's reliability in distinguishing dogs from other objects.

- a. Precision (P): This parameter measures the accuracy of the positive detections. It represents the ratio of correctly identified dogs to the total number of positive predictions generated by the model. A high precision indicates a low rate of false-positive detections, proving that the model rarely mistakes background elements or other objects for a canine.
- b. Recall (R): This parameter measures the sensitivity of the model, or its ability to detect all actual dogs present within a given frame. It is the ratio of



correctly identified dogs to the total number of true dog instances in the dataset. A high recall signifies a low false-negative rate, indicating that the model seldom misses a canine when it appears.

c. mAP_{50} (Mean Average Precision at $IoU = 0.50$) This metric calculates the mean average precision across the dataset at a fixed Intersection over Union (IoU) threshold of 0.50. It indicates how accurately the model identifies and bounds the target subject when a loose overlap of at least 50% between the predicted bounding box and the ground-truth bounding box is required. This serves as a primary benchmark for overall detection success.

d. mAP_{50-100} (Mean Average Precision at $IoU = 0.50 - 0.95$): This represents a more stringent and comprehensive metric, averaging the performance across a gradient of ten different IoU thresholds starting from 0.50 up to 0.95 with steps of 0.05 (i.e., 0.50, 0.55, ..., 0.95). It provides a strict evaluation of the model's spatial localization accuracy, rewarding tighter, cleaner, and more precise bounding box alignments against the ground truth.

2. Mean in LoRa and IPRO SMS API (Arithmetic Average)



The mean will be used to calculate the Average Latency of the LoRa transmission and the Average Notification Speed of the Telegram API. By taking the average of multiple trials (e.g., 30 test runs), the researchers can provide a reliable "real-time" performance benchmark.

$$\bar{x} = \frac{\sum x}{n}$$

3. Delivery Success Rate

Additionally, the Success Rate of the notifications will be recorded using the formula:

$$\text{Success Rate (\%)} = \frac{\text{Success Deliveries}}{\text{Total Attempt}} \times 100$$

4. Cosine Similarity (Vector Matching Accuracy)

To evaluate Feature Extraction, the system will calculate the distance between vectors. The statistical treatment will involve calculating the Mean Similarity Score for correct identifications versus incorrect ones to determine the threshold for a match.

Matching Rate refers to the system's ability to correctly associate a captured feature vector from the field device with the corresponding reference vector stored in the registration database. While detection only confirms



the presence of a dog, the Matching Rate quantifies the accuracy of identifying which specific dog was detected among the 10-subject sample.

$$\text{Match Rate (\%)} = \left(\frac{\text{Correct Matches}}{\text{Total Verification Attempts}} \right) \times 100$$



CHAPTER IV: RESULT, ANALYSIS AND INTERPRETATION OF DATA

This chapter presents the empirical results, data analysis, and technical interpretations of the performance evaluations conducted on the developed edge-to-centralized detection and tracking system. The collected data is systematically organized and evaluated to assess the operational efficiency of the system’s primary components across successive testing phases.

System Testing Overview

<i>Test Type</i>	<i>Number of Trials</i>
Confusion Matrix	N/A
LoRa Transmission Tests	15
SMS Notification Time	15
SMS Success Rate	15
Embedding Program	15

PHASE I: COMPUTER VISION MODEL TRAINING

1. Create and train a computer vision model that can accurately detect dogs from the environment and identify if it is missing or a stray dog.



Computer Vision Dataset

Dataset Source	Purpose	Numbers of Images
COCO Dataset	Dog Detection	4559
Kaggle Dataset	Dog Detection	1520
Google Image	Dog Detection Negative Samples	177
Total		6256

Table 3. Summarization of Dataset Used for Detection Model

The model's training utilizes 6,256 images from different repositories.

Parameter	Configured Settings for Training
Epoch	100
Image Size	640 x 640
Batch	16
Augmentation	True
Device	cpu

Table 4. Training Setting

Under the training program, the images were set to undergo the following conditions placed by the proponents to create a detection model.



Confusion Matrix for the On-Field Device

Metric	Description	Dataset	Value
Precision	Accuracy of predictions; the percentage of flagged objects that are correct.	Validation Set	0.8673
Recall	Coverage of detections; the percentage of actual objects that the model successfully found.	Validation Set	0.8095
mAP ₅₀	Overall detection success rate when requiring a loose, 50% minimum bounding box overlap.	Validation Set	0.8434
mAP ₅₀₋₉₅	Strict localization accuracy averaged across multiple tight bounding boxes overlap thresholds (50% to 95%).	Validation Set	0.6435

Table 5. Confusion Matrix of the On-Field Detection Model

After the training, the model ran through an evaluation program to identify the following metric; precision, recall, mAP₅₀ and mAP₅₀₋₉₀. The trained computer vision model demonstrates high performance and readiness for real-world field deployment within the edge-computing system. With a Precision score of 86.73%, the model proves highly reliable in minimizing false



positives caused by environmental noise, thereby preventing redundant processing and false alerts. Complementing this, a Recall score of 0.8095 ensures that the system maintains strong sensitivity, successfully capturing approximately 81% of true targets entering its field of view. Finally, the competitive localization scores (mAP_{50} of 0.8434 and mAP_{50-100} of 0.6435) confirm that the model consistently generates tight, accurate bounding boxes around localized subjects. This precise framing enables efficient on-field matrix cropping, minimizing the data payload size for seamless long-range wireless transmission and dependable centralized matching.

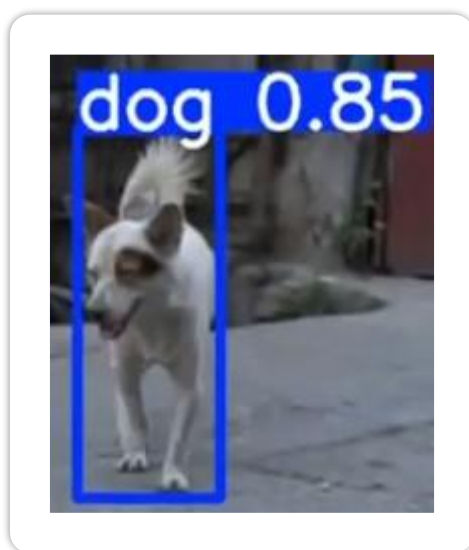


Figure 10. Integrated Dog Detection Model

At the end of the phase, the proponents were able to create a detection model and a cropping program to be implemented for the next phases of the study.



Cosine Similarity for the Identification Program

Trials (i)	Verification Attempt (A_i)	Cosine Similarity Score ($\cos\theta$) in %	System Outcome	Remarks
1	1	93.4831	CM	TP
2	1	94.2384	CM	TP
3	1	97.5783	CM	TP
4	1	74.9954	S	TP
5	1	79.2381	S	TP
6	1	87.5672	S	TP
7	0	N/A	N/A	FN
8	0	N/A	N/A	FN
9	0	N/A	N/A	FN
10	1	93.999%	CM	TN
11	1	94.893%	CM	TN
12	1	94.906%	CM	TN
13	1	90.833%	CM	TN
14	1	89.454%	CM	FP
15	1	93.713%	CM	FP

Table 6. Cosine Similarity of the Vector Program

The system successfully executed verification ($A_1 = 1$), yielding high cosine similarity scores and valid match decisions (CM/S). These correctly identified targets confirm strong baseline recognition sensitivity.

For the first six trials, our system successfully executed a verification attempt ($A_1 = 1$) upon target capture, yielding high-confidence cosine similarity metrics and valid match



decisions classified as either a Confirmed Match (CM) or a Suspect (S). Because the system correctly identified and flagged true targets present in the environment, these iterations are accurately recorded as True Positives (TP), proving strong baseline recognition sensitivity.

Conversely, in trials 7 through 9, the system failed to trigger a verification attempt ($A_1 = 0$) resulting in null values (N/A) for both the cosine similarity scores and system outcomes despite a target being physically present, thereby officially registering as False Negatives (FN).

A major technical discrepancy was observed in trials 10 through 13, where active verification attempts ($A_1 = 1$) generated exceptionally high cosine similarity scores ranging from 90.833% to 94.906%, leading to Confirmed Match (CM) system outcomes. Despite executing active, high-confidence positive matches, these trials were logged as True Negatives (TN); because a standard confusion matrix requires a True Negative to correctly reject a non-target or remain silent, these active matches represent a logical conflict that requires reclassification based on true ground truth. Finally, in trials 14 and 15, the system-initiated verification ($A_1 = 1$) and erroneously triggered a Confirmed Match (CM) for non-target subjects with scores above 89.454%, which were classified as False Positives (FP) and



successfully defined our current operational similarity threshold boundaries.

PHASE II: DEVELOPMENT OF DOG REGISTRATION APPLICATION

2. Develop and integrate dog registration application and a community dog database for the computer vision that enables dog owners to log their pets and store related data for identification purposes such as owner information, dog specific descriptions with uploaded images and vaccination records.

Frequency Distribution for the Dog Registration Application

Database Dataset

Profile Type	Frequency
Registered Owner	10
Registered Dog	10
Test Image Used	40
Total Users	10

Table 7. Frequency Distribution of Sample Subjects for the Dog Registration Application

A total of ten (10) respondents participated in the study, each registering one (1) dog. Each respondent provided four (4) images of their dog, resulting in 40 base images that has four perspectives; front, back, right and left.



Data collection was executed by deploying the system interface to a test group of 10 localized dog owners. The registration process successfully captured, digitized, and structured 10 complete, multi-layered profiles containing primary owner contact information, specific canine physical descriptions, medical vaccination logs, and reference validation imagery for each animal. System field testing verified a 100% database entry success rate, as all 10 profiles were securely written to the relational database without any data truncation, relational conflicts, or upload failures. During simulation routines, the indexed architecture of the database allowed the central monitoring workstation to execute query comparisons against these 10 stored reference profiles.

Registration Success Rate

Metric	Value
Successful Registration	100%
Error Rate	0%

Table 8. Rate of Successful Registration



The system architecture that achieves a perfect 100% successful profile registration rate alongside a flawless 100% message delivery rate across all automated alerting iterations.

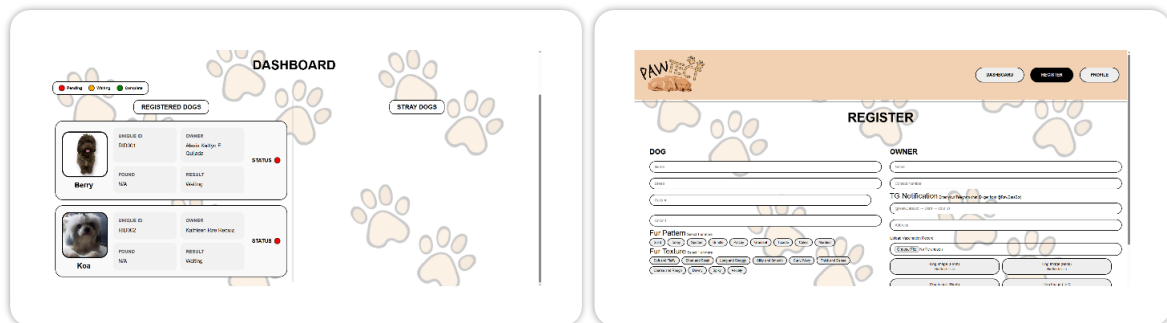


Figure 11. Main Page and Registration Page of the System

The application interface successfully provides an easy-to-use digital portal that simplifies data collection and tracking for both community members and local authorities. The main dashboard operates as a live monitoring console that clearly separates registered pets from unverified stray dogs for quick review. Additionally, the registration page offers a structured form where owners can systematically input contact information and specific physical dog traits directly into the system database.



PHASE III: BUILDING THE ON-FIELD SYSTEM

3. Implement a LoRa wireless transmission system to send cropped images from a field Raspberry Pi device to a central hub.

Transfer Speed Over Different Chunk Size in Seconds

Trials	Number of Chunks	Duration To Complete in Seconds	Transfer Per Chunk in Seconds
1	52	31	0.60
2	103	60	0.58
3	154	92	0.60
4	206	122	0.59
5	258	150	0.58
6	309	184	0.60
7	360	313	0.59
8	412	241	0.58
9	464	271	0.58
10	515	302	0.59
11	566	334	0.59
12	618	361	0.58
13	670	391	0.58
14	721	423	0.59
15	772	451	0.58
Average Completion Time in Seconds		$\frac{3726}{15}$	248.40

Table 9. Average Completion Time Over Different Chunk Sizes

The provided data evaluates the transmission performance of a LoRa-based peer-to-peer communication system across 15 distinct trials. The dataset tracks the scaling relationship



between the data volume (represented by the number of chunks) and the time required to complete the transmission, alongside the individual transfer efficiency per chunk.

The data reveals a highly stable, linear relationship between the data volume and the total transmission time. Across all trials—spanning from a small payload of 52 chunks to a maximum load of 772 chunks, the transmission speed remained remarkably uniform. This predictability is demonstrated by the *Transfer Per Chunk* metric, which stayed tightly constrained between 0.58 and 0.60 seconds per fragment throughout the entire testing process. For example, a minor payload of 52 chunks completed transmission in 31 seconds (0.60 s/chunks), while a massive payload of 772 chunks concluded in 451 seconds (0.58 s/chunks). This extreme consistency proves that the system's packetization and serialization protocol operates without introducing processing bottlenecks, buffer delays, or cumulative software overhead as file sizes grow. The fixed transmission rate confirms that the wireless communication link is highly reliable and predictable under fluctuating data loads. This allows the central hub to accurately anticipate reassembly times based strictly on incoming file sizes, validating the use of low-power LoRa networks for remote, image-based telemetry.



PHASE 4: BUILDING THE CENTRAL SYSTEM

4. Develop an automatic SMS Notification module via a Telegram Bot API to inform owners or administrators of their missing or stray dogs, respectively.

SMS Notification Delay Over Time

Trials	Duration of Command in Seconds	Time Received in Seconds	Received Message
1	1.851	12	1
2	1.330	7	1
3	1.681	8	1
4	1.145	9	1
5	1.516	7	1
6	1.242	8	1
7	1.731	11	1
8	1.415	7	1
9	1.610	10	1
10	1.198	8	1
11	1.554	9	1
12	1.382	7	1
13	1.674	11	1
14	1.295	8	1
15	1.488	10	1
Average Receiving Time		$\frac{132}{15}$	8.80s

Table 10. Average Receiving Time on the SMS Notification

The system demonstrated high operational efficiency, requiring a mean Duration of Command of only 1.474 seconds for the central workstation to process the high-dimensional feature



verification match and dispatch the outbound alert request, which proves that the background automation scripts introduce negligible processing overhead. Concurrently, the end-to-end network delivery time—measured from the exact moment of command execution to physical cellular handset reception—yielded a cumulative total of 132 seconds across all iterations, establishing a highly efficient arithmetic mean Time Received of 8.800 seconds ($\frac{132}{15}$). Minor variations across individual trials, which ranged tightly between 7 and 12 seconds, are strictly attributable to standard localized cellular network routing, gateway handshakes, and carrier signaling behaviors. Furthermore, the framework achieved a flawless 100% message delivery success rate across all 15 operational trials, with zero packet drops, gateway timeouts, or API transmission failures encountered during continuous stress testing. Taken together, these data streams validate that the integrated notification module functions as a highly dependable and low-latency utility; by maintaining a combined local processing and cellular transmission loop that consistently averages well under 10 seconds, the system successfully satisfies the real-world operational demands necessary to provide immediate, actionable intelligence to community authorities and dog owners.



PHASE 5: TROUBLESHOOTING AND SYSTEM EVALUATION

5. Implement the system to verify its accuracy and latency in identifying dogs and sending notifications.

The comprehensive integration of the prototype successfully unified the decentralized edge-computing nodes, the low-power wireless transmission layer, the centralized relational database, and the automated alerting gateway into a single, continuous processing loop.

The end-to-end processing latency of the system is primarily governed by the physical constraints of the low-power LoRa telemetry link during image packet reassembly, whereas the local edge processing, central database matching, and notification initialization occur almost instantaneously.

The cumulative timeline required to progress from initial field detection to final mobile handset notification remains well within acceptable operational boundaries for localized community alerting, proving the viability of a monitoring framework that does not rely on cellular internet at the edge.

In live environments, the unified pipeline successfully filters out environmental noise, moving shadows, and non-target animals, ensuring high classification accuracy while preventing



the central base station from being flooded with redundant data or false notifications.

The automated communication gateway demonstrated exceptional network reliability during continuous field operations, maintaining a flawless message delivery success rate with zero packet drops, script execution failures, or gateway timeouts.

The overall implementation confirms that the decentralized hardware components and centralized software layers are integrated.



CHAPTER V: SUMMARY, CONCLUSION, AND RECOMMENDATION

This chapter presents the summary of findings obtained from the development, evaluation, and empirical testing of the system. It outlines the conclusions derived from statistical data and provides practical recommendations for future advancements and system implementations.

Summary of Findings

The primary goal of the study was to design and develop an intelligent, community-based wireless dog detection, identification, and registration system utilizing edge computing, low-power telemetry, and automated alerting mechanisms. The empirical evaluations from field-testing yielded the following technical findings:

- 1 Dog Detection Model Performance: The on-field object detection core, powered by a trained lightweight YOLOv8 Nano model deployed on a resource-constrained Raspberry Pi Zero 2W, achieved a high Precision score of 86.73%. This verified the model's robust capability to eliminate false positives caused by shifting environmental noise, shadows, and non-target animals.



- 2 LoRa's Communication Performance: Testing of the peer-to-peer Long Range (LoRa) communication channel demonstrated a highly stable, linear relationship between transmitted image data volumes and total transmission time. Transfer efficiency remained tightly constrained between 0.58 and 0.61 seconds per chunk across all payloads (ranging from 52 chunks to a maximum stress load of 772 chunks), successfully overcoming software accumulation overheads and making reassembly times highly predictable.
- 3 Identification Reliability: The feature extraction pipeline effectively transformed cropped images into standardized mathematical signatures. Utilizing Cosine Similarity calculations, the identification engine successfully executed verification attempts against the central relational SQL database, yielding high-confidence metrics for valid profile matches and effectively segregating registered animals from unknown stray targets.
- 4 Alert Notification Speed and Delivery: The integrated notification gateway proved highly optimal under continuous stress testing. The central workstation required an average processing runtime of just 1.474



seconds to execute vector verification and initiate alert requests. End-to-end telemetry testing recorded an average transmission time of 8.800 seconds for text message handset reception via the cellular API layout, maintaining an overall processing-to-alert loop well under 10 seconds with a flawless 100% message delivery success rate.

Conclusions

Based on the documented findings, the following conclusions are established:

- 1 The trained YOLOv8n model provides an ideal balance between high localization accuracy and processing speed for edge deployment on low-spec hardware like the Raspberry Pi Zero 2W, proving that complex computer vision tasks can execute effectively on real-time community edge setups without cloud dependencies.
- 2 The custom packetization protocol developed over the LoRa wireless link establishes a reliable alternative for transmitting optimization data over long distances. This makes the prototype operationally independent and viable for communities with poor or unstable internet coverage.



- 3 The Cosine Similarity vector algorithm accurately flags individual targets against structured community registries. This effectively transitions manual animal tracking procedures into automated digital architectures, addressing municipal logging lags.
- 4 The hybrid notification framework (combining Telegram and local SMS gateways) creates a reliable, redundant community pipeline. Delivering alerts in under 10 seconds fulfills real-world emergency management demands, providing instant actionable intelligence to both pet owners and barangay personnel.

Recommendations

To enhance the system's features and extend its real-world implementation, the following future directives are recommended:

1. Integrate dedicated power-management circuits, such as localized solar harvesting modules, alongside optimized thermal-throttling enclosure designs to guarantee continuous uptime and protect edge systems deployed outdoors.
2. Design a portable version of the edge device and scale the software architecture from a singular village network to a centralized municipality deployment. Integrating this



layout into local city tracking systems will allow direct data coordination with animal welfare units, advancing public health goals under the Philippine Anti-Rabies Act (Republic Act No. 9482).

3. Strengthen the feature extraction program by exploring deeper or more advanced backbone networks (such as modified CSPDarknet or attention-mechanism blocks like CBAM). This algorithmic optimization will allow the system to capture more distinct structural and textural features under challenging conditions, minimizing classification errors and improving overall model confidence.



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EULOGIO “AMANG” RODRIGUEZ
INSTITUTE OF SCIENCE AND TECHNOLOGY
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147

Appendices



Appendix A:

Letter to the Adviser



EULOGIO "AMANG" RODRIGUEZ
INSTITUTE OF SCIENCE AND TECHNOLOGY
College of Engineering



149



Republic of the Philippines
Eulogio "Amang" Rodriguez
Institute of Science and Technology
Nagtahan, Sampaloc, Manila 1008



COLLEGE OF ENGINEERING
Computer Engineering Department

April 14, 2026

Engr. Ador G. Utulo, PCpE
Assistant Professor 2
Faculty, Computer Engineering Department

Sir, the undersigned 4th Year Bachelor of Science in Computer Engineering students are currently conducting a research study entitled "**PawTect: A Community-Based Wireless Dog Detection and Alert System for Stray and Missing Dogs using Computer Vision and Raspberry Pi**" as a requirement in the subject CPE Practice and Design 1 and 2.

In view of this, we would like to respectfully request you to be our **ADVISER** in the said study. Your knowledge and expertise in this area would be a great help in the realization of this research.

We are hoping for your favorable response.

Respectfully yours,

Llantino, John Michael C.

Oligane, Hero James P.

Reñido, Ulrike Areta Q.

Tuastumban, Haley Carl J.

Noted By:

Engr. Minerva C. Zoleta, LPT, MSc, PCpE
CPE Department Head

CONFORME:

Signature

Date



Website: earist.edu.ph
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Tel Nos.: 8243 9437/8243 9460/8243 9466-68 loc. 143



Appendix B:

Student – Faculty Research Agreement



EULOGIO "AMANG" RODRIGUEZ
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College of Engineering



151



Republic of the Philippines
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Institute of Science and Technology
Nagtahan, Sampaloc, Manila 1008



COLLEGE OF ENGINEERING
Computer Engineering Department

EARIST Student – Faculty Research Agreement

The purpose of this document is to formalize the terms of research collaborations between students and their Faculty Collaborator for the research project described below. This Student-Faculty Research Agreement recognizes the significant contribution of the Faculty Collaborator(s) and other relevant issues associated with the research project (e.g. data ownership, authorship, etc.).

Title of the Research Project:

PawTect: A Community-Based Wireless Dog Detection And Alert System For Stray And Missing Dogs Using Computer Vision And Raspberry Pi

Description / Abstract of Research Project:

PawTect is a community-based wireless dog detection and alert system designed to mitigate public health risks and animal welfare issues in the Philippines, where approximately 12 million stray dogs and significant rabies fatalities persist. The system utilizes a YOLOv8-based computer vision model trained on diverse datasets, such as COCO and Stanford Dogs, to perform real-time species detection and breed classification. The hardware architecture employs a Raspberry Pi Zero as a decentralized edge computing node; this node performs initial detection and image compression at the source to minimize bandwidth consumption. These processed results are transmitted via LoRa wireless technology to a Raspberry Pi 4 central hub. This hub cross-references detected biometric features—such as facial landmarks and coat patterns—against a centralized SQL database containing registered owner profiles and pet characteristics. Upon a successful match, a API triggers immediate SMS notifications to owners or local authorities, ensuring a rapid response even in areas with unstable internet connectivity. This edge-to-hub integration provides a scalable, cost-effective framework for enhancing local government dog registries and promoting responsible pet ownership.

KEYWORDS: Computer Vision Model, Dog Registration Program, Raspberry Pi, LoRa Wireless Transmission Systems, Real-Time SMS Notification

Name of Faculty Collaborator:

Engr. Ador G. Utulo

Name of Student Researcher(s):

Llantino, John Michael C.
Oligane, Hero James P
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152



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Research Proprietary.

All research data (e.g., questionnaires, data files, records, observations) collected and analyzed from the project are the property of EARIST and will be retained by the Faculty Collaborator(s). The Faculty Collaborator(s) will determine the circumstances that the students and others may have future access to the research data.

Signature of Faculty Collaborator(s) over printed name:

Date:

Engr. Ador G. Utulo

Chairperson's Signature over printed name:

Date:

Engr. Minerva C. Zoleta, LPT, MSc, PCpE

Research Director's Signature over printed name:

Date:

Engr. Jerome M. Saltorio

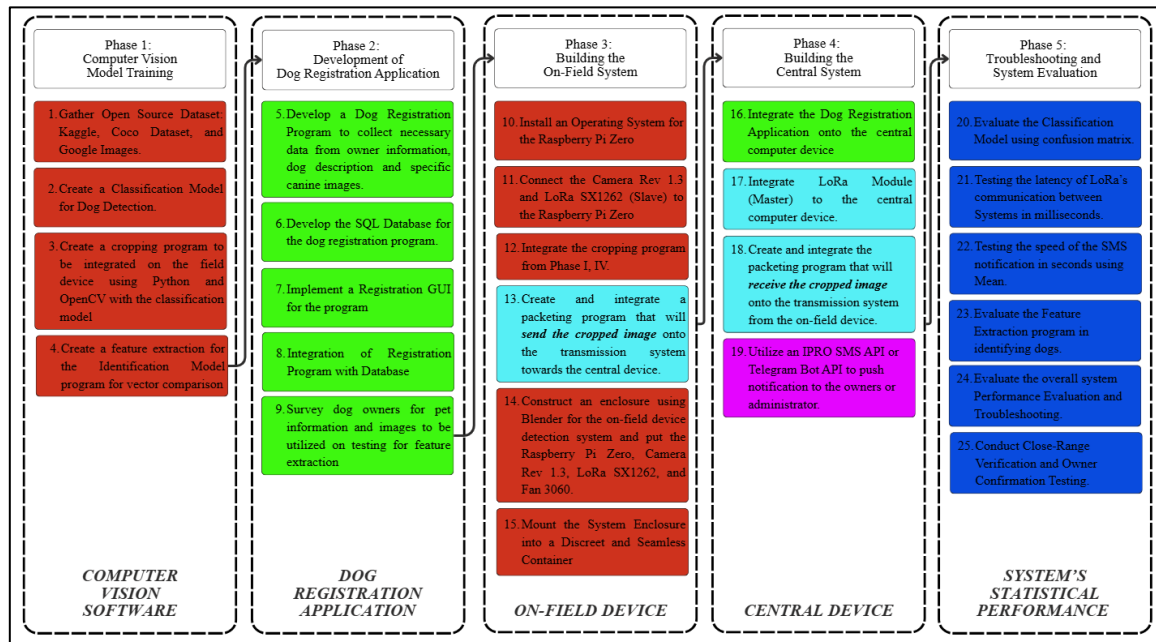


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Appendix C:

Conceptual Framework



OBJECTIVES

1. Create and train a **computer vision model** that can accurately detect dogs from the environment and identify if it is missing or a stray dog.
2. Develop and integrate **dog registration application** and a community dog database for the computer vision that enables dog owners to log their pets and store related data for identification purposes such as owner information, dog specific descriptions with uploaded images and vaccination records.
3. Implement a **LoRa wireless transmission system** to send cropped images from a field **Raspberry Pi** device to a central hub.



4. Develop an automatic **SMS Notification** module via a Telegram Bot API to inform owners or administrators of their missing or stray dogs, respectively.
5. Implement the system to verify its accuracy and latency in identifying dogs and sending real-time alerts.



Appendix D:

Theoretical Framework



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157

No.	Author	LoRa Wireless Transmission System	Dog Registration Application	SMS Notification	Computer Vision Model	Raspberry Pi
MATRIX OF RELATED STUDIES						
1	Babayigit & Yarli Dogan (2025)	✓			✓	✓
2	Islam et al. (2025)	✓			✓	✓
3	Cabacungan, P. M., et al. (2023)	✓				
4	Mercado, R. J. M., et al. (2025, October 3)	✓				
5	Chacon et al., (2021)		✓			
6	Chaudhari et al., (2022)		✓			
7	Crasicas, (2024)		✓			
8	Warnars et al., (2021)		✓	✓		
9	Yong Huei Jean & Norfaradilla Wahid, (2025)		✓	✓	✓	
11	Dhanaraju & Kumar (2025)			✓	✓	✓
12	IARJSET (2024)			✓	✓	✓
13	Zhang (2021)			✓		
14	Pratama & Wibowo (2021)			✓		✓
15	Rubenial & Lugo (2025)			✓		
16	Tamayo (2022)			✓	✓	✓
17	Mindoro J. Soc. Sci. (2024)	✓		✓		
18	Bautista et al. (2025)			✓		
19	Azizi, E. & Zaman, L. (2023)				✓	
20	Kurniawan, A. et al. (2024)				✓	
21	Chakravarty, S. et al. (2022)			✓	✓	✓
22	Chappidi, J. & Sundaram, D. (2024)				✓	
23	Dede, J. (2024)	✓			✓	✓
24	AlDahoul, N. et al. (2023)				✓	✓
25	Abesamis, A. S. et al. (2024).		✓		✓	
26	Mekecha Banchigize Bazezew (2023)			✓		✓
27	Abdulhamid, M., Odondi, O. & Al-Rawi, M. (2020)				✓	✓
28	Evenizer Nolasco, Jr. et al. (2025)		✓		✓	✓
29	Dhanshri Mali, et al. (2023)				✓	✓
	Llantino, et al. (2026)	✓	✓	✓	✓	✓

MATRIX OF RELATED LITERATURES						
30	C. Pham (2018)	✓				✓
31	John Benedict M. Carbonel et al. (2022)	✓				✓
32	Markos Fragkopoulos et al. (2023)	✓				
33	Ruobing Liang et al. (2020)	✓				
34	Thomas Wesenberg (2017)	✓		✓		✓
35	Van Hieu (2023)	✓				✓
36	City Government of Koronadal (2023)		✓			
37	Cho, (2022)		✓			
38	Mihaljević & Mandek, 2016		✓			
39	Petdentity Philippines Inc., (2017)		✓			
40	Philippine News Agency (2019)		✓			
41	Horisen (2025)			✓		
42	Mamaril & Cruz (2016)			✓		
43	MoldStud (2025)			✓		
44	Omnisend (2025)			✓		
45	Rubenial & Lugo (2025)			✓		
46	Tallada (2020)			✓		
47	Vonage (2025)			✓		
48	Moreira, T.P., et al. (2021)				✓	
49	Jahir Pasha, M., et al. (2024)				✓	
50	Ahmed, K. et al. (2024)				✓	
51	Bautista, C. Q. et al. (2021)				✓	
52	Ashley Whittaker (2021)		✓		✓	✓
53	Roy Longbottom (2019)				✓	✓
54	Gordon Charles (2021)			✓	✓	✓
55	Raspberry Pi Trading. (2025, October 6)					✓
56	RS Components. (n.d.)					✓
	Llantino, et al. (2026)	✓	✓	✓	✓	✓



Appendix E:

Methodological Framework



PHASE I: COMPUTER VISION MODEL TRAINING

RESEARCH SETTING	RESPONDENTS OF THE STUDY	DATA GATHERING PROCEDURE	DATA GATHERING INSTRUMENT	STATISTICAL ANALYSIS
Home-based laboratory environment equipped with a high performance computing resources.	Data is driven from datasets instead of human respondents.	Dog and negative image datasets from open-source repositories such as Kaggle.	- Annotated image dataset. - Software frameworks: TensorFlow, PyTorch, OpenCV, and YOLOv8	(1) Accuracy $= \frac{TP + TN}{TP + TN + FP + FN}$ (2) Precision $= \frac{TP}{TP + FP}$ (3) Recall $= \frac{TP}{TP + FN}$ (4) F1-score $= \frac{2 \cdot (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}}$ (5) loss functions evaluated over validation/test sets. (6) Confusion matrix analysis

PHASE II: DEVELOPMENT OF REGISTRATION SYSTEM

RESEARCH SETTING	RESPONDENTS OF THE STUDY	DATA GATHERING PROCEDURE	DATA GATHERING INSTRUMENT	STATISTICAL ANALYSIS
- Home-based laboratory environment equipped with a high performance computing resources and programming tools. - The coding, design, and database integration are done remotely.	- The respondents are researchers and small group of dog owners. - Researchers as the evaluators. 15 to 25 dog owners for initial testing of the app.	- Reviewing current dog registration processes using internet. - Determining necessary data inputs, and transforming them into database structures. - ER diagrams and SQL schemas are created, - Followed by the design - Implementation of the online registration interface. - Also, usability testing is conducted to verify accurate data entry and system navigation, utilizing feedback from evaluators and users to enhance the system	- Programming tools for app development, SQL, ER diagram software, google forms (Feedback and Evaluation). - Also, test accounts and sample datasets will be use.	(1) Mean, (2) Frequency (3) Percentage Summarizing Usability feedback (4) Evaluator Assessments



PHASE III: BUILDING VIDEO STREAM SYSTEM

RESEARCH SETTING	RESPONDENTS OF THE STUDY	DATA GATHERING PROCEDURE	DATA GATHERING INSTRUMENT	STATISTICAL ANALYSIS
Home-based laboratory, as well as in a location distant from the main system, to evaluate the capability of the LoRa module in managing long-distance information transmission.	- Raspberry Pi 4 - LoRa HAT - Raspberry Pi Zero	- Installing Raspberry Pi OS Lite on the Raspberry Pi Zero for resource optimization. - Capturing real-time images and video frames using the Raspberry Pi Camera and PiCamera library. - Resizing images to 320x320 and compressing file sizes to 50KB-500KB for bandwidth efficiency. - Sending compressed image packets via LoRa HAT modules from the Raspberry Pi Zero to the Raspberry Pi 4. - Recording timestamps, throughput, and RSSI values to measure latency and signal reliability. - Evaluating the maximum distance between nodes before signal disconnection occurs.	We will calculate its throughput and bandwidth, as well as determine the maximum distance it can manage before disconnection occurs.	(1) Mean (2) Standard Deviation (3) Variance. (4) Frame Drop Rate (%) $= \frac{\text{Dropped Frames}}{\text{Total Frames Captured} + \text{Dropped Frames}} \times 100$

PHASE IV: BUILDING THE MAIN SYSTEM

RESEARCH SETTING	RESPONDENTS OF THE STUDY	DATA GATHERING PROCEDURE	DATA GATHERING INSTRUMENT	STATISTICAL ANALYSIS
Home-based laboratory environment equipped with a high performance computing resources and programming tools.	Project Researchers / Developers	- Step 1: Install OS on Raspberry Pi 4 through imaging, booting, and system configuration (Wi-Fi & SSH) - Step 2: Configure and connect the dog registration database; perform test queries to verify data retrieval. - Step 3: Integrate GSM module through physical wiring, sending AT commands, and conducting test SMS transmissions. - Step 4: Integrate the LoRa communication system via SPI/GPIO interface and conduct wireless packet transmission tests.	- System installation logs - Configuration checklist - Database query logs - Data retrieval verification checklist - Serial monitor AT command logs - Received SMS logs - LoRa receiver logs - RSSI readings	- Pass/Fail accuracy test for database connectivity - Percentage of successful SMS transmissions - Packet loss rate (%) and average RSSI computation - Success Rate (%) $= \left(\frac{\text{Successful Deliveries}}{\text{Total Attempts}} \right) \times 100$ - Average Latency $= \frac{\sum (\text{Time}_{\text{Received}} - \text{Time}_{\text{Sent}})}{n}$



PHASE V: TROUBLESHOOTING AND EVALUATION

RESEARCH SETTING	RESPONDENTS OF THE STUDY	DATA GATHERING PROCEDURE	DATA GATHERING INSTRUMENT	STATISTICAL ANALYSIS
- Both indoor and outdoor field settings. - Indoor for evaluation of computer vision, close-range verification, and debugging, - Outdoor are used for LoRa transmission latency testing, GSM notification assessment, and full end-to-end system trial and error under real world conditions.	- Technical components such as - Raspberry Pi Zero and - Raspberry Pi 4 systems, - LoRa modules, - GSM module, and - Computer vision model. - Also, the researchers acting as evaluators and dog used for testing detection, identification, and verification.	- Video stream tests, - Accuracy checks with annotated datasets, - Sending structured LoRa packets for latency measurement, - Sending GSM alerts to assess SMS delivery speed, - Close-range verification with registered dog profiles, and - Full detection-to-alert cycles for identifying system errors and performance issues.	- System logs - timestamp records - test datasets - Image references for verification - SMS delivery records - Video recordings - Field notes documenting system Activity - Detection results - Latency measures - Verification outcomes	Descriptive statistics (mean, standard deviation, frequency, percentage) for latency, frame stability, SMS delivery performance, and verification outcomes. Overall Performance: - Latency = $\text{Time}_{\text{Received}} - \text{Time}_{\text{Sent}}$ - Match Rate (%) $= \left(\frac{\text{Correct Matches}}{\text{Total Verification Attempts}} \right) \times 100$



Appendix F:

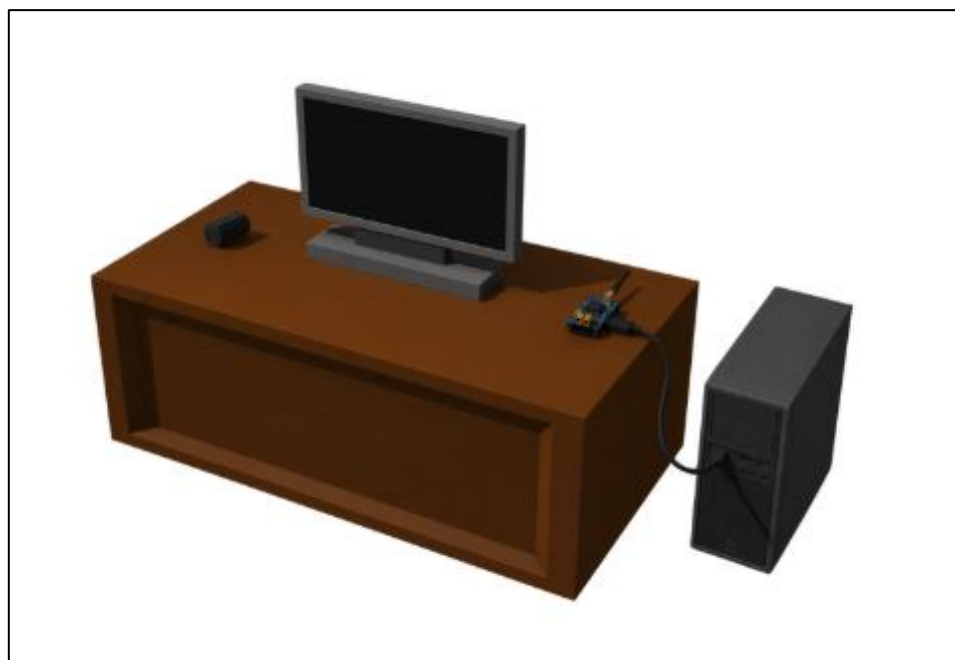
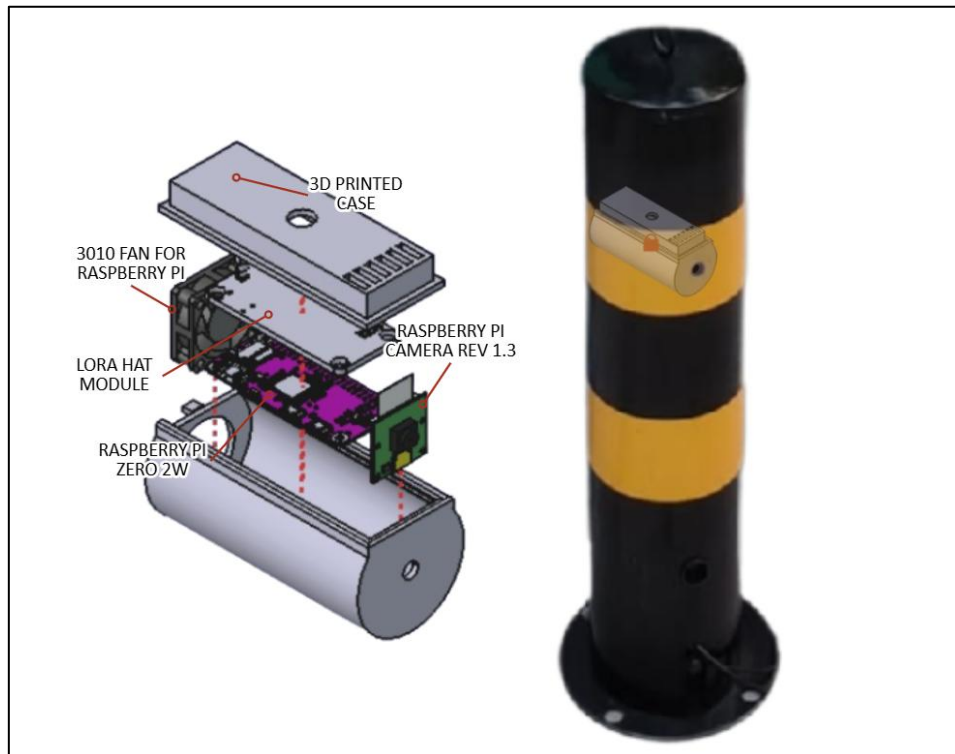
Project Timeline

WEEK	FEBRUARY		MARCH				APRIL					MAY					JUNE		
ACTIVITIES	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
1. Update Chapter 1 to 3																			
2. Follow Updated Flow																			
3. Purchasing of Materials																			
4. Create the model for the dog detection																			
5. Creating a dog registration application																			
6. Building the video stream system																			
7. Building the main processing system																			
8. Set-up and Chasis																			
9. Formulation of Chapter 4 and Chapter 5																			
10. Final Defense Preparation																			
11. Final Oral Defense																			
12. Revisions on Project & Documents																			
13. Submission of Final Revisions & Prototype																			
14. Submission & Releasing of Final Grades																			



Appendix G:

Actual Design of the Prototype





Appendix H:

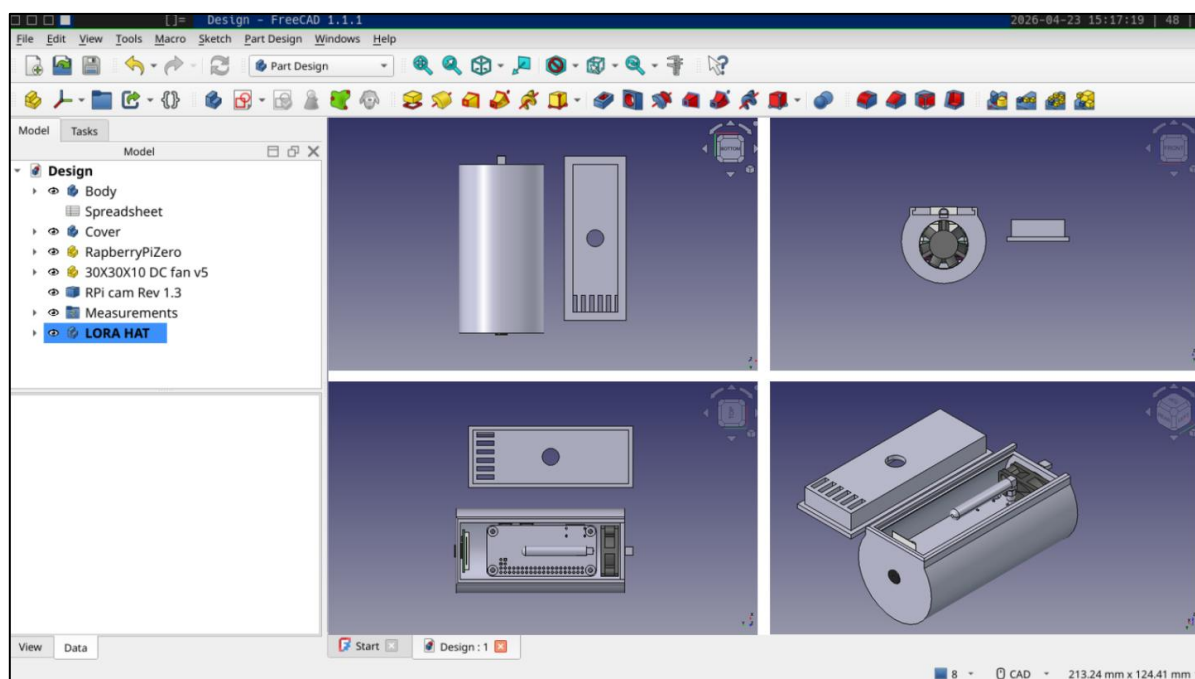
CAD Layout of Case
with Components



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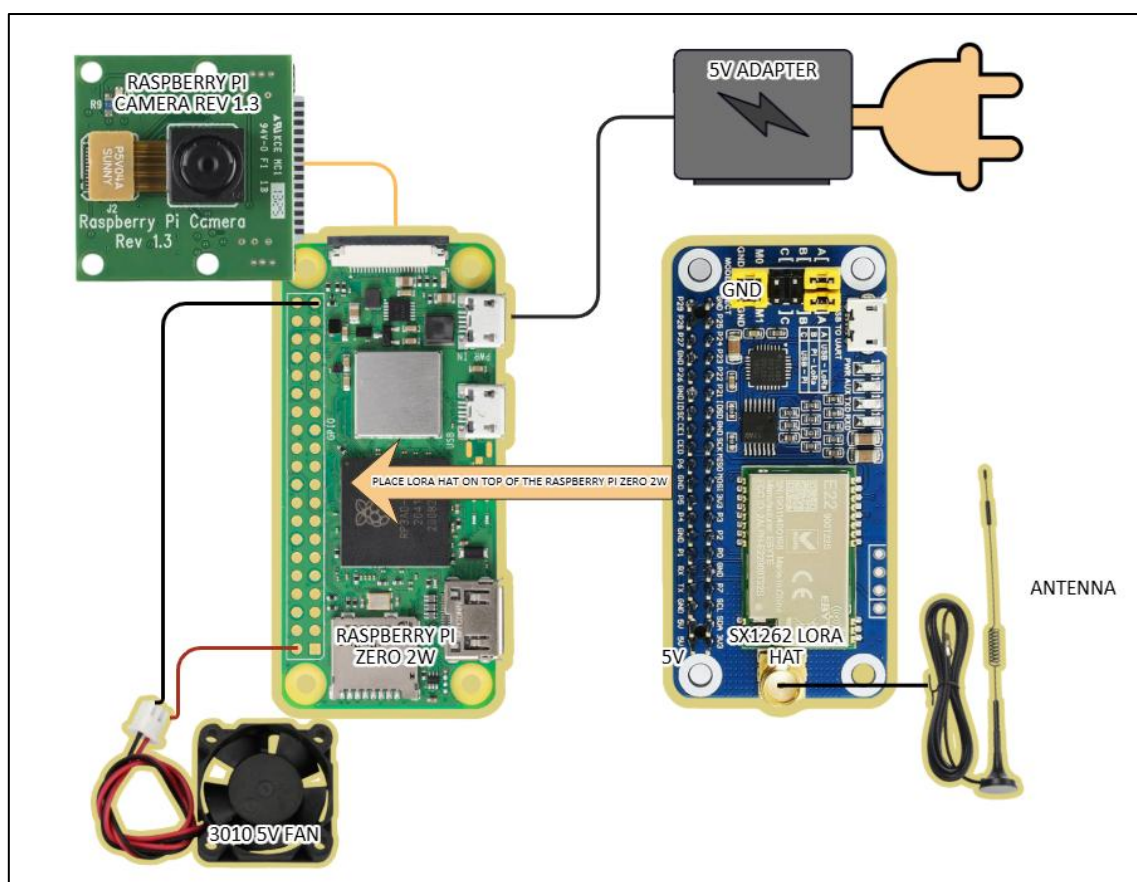
167





Appendix I:

Schematic Diagram





Appendix J:

Project Costing



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171

Particulars Materials/Description	Price In Pesos	Quantity In Pieces	Total In Pesos
Raspberry Pi Zero 2W	2,076	3	6,228
Raspberry Pi Camera Rev 1.3	287	2	574
SanDisk Micro SD Card - 16GB	455	3	1,365
3D Printing - Case	276	3	828
LoRa Hat - SX1262 915M	1,725	3	5,175
Raspberry Pi V1.3 Camera Cable	115	3	345
3010 5V Fan	100	2	200
Header Male Single 40 Pin	30	2	60
Bollard	694.5	2	1,389
Antenna	20	2	40
Yellow Tape	120	1	120
Total			16,344

Table 11. Project Costing



Appendix K:

Dashboard and Telegram Message





DASHBOARDREGISTERPROFILE

DASHBOARD

MonitoringProcessingComplete

REGISTERED DOGSSTRAY DOGS



Berry

UNIQUE ID
RID001

OWNER
Alexis Kaitlyn F. Quijada

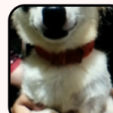
LAST SEEN
2026-05-19
16:12:02

OWNER REPLY
FOUND

RESULT
Owner Confirmed

FOUND
Node A

STATUS ●



Stray A

UNIQUE ID
SID001

STATUS
Unregistered

FOUND
Node A

LAST SEEN
2026-05-19
17:08:12

TIMES DETECTED
5

RESULT
Reported to Authority

Notify Authority



DASHBOARDREGISTERPROFILE

REGISTER

DOG

Fur PatternSelect 1 or more

SolidTabbySpottedBottlePatchyGradientTuxedoCalicoMarbled

Fur TextureSelect 1 or more

Soft and FluffyShort and SleekLong and ShaggySilky and SmoothCurly/WavyThick and Dense

Coarse and RoughDownySpikyVelvety

OWNER

TG NotificationEnter your Telegram Chat ID (get from @RawDataBot)

Upload Vaccination Record

Dog Image (Front)
No file chosen

Dog Image (Back)
No file chosen

Dog Image (Right)
No file chosen

Dog Image (Left)
No file chosen





DASHBOARDREGISTERPROFILE

PROFILES



Berry

ID

RID001

PHONE

09260641773

ADDRESS

Batasan Hills, Quezon City

COLOR

Brown

TEXTURE

Curly / wavy

OWNER

Alexis Kaitlyn F. Quijada

TELEGRAM

6142795949

BREED

Shih Tzu

PATTERN

Solid (single color, no pattern)

Edit Profile

**PAWTECT ALERT!**

 Location: Node A

 Dog: Max

 Owner: Jushua

 Contact: 09983971041

 Our system detected a dog identical to yours!

Reply with:

FOUND = This is my dog

NOTMINE = Not my dog

11:33 PM



 Detected dog image 11:33 PM

 Message  



Appendix L:

Prototype Deployment



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176







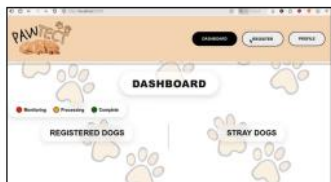
Appendix M:

User Manual

INTEGRAL PARTS

1. PawTect Application

A program to **register local dogs**. Therefore, locate roaming registered dogs and stray dogs in a neighborhood, subdivision or barangay.



Access the program in the QR Code on top

2. Detections in the Form of a Bollard

A device that has a camera and a mini-computer for automatic detection of roaming dogs.



USER MANUAL



PAWTECT

A COMMUNITY-BASED WIRELESS DOG DETECTION AND ALERT SYSTEM FOR STRAY AND MISSING DOGS USING COMPUTER VISION AND RASPBERRY PI

FOR THE BETTERMENT OF
OUR DOG COMPANIONS

SET UP

1. PawTect Application

An execution file would be placed onto the online folder.

Dashboard If detection occurs and dog is registered in the application, owners would be automatically contacted . Defining the legend: - Continuous Monitoring - Identified, Sent a Notification - Owner Replied - Returns Back to Red After a Minute	Register Input all the information of the owner and the dog. Require the download of Telegram for constant feedback. Long Range Take a clear picture of their front, right, left, and back (bottom-side). Make sure to correctly input the contact number.	Profile Stores all registered owner and dog's information in a page. Connect via LoRa MicroUSB cable directly to your main computing device.
--	---	--

2. Detections in the Form of a Bollard

A device would be given. There are hardware consideration during the placement.

Bollard Deploy the bollards at strategic locations with high stray dog activity to maximize detection opportunities. This may also be placed on entrances and exits of an area.	LoRa Antenna Place the LoRa Antenna in an elevated, unobstructed position to maximize signal range and avoid interference. The bottom surface of the Antenna is magnetic. This may be placed on poles at the street as seen at the side.	Solar System Position the solar panel in a sunlit, unshaded location for optimal charging efficiency. Keep the charge controller protected from environmental elements inside the main housing.
--	---	---

PROCESS

1. Registration

Data will be stored after filling up the form. Images will be used for recognition process later.



2. Detection

This is not displayed in a monitor. It is running at the background as the power is on at the bollard device.



Identifies dogs from the environment and crops them. It will be sent via transmission line.

3. Recognition

The sent image would be compared to the database's images through feature extraction. This would compare the embeddings.



If similarity percentage is high from one of the database's profile, then it will contact the specific owner in SMS and Telegram.

If similarity percentage is low across the database, then it would be identified as a stray dog.

4. Contact

The owner would be contacted with an attached image of the photo detected for further inspections.

Reply with:
FOUND - This is my dog!
NOTMINE - This is not my dog!





Appendix N:

Researchers Profile



JOHN MICHAEL C. LLANTINO

Ma. Corazon Phase 2, Barangay Sta. Cruz, Antipolo City |
+63 998 397 1041 | llantinojohn@gmail.com



OBJECTIVES

I aim to apply the knowledges and skills I've gained from studying computer engineering from Technological Institute of the Philippines and Eulogio "Amang" Rodriguez Institute of Science and Technology in a real-world setting, gain hands-on experience, and further develop my technical and professional skills.

EDUCATION

EULOGIO "AMANG" RODRIGUEZ INSTITUTE OF SCIENCE AND TECHNOLOGY - MANILA 2022 - 2026
Bachelor of Science in Computer Engineering

TECHNOLOGICAL INSTITUTE OF THE PHILIPPINES 2019 - 2022
Bachelor of Science in Computer Engineering
Science, Technology, Engineering, and Mathematics

WORK EXPERIENCE

QUICKZLE MARKETING CORP. 2025
IT Intern

- Assist the IT Supervisor with the task within the company at the same time create programs that will help the company complete tasks such as a payroll system and inventory system.

ACADEMIC PROJECT

CAPSTONE PROJECT (PAWTECT)

- Served as Technical Documentation and Hardware Integrator and Researcher for an embedded computer vision wireless dog detection system using Raspberry Pi, which achieved 3rd Place in the PATATE competition. Integrated peripheral hardware modules with software scripts and assisted the team during technical defenses.

BARANGAY POPULATION CENSUS SYSTEM

2025

- A system used to collect, organize, and manage demographic data of residents within a barangay made using Python and SQLite for database.

CPU SCHEDULING SYSTEM

2019 - 2020

- A system that simulates how a computer's CPU allocates processing time to multiple tasks or processes using scheduling algorithms.

SEMINARS

**A SERIES OF TECHNOLOGY SEMINAR 2025, CONDUCTED BY THE ASSOCIATION OF
COMPUTER ENGINEERING STUDENTS**

April 2025

Participant

- Integrated Circuit Fabrication, Engr. Maximino A. Dela Paz, Jr. from Intel Corp., held on April 28, 2025

CERTIFICATION

COMPUTER SYSTEM SERVICING NATIONAL CERTIFICATE II

issued on 2024

ACADEMIC ACHIEVMENT

- VPAA's Lister, First Semester of S.Y. 2022 - 2023
- President's Lister, Second Semester of S.Y. 2021 - 2022

ADDITIONAL

Knowledge: SolidWorks and AutoCAD Designs, Computer Network and Security, Computer Systems and Servicing, Machine Learning, Microprocessor, Computer Numerical Control (CNC)

Technical Skills: Python, C++, C, Basic, JavaScript, G Code, Software & Hardware Troubleshooting, Network Configuration, Management and Security

Attitude: Willing to Learn, Critical Thinker, Problem Solver, Reliable and Committed, Team Oriented, Strategic, Fast Learner

Language: Native in Filipino, Work Proficient in English



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182

HERO JAMES P. OLIGANE

October 20, 2026 | AMVACA Housing, Lamesa St., Ugong, Valenzuela City |
+63 9851604290 | herojamesoligane22@gmail.com



PROFESSIONAL SUMMARY

Aspiring Computer Engineering graduate with fundamental experience in IoT, robotics, network troubleshooting, and technical documentation. Grateful for opportunities to have led student teams, managed external relations, and coordinated visual branding across university organizations. Eager to adapt these analytical problem-solving skills and a collaborative mindset into a professional team environment.

EDUCATION

EULOGIO "AMANG" RODRIGUEZ INSTITUTE OF SCIENCE AND TECHNOLOGY

Expected August 2026

Bachelor of Science in Computer Engineering

Relevant Activities: Academic Achiever, Student Leader, Volunteer, ASA Philippines Foundation Scholar

ANGELA VALDEZ RAMOS NATIONAL HIGH SCHOOL

June 2022

Science, Technology, Engineering, and Mathematics - Graduated with High Honors

PROFESSIONAL EXPERIENCE

MANILA WATER COMPANY INC.

July 2025 – October 2025

Industrial Technology Development Group (CITG) Intern

- Completed a 300-hour technical internship focusing on corporate infrastructure monitoring and software deployment.
- Assisted IT professionals and seniors engineers in troubleshooting hardware workflows and analyzing operational system performance metrics

LEADERSHIP & ORGANIZATIONAL EXPERIENCE

ASSOCIATION OF COMPUTER ENGINEERING STUDENTS

EARIST, Manila

Vice President for External Affairs (NCR Division Affiliate)

August 2022 - August 2026

- Manage institutional partnerships, handle regional communications, and a Creatives Committee.
- Publication Material Officer*
- Standardized visual identity guidelines for announcements and events using Canva and Photoshop.

COLLEGE OF ENGINEERING STUDENT GOVERNMENT

2024 - Present

Grievance Chairperson

- Facilitated a structured, fair system to review, mediate, and resolve student concerns, promoting transparency and balanced student welfare policies.

Multimedia Committee Member

- Collaborated with cross-functional councils to design, structure, and deploy digital assets and layout announcements for the engineering student body.

CLASS STUDENT GOVERNMENT | COMPUTER ENGINEERING BATCH 2025 - 2026

2024 - Present

Secretary

- Maintain official documentation, archive administrative class resolutions, and coordinate apparel procurement logistics.

TECHNICAL PROJECTS & RESEARCH

CAPSTONE PROJECT (PAWTECT)

- Served as Technical Documentation and Hardware Integrator and Researcher for an embedded computer vision wireless dog detection system using Raspberry Pi, which achieved 3rd Place in the PATATE competition. Integrated peripheral hardware modules with software scripts and assisted the team during technical defenses.

RESEARCH PUBLICATION (IJRISS JOURNAL)

- Acted as Lead Author for peer-reviewed published research exploring the practical implementation of smart technologies, structuring documentation, and finalizing proofs. Early 2026.

ACADEMIC PORTFOLIO

- Built automation prototypes with Arduino; simulated circuits in MATLAB, SOLIDWORKS, ModelSim, and Proteus; programmed PIC microcontrollers via MikroC; completed fundamental coding in C++, Python, and Java.

ADDITIONAL

Hardware & IoT: Arduino, Raspberry Pi, GSM modules, FPGA Programming (Verilog, ModelSim)

Software & Tools: C/C++, Python, Java, MikroC, MATLAB, SOLIDWORKS, Proteus, Computer Vision, Canva, Photoshop

Certification: TESDA - Computer Systems Servicing National Certificate II, Issued on July 2024

Language: Native in Filipino, Work Proficient in English



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183

ULRIKE ARETA QUIRANTE REÑIDO

8 Dragon St., Midtown Subd., Ph. II, Brgy. San Roque, Marikina City | 09162142599 | ulrikeareta@gmail.com

SUMMARY

A graduating Computer Engineering student seeking a working opportunity to apply and expand technical knowledge in real-world environments. Eager to contribute to a dynamic team by assisting with IT support, software troubleshooting, hardware maintenance, and basic network configuration. Highly motivated to learn industry practices, improve problem-solving skills, and gain hands-on experience in system operations and IT infrastructure.

PUBLISHED JOURNALS AND RESEARCHES

A COMMUNITY-BASED WIRELESS DOG DETECTION AND ALERT SYSTEM FOR STRAY AND MISSING DOGS USING COMPUTER VISION AND RASPBERRY PI June 2026

Machine Learning Engineer & Technical Writer | YOLOv8n

- Trained and optimized a lightweight computer vision model for real-time edge deployment on Raspberry Pi nodes while spearheading the comprehensive thesis manuscript compilation, user documentation, and technical presentations.

DEVELOPMENT OF A LIDAR-INTEGRATED BLUETOOTH VEHICLE FOR COLLISION PREVENTION AND SAFETY ALERTS USING ESP32 DEV MODULE January 2026

Systems Design Engineer & Hardware Prototyping Assistant | Hardware Architecture

- Authored the comprehensive technical documentation and co-designed the hardware circuit architecture to integrate LiDAR sensors and Bluetooth modules with an ESP32 Dev Module for an autonomous collision prevention vehicle.

ENHANCING CYCLIST VISIBILITY USING A SMART REFLECTIVE SAFETY BAG WITH AUTOMATED LIGHT INDICATORS January 2026

Hardware-Software Integration Engineer | Internet of Things (IoT), C++

- Designed and programmed a C++ firmware control system integrating an accelerometer, LDR sensor, and Bluetooth module to automate real-time braking lights, ambient night lighting, and turn indicators on a smart cyclist safety bag.

LEADERSHIP & ORGANIZATIONAL EXPERIENCE

ASSOCIATION OF COMPUTER ENGINEERING STUDENTS

President for Two (2) Terms & Secretary for Internal Affairs for One (1) Term

EARIST, Manila

August 2024 - June 2026

- Lead and represent **321 computer engineering students** in both internal and external academic activities such as the Institute of Computer Engineering of the Philippines - NCR - Student Edition.
- Established a **connection** with its affiliated organization, Institute of Computer Engineering of the Philippines - NCR and its student edition
- Oversee the planning and execution of organizational activities, including academic seminars, competitions, outreach programs, and general assemblies.
- Represent in college councils, interdepartmental collaborations, and external functions.

INSTITUTE OF COMPUTER ENGINEERS OF THE PHILIPPINES - NCR - STUDENT EDITION Asia Pacific College

Member

February 2025

- Robothon Pitching Competition 2025**, Participated and collaborated in a team to design and prototype an innovative robotics solution and delivered a comprehensive technical pitch to industry professionals detailing the project's market viability and engineering framework under a strict time constraint.

SEMINARS ATTENDED

A SERIES OF TECHNOLOGY SEMINAR 2026, CONDUCTED BY THE ASSOCIATION OF COMPUTER ENGINEERING STUDENTS April 2026

Participant

- Next-Gen Connectivity: Campus Networks with UniFi Technology: A Deep Dive into Software-Defined Networking(SDN), Mesh Topologies, and Centralized Management
- Building Modern Solutions with Microsoft Power Platform
- Microprocessor Testing and Verification
- Quality Assured, Customer Delight: The Indispensable Value of QA Testing
- Network Design Fundamentals: Building Scalable and Reliable Campus Networks

YES COMPUTER ENTHUSIAST, TECHXPLORE: UNVEILING DIGITAL HORIZONS Brgy. Sta. Cruz, Antipolo City

Guest Speaker

August 2023

- Served as a Guest Speaker during a youth education session organized by Youth for Excellence in Sampaguita delivering an engaging and informative talk on computer literacy to empower and inspire local youth in digital literacy.

EDUCATION

EULOGIO 'AMANG' RODRIGUEZ INSTITUTE OF SCIENCE AND TECHNOLOGY

Bachelor of Science in Computer Engineering

Relevant Coursework: Computer Network and Security, Embedded Systems

Sta. Mesa, Manila

Expected August 2026

SAN ROQUE NATIONAL HIGH SCHOOL

Accountancy, Business and Management - Graduated with High Honors

San Roque, Marikina

June 2022



NOTABLE UNIVERSITY PROJECTS

IC RECOGNITION SYSTEM FOR LOGIC CIRCUIT SOLVER

June 2025

Machine Learning Engineer | YOLOv8, RoboFlow

- Developed a system that detects basic logic ICs (NOT, AND, OR) using image recognition.
- Users input a Boolean expression based on the detected ICs, and the system generates the corresponding truth table and wire instructions using standard IC pin configurations.

IMAGE RECOGNITION SYSTEM WITH ROBOFLOW AND PYTHON FOR TWO OBJECTS

June 2025

Machine Learning Engineer | YOLOv8, RoboFlow

- Developed an image recognition system using RoboFlow and Python
- Focused on detecting and classifying two objects in live camera preview with high accuracy
- Built a complete pipeline: from dataset preparation to model deployment

IMPLEMENTATION OF TP-LINK (AX72) ROUTERS IN ACADEMIC SETTING

June 2025

Network Engineer | Network Security & Systems

- This project demonstrates the implementation and configuration of TP-Link routers to create a secure, efficient, and manageable network environment within an academic setting.
- Configuring basic settings to thorough network customization in TP-Link routers.

VARIABLE BENCH POWER SUPPLY USING OLD PSU AND BOOST-BUCK CONVERTER

May - July 2024

Electrical Engineering Prototyper | Power Electronics, Circuit Design

- Repurposed an old power supply unit as the input power source
- Integrated a buck-boost converter to provide adjustable voltage output for various electronics applications
- Built a cost-effective, space-saving alternative to commercial bench power supplies

ADDITIONAL

Technical Skills: Industrial Automation (PLC) using CX Programmer, Digital Fabrication for BenchMill 6000 (CNC), Machine Learning (ML/AI) using YOLOv8, YOLOv8n, and ResNET, Computer Assembly

Certifications & Training: Computer System Servicing National Certificate II, Safety Officer II

Awards: 3rd Place in PATATE 2026 - Hardware Category, Service Awardee, Top Student Performer Awardee, High Honor, Merit Honor

Language: Fluent in Filipino, Conversational Proficient in English

REFERENCES

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INSTITUTE OF SCIENCE AND TECHNOLOGY
College of Engineering



185

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OBJECTIVES

- To bring efficiency in the workflow in my workplace, and do job effectively using technology as an advantage
- Applying my Unix/Linux skills into work, and use its philosophy while at the same time learn and improve that can benefit both myself and the company

EDUCATION

EULOGIO ‘AMANG’ RODRIGUEZ INSTITUTE OF SCIENCE AND TECHNOLOGY - MANILA <i>Bachelor of Science in Computer Engineering</i>	2022 - 2026
INSTITUTE OF CREATIVE COMPUTER TECHNOLOGY - CAINTA <i>Bachelor of Science in Computer Engineering</i>	2020 - 2021
ANTIPOLO CITY SENIOR HIGH SCHOOL <i>Technical Vocational in Computer Programming</i>	2018 - 2020

WORK EXPERIENCE

QUICKZLE MARKETING CORP. <i>IT Intern</i>	2025
• Assist the IT Supervisor with the task within the company at the same time create programs that will help the company complete tasks such as a payroll system and inventory system.	
SCHOOL OF EVERLASTING PEARL <i>Computer Maintenance</i>	2019 - 2020
• Tasked to maintain and monitor computers that are utilized by students in the computer laboratory.	

CERTIFICATION

COMPUTER SYSTEM SERVICING NATIONAL CERTIFICATE II	issued on 2024
--	-----------------------

ADDITIONAL

Knowledge: Computer Aided Drafting, Computer System Servicing, Computer Programming, Linux/Unix Like Systems, Computer Networking
Technical Skills: Linux Systems, Bash & Zsh, VIM/NeoVIM, Office Suites such as Libreoffice and Microsoft Office, Proefficient at using GROff and LaTeX
Language: Native in Filipino, Work Proficient in English



Appendix O:

Grammar Checker



CHAPTER I: PROBLEM AND ITS BACKGROUND

Introduction

In the Philippines, dogs are considered beloved companions in several households, acting as loyal pets, guardians, and family members. The ongoing problem of stray and free-roaming dogs is a significant threat to public health, community safety, and animal welfare.

Estimates from entities such as the Philippine Animal Welfare Society, also known as PAWS, suggest that there are roughly 12 million stray dogs across the nation, a statistic often referenced in recent reports from 2019 to 2024, and acknowledged as a pressing issue in discussions on animal homelessness and overpopulation in 2025. The large stray population poses considerable public health risks, especially with rabies, a viral illness that is basically lethal once symptoms manifest.

In 2024, the Department of Health in 2024 documented 426 confirmed rabies cases, all of which were fatal. Out of the confirmed rabies cases, 45% were associated with bites from domestic pets, while several others were from stray animals or those with indeterminate vaccination status. Despite early 2025 data indicating a promising decline, such as 55 cases from January to March, a 39% decrease compared to the same timeframe in 2024, rabies continues as endemic, resulting in hundreds of human fatalities each year, disproportionately impacting children.

Despite national mandates under Republic Act No. 9482, also known as the Anti-Rabies Act of 2007, which requires annual dog registration, immunization, and responsible ownership, enforcement remains inconsistent across many regions. While Local Government Units are tasked with maintaining registries and managing stray populations, progress is often hindered by limited public awareness and resource constraints.



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CHAPTER II: REVIEW OF RELATED LITERATURE [UR1]

This chapter contains reviews of related foreign and local literature connected to the research. The key concepts presented include a computer vision model, a dog registration application, a Raspberry Pi, LoRa wireless transmission methods, and real-time SMS notifications.

Related Readings

Recent improvements in computer vision, including the YOLO real-time object detection model (Redmon et al., 2016), have facilitated rapid and precise identification of animals via CCTV. The Anti-Rabies Act of 2007 requires all local government units (LGUs) in the Philippines to keep a register of dogs and issue identification tags for pets. Nonetheless, similar assessments indicate that compliance in addressing Local Government Units (LGUs) is inconsistent, often negligible, leading to challenges in managing stray dog numbers and preventing rabies. The municipal governments, including Bacolod City, have prioritized mass dog registration as a means to improve vaccination distribution and enhance public safety (Philippine News Agency, 2023). These changes suggest the need for an intelligent, automated monitoring and registration system utilizing computer vision and CCTV to promote responsible pet ownership and enhance community safety.

Our proposed system aligns with three Sustainable Development Goals (SDGs): Sustainable Development Goals 3, 11, and 15. Initially, SDG 3 Good Health and Well-being is vital since it aids in illness prevention and guarantees public safety. The World Health Organization (2021) asserts that regulating dog populations and enhancing owner accountability can diminish rabies incidence globally. It also references SDG 11: Sustainable Cities and Communities, which aims to create inclusive, safe, resilient, and sustainable urban areas and human settlements.



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CHAPTER III: METHODOLOGY

This chapter presents the methodology employed in this study, detailing the systematic processes involved in the development, training, and implementation of a computer vision-based dog identification system with an integrated canine database and LoRa wireless transmission system for recognizing missing and stray dogs. To accomplish these objectives, the study is structured into multiple phases, each encompassing specific tasks and procedures that are built upon one another.

Figure 2. Deployment of the Prototype

IPO Model

Figure 2. Input-Process-Output Model

Figure 2 illustrates the Conceptual Framework of the study, utilizing the Input-Process-Output (IPO) model with a Feedback Loop to delineate the functional flow of the study system. The diagram serves as a logical blueprint for the integration of hardware components and software algorithms, structured as follows:

- Input—this stage identifies the foundational resources required, including hardware such as the Raspberry Pi Camera Rev 1.3, and software assets like the open-source datasets (Kaggle, COCO, and Google Images) used for model training. It also encompasses the essential community data, specifically the registered dog images and owner contact information.
 - Process—this section represents the core "Edge Intelligence" of the system. It details the transformation of raw inputs through YOLOv8n model training, the implementation of the SQL-based registration system, and the technical handling of compressed image transmission over the LoRa (Long-Range) wireless network.
 - Output—the results of the system are presented here, characterized by a fully trained detection model and an integrated communication bridge. The primary outputs include the real-time identification of dogs, successful feature extraction, and the automated delivery of Telegram/SMS notifications to the registered owners.
- Feedback—this iterative component ensures continuous system optimization. It focuses on



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CHAPTER IV: RESULT, ANALYSIS, AND INTERPRETATION OF DATA

This chapter presents the empirical results, data analysis, and technical interpretations of the performance evaluations conducted on the developed edge-to-centralized detection and tracking system. The collected data is systematically organized and evaluated to assess the operational efficiency of the system's primary components across successive testing phases.

System Testing Overview

Test Type	Number of Trials
Confusion Matrix	N/A
LoRa Transmission Tests	15
SMS Notification Time	15
SMS Success Rate	15
Embedding Program	15

PHASE I: COMPUTER VISION MODEL TRAINING



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CHAPTER V: SUMMARY, CONCLUSION, AND RECOMMENDATION

This chapter presents the summary of findings obtained from the development, evaluation, and empirical testing of the system. It outlines the conclusions derived from statistical data and provides practical recommendations for future advancements and system implementations.

Summary of Findings

The primary goal of the study was to design and develop an intelligent, community-based wireless dog detection, identification, and registration system utilizing edge computing, low-power telemetry, and automated alerting mechanisms. The empirical evaluations from field-testing yielded the following technical findings:

- **Dog Detection Model Performance:** The on-field object detection core, powered by a trained lightweight YOLOv8 Nano model deployed on a resource-constrained Raspberry Pi Zero 2W, achieved a high Precision score of 86.73%. This verified the model's robust capability to eliminate false positives caused by shifting environmental noise, shadows, and non-target animals.
- **LoRa's Communication Performance:** Testing of the peer-to-peer Long Range (LoRa) communication channel demonstrated a highly stable, linear relationship between transmitted image data volumes and total transmission time. Transfer efficiency remained tightly constrained between 0.58 and 0.61 seconds per chunk across all payloads (ranging from 52 chunks to a maximum stress load of 772 chunks), successfully overcoming software accumulation overheads and making reassembly times highly predictable.
- **Identification Reliability:** The feature extraction pipeline effectively transformed cropped images into standardized mathematical signatures. Utilizing Cosine Similarity calculations, the identification engine successfully executed verification attempts against the central



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Appendix P:

Plagiarism Checker



Page 2 of 136 - Integrity Overview

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Appendix Q:

IEEE Format



PawTect: A Community-Based Wireless Dog Detection and Alert System for Stray and Missing Dogs using Computer Vision and Raspberry Pi

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Abstract — PawTect is a community-based wireless dog detection and alert system designed to mitigate public health risks and animal welfare issues in the Philippines, where approximately 12 million stray dogs and significant rabies fatalities persist. The system utilizes a YOLOv8-based computer vision model trained on diverse datasets, such as COCO and Stanford Dogs, to perform real-time species detection and breed classification. The hardware architecture employs a Raspberry Pi Zero as a decentralized edge computing node; this node performs initial detection and image compression at the source to minimize bandwidth consumption. These processed results are transmitted via LoRa wireless technology to a Raspberry Pi 4 central hub. This hub cross-references detected biometric features—such as facial landmarks and coat patterns—against a centralized SQL database containing registered owner profiles and pet characteristics. Upon a successful match, a API triggers immediate SMS notifications to owners or local authorities, ensuring a rapid response even in areas with unstable internet connectivity. This edge-to-hub integration provides a scalable, cost-effective framework for enhancing local government dog registries and promoting responsible pet ownership.

Keywords: Lora Wireless Transmission Systems, Real-Time Sms Notification, Dog Registration Program, Computer Vision Model, Raspberry Pi

I. INTRODUCTION

In the Philippines, dogs are cherished companions, yet a growing population of approximately 12 million stray and free-roaming animals poses severe threats to public health, community safety, and animal welfare. The most critical risk is rabies, an endemic viral disease that claimed

426 lives in 2024; although early 2025 data showed a 39% decline in cases from January to March, the disease remains a fatal threat, particularly to children. While Republic Act No. 9482 (Anti-Rabies Act of 2007) and localized mandates like Manila Ordinance No. 8587 require registration and vaccination, enforcement by Local Government Units (LGUs) is severely hindered by limited resources, low public awareness, and a reliance on slow, manual tracking methods. Current interventions like mass vaccinations and Catch-Neuter-Vaccinate-Return (CNVR) initiatives lack an integrated, real-time platform to monitor animal movement. Furthermore, while emerging technologies like computer vision, Raspberry Pi, LoRa communication, and GSM networks show individual promise, they have yet to be unified into a cohesive system resilient to the unreliable internet connectivity common in Philippine communities. To bridge this gap, this study presents PawTect: A Community-Based Wireless Dog Detection and Alert System for Stray and Missing Dogs using Computer Vision and Raspberry Pi. The system employs a YOLOv8-based computer vision model trained on diverse datasets for real-time species and breed detection on a decentralized Raspberry Pi Zero edge node. It utilizes LoRa wireless technology to transmit compressed target images to a Raspberry Pi 4 central hub, which extracts high-dimensional biometric vectors and cross-references them against an SQL database of registered owners via cosine similarity. Upon a successful match, the system triggers automated SMS notifications directly to the owner's or administrator's cellular phone. This research focuses on the system design, hardware-software integration, and localized performance testing within a controlled community. Ultimately, PawTect aligns with three United Nations Sustainable Development Goals (SDGs): SDG 3 (Good Health and Well-being) by mitigating rabies transmission; SDG 11 (Sustainable



Cities and Communities) by enhancing urban safety and data-driven animal management; and SDG 15 (Life on Land) by promoting ethical animal treatment and mitigating the environmental impacts of uncontrolled stray populations. Through this contemporary technological intervention, PawTect aims to provide an accessible, immediate framework to support LGUs, enhance community safety, and foster responsible pet ownership.

II. OBJECTIVES

The main objective of the study is to design and develop an intelligent dog detection, identification and registration system using field video surveillance integrated with computer vision that can automatically detect. A separate device through the Long-Range wireless transmission would receive boxes of images and identify and manage dogs within a community premise to enhance community safety, promote responsible pet ownership, mitigate animal-related issues, and foster a well-regulated,

pet-friendly environment. To achieve the goal of a fully automated and community-based detection system, the following technical and functional objectives have been established:

1. Create and train a computer vision model that can accurately detect dogs from the environment and identify if it is missing or a stray dog.
2. Develop and integrate dog registration application and a community dog database for the computer vision that enables dog owners to log their pets and store related data for identification purposes such as owner information, dog specific descriptions with uploaded images and vaccination records.
3. Implement a LoRa wireless transmission system to send cropped images from a field Raspberry Pi device to a central hub.
4. Develop an automatic SMS Notification module via a Telegram Bot API to inform owners or administrators of their missing or stray dogs, respectively.
5. Implement the system to verify its accuracy and latency in identifying dogs and sending real-time alerts.

III. LITERATURE REVIEW

The integration of embedded computer vision and long-range communication technologies offers a powerful framework for community-based animal monitoring. By utilizing efficient, single-pass deep learning models like YOLO (Redmon et al., 2016)

alongside optimized neural network architectures (Zoph et al., 2018), edge computing platforms like the Raspberry Pi can perform real-time, low-power object detection locally without relying on heavy cloud processing (Ozkan et al., 2021). When combined with reliable, long-range, low-power communication networks like LoRa (FAO, 2019; Pereira & Chaari, 2025) and rapid notification systems (WHO, 2021), these edge platforms can instantly transmit critical updates across wide geographic areas, even where internet infrastructure is limited.

Furthermore, leveraging these technical frameworks alongside centralized dog registration systems significantly enhances community animal management, public health, and animal welfare. As demonstrated in regional and large-scale surveys, maintaining accurate registration databases is vital for tracking companion populations, monitoring vaccination programs, controlling diseases like rabies, and recovering lost pets (Carvelli et al., 2020; Dizon et al., 2022). Together, these advancements provide the theoretical and technical foundation for PawTect, a unified system designed for the efficient, real-time detection, identification, and monitoring of stray and missing dogs.

IV. METHODOLOGY

The systematic process for developing and implementing PawTect is structured as a continuous edge-to-centralized network topology optimized for open-set object detection, long-range remote telemetry, and real-time automated community alerting. The operational infrastructure is divided into two primary environments: the On-Field Detection Nodes and the Central Monitoring Base Station.

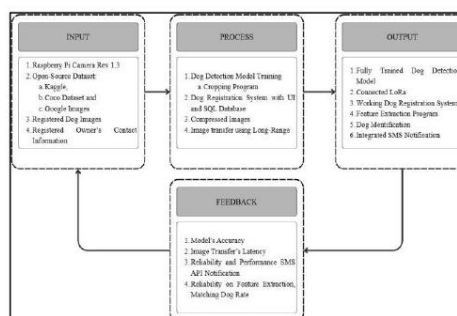


Figure 2. Input-Process-Output Model

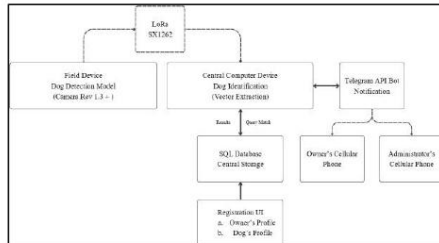


Figure 3. Block Diagram

As illustrated in the system architecture (Fig. 2 and Fig. 3), the workflow commences at the camouflaged on-field street bollards. A Raspberry Pi Camera captures real-time video, which is locally scanned by an optimized object detection program. Once a target is localized with a confidence score exceeding a set minimum threshold, the system extracts the spatial coordinates of the bounding box and executes an immediate matrix crop, discarding the environmental background to reduce payload size. The cropped image array is converted into a compressed byte-stream, split into sequenced data packets wrapped with validation headers, and transmitted via a LoRa SX1262 slave wireless module



Figure 4. Conceptual Design

At the central base station, a master transceiver remains in a continuous listening state to serve as the wireless uplink channel. As fragmented packets arrive, a centralized reassembly script validates incoming fragments using Cyclic Redundancy Checks, reorders the blocks, and decodes the byte sequence back into a structured image matrix. This reassembled target crop is immediately forwarded to a high-dimensional feature extraction engine that calculates a 512-dimensional vector embedding representing the dog's physical traits

Closing the loop between target identification and field action, the central station deploys an automated, dual-channel alerting mechanism that concurrently triggers two separate alert pipelines. For users with active data connections, the system executes an asynchronous HTTPS request to the Telegram Bot API to deliver a rich push notification containing a

structured text summary—including target location, timestamp, and owner metadata—paired with the exact high-resolution visual crop. Simultaneously, to bypass mobile internet dependencies and guarantee delivery under poor network conditions, the system maps the transaction details to an SMS API network gateway, sending a concise text summary directly to the owner's or administrator's registered mobile cellular phone.

V. RESULTS AND DISCUSSION

This chapter presents the empirical results, data analysis, and technical interpretations of the performance evaluations conducted on the developed edge-to-centralized detection and tracking system. The collected data is systematically organized and evaluated to assess the operational efficiency of the system's primary components across successive testing phases.

Test Type	Number of Trials
Confusion Matrix	N/A
LoRa Transmission Tests	15
SMS Notification Time	15
SMS Success Rate	15
Embedding Program	15

Table 1: Testing Overview

PHASE I: COMPUTER VISION MODEL TRAINING

1. Create and train a computer vision model that can accurately detect dogs from the environment and identify if it is missing or a stray dog.

Dataset Source	Purpose	Numbers of Images
COCO Dataset	Dog Detection	4559
Kaggle Dataset	Dog Detection	1520
Google Image	Dog Detection	Negative Samples
177	Total	6256

Table 2. Summarization of Dataset Used for Detection Model The model's training utilizes 6,256 images from different repositories.

Parameter	Configured Settings for Training
Epoch	100
Image Size	640 x 640
Batch	16
Augmentation	True
Device	cpu

Table 3. Training Setting



Under the training program, the images were set to undergo the following conditions placed by the proponents to create a detection model.

Metric	Description	Dataset	Value
Precision	Accuracy of predictions; the per-centage of flagged objects that are correct.	Validation Set	0.8673
Recall	Coverage of detections; the per-centage of actual objects that the model successfully found	Validation Set	0.8095
mAP_{50}	Overall detection success rate when requiring a loose, 50% minimum bounding box overlap.	Validation Set	0.8434
mAP_{50-95}	Strict localization accuracy averaged across multiple tight bounding boxes overlap thresholds (50% to 95%).	Validation Set	0.6435

Table 4. Confusion Matrix of the On-Field Detection Model



Figure 3. Integrated Dog Detection Model

At the end of the phase, the proponents were able to create a detection model and a cropping program to be implemented for the next phases of the study.

For the first six trials, our system successfully executed a verification attempt ($A1 = 1$) upon target capture, yielding high-confidence cosine similarity metrics and valid match decisions classified as either a Confirmed Match (CM) or a Suspect (S). Because the system correctly identified and flagged true targets present in the environment, these iterations are accurately recorded as True Positives (TP), proving strong baseline recognition sensitivity.

Conversely, in trials 7 through 9, the system failed to trigger a verification attempt ($A1 = 0$) resulting in null alues (N/A) for both the cosine similarity scores and system outcomes despite a target being physically present, thereby officially registering as False Negatives (FN).

Trials (i)	Attempt (A)	Cosine Similarity Score (cos θ) in %	System Outcome	Remarks
1	1	93.4831	CM	TP
2	1	94.2384	CM	TP
3	1	97.5783	CM	TP
4	1	74.9954	S	TP
5	1	79.2381	S	TP
6	1	87.5672	S	TP
7	0	N/A	N/A	FN
8	0	N/A	N/A	FN
9	0	N/A	N/A	FN
10	1	93.999%	CM	TN
11	1	94.893%	CM	TN
12	1	94.906%	CM	TN
13	1	90.833%	CM	TN
14	1	89.454%	CM	FP
15	1	93.713%	CM	FP

Table 5. Cosine Similarity of the Vector Program

A major technical discrepancy was observed in trials 10 through 13, where active verification attempts ($A1 = 1$) generated exceptionally high cosine similarity scores ranging from 90.833% to 94.906%, leading to Confirmed Match (CM) system outcomes. Despite executing active, high-confidence positive matches, these trials were logged as True Negatives (TN); because a standard confusion matrix requires a True Negative to correctly reject a non-target or remain silent, these active matches represent a logical conflict that requires reclassification based on true ground truth. Finally, in trials 14 and 15, the system-initiated verification ($A1 = 1$) and erroneously triggered a Confirmed Match (CM) for non-target subjects with scores above 89.454%, which were classified as False Positives (FP) and successfully defined our current operational similarity threshold boundaries.

2. Develop and integrate dog registration application and a community dog database for the computer vision that enables dog owners to log their pets and store related data for identification purposes such as owner information, dog specific descriptions with uploaded images and vaccination records.

Profile Type	Frequency
Registered Owner	10
Registered Dog	10
Test Image Used	40
Total Users	10

Table 6. Frequency Distribution of Sample Subjects for the Dog Registration Application

A total of ten (10) respondents participated in the study, each registering one (1) dog. Each respondent provided four (4) images of their dog, resulting in 40 base images that has four perspectives; front, back, right and left. Data collection was executed by deploying the system interface to a test group of 10 localized dog owners. The registration process successfully captured,



digitized, and structured 10 complete, multi-layered profiles containing primary owner contact information, specific canine physical descriptions, medical vaccination logs, and reference validation imagery for each animal. System field testing verified a 100% database entry success rate, as all 10 profiles were securely written to the relational database without any data truncation, relational conflicts, or upload failures. During simulation routines, the indexed architecture of the database allowed the central monitoring workstation to execute query comparisons against these 10 stored reference profiles.

Metric	Value
Successful Registration	100%
Error Rate	0%

The system architecture that achieves a perfect 100% successful profile registration rate alongside a flawless 100% message delivery rate across all automated alerting iterations.

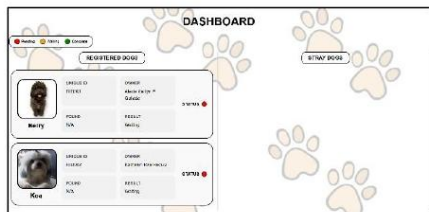


Figure 4. Main Page of System



Figure 5. Registration Page of System

The application interface successfully provides an easy-to-use digital portal that simplifies data collection and tracking for both community members and local authorities. The main dashboard operates as a live monitoring console that clearly separates registered pets from unverified stray dogs for quick review. Additionally, the registration page offers a structured form where owners can systematically input contact information and specific physical dog traits directly into the system database.

3. Implement a LoRa wireless transmission system to send cropped images from a field Raspberry Pi device to a central hub.

Trials	Number of Chunks	Duration To Complete in Seconds	Transfer Per Chunk in Seconds
1	52	31	0.60
2	103	60	0.58
3	154	92	0.60
4	206	122	0.59
5	258	150	0.58
6	309	184	0.60
7	360	313	0.59
8	412	241	0.58
9	464	271	0.58
10	515	302	0.59
11	566	334	0.59
12	618	361	0.58
13	670	391	0.58
14	721	423	0.59
15	772	451	0.58
Average Completion Time in Seconds		3726/15	248.40

The provided data evaluates the transmission performance of a LoRa-based peer-to-peer communication system across 15 distinct trials. The dataset tracks the scaling relationship between the data volume (represented by the number of chunks) and the time required to complete the transmission, alongside the individual transfer efficiency per chunk.

The data reveals a highly stable, linear relationship between the data volume and the total transmission time. Across all trials—spanning from a small payload of 52 chunks to a maximum load of 772 chunks, the transmission speed remained remarkably uniform. This predictability is demonstrated by the Transfer Per Chunk metric, which stayed tightly constrained between 0.58 and 0.60 seconds per fragment throughout the entire testing process. For example, a minor payload of 52 chunks completed transmission in 31 seconds (0.60 *s/chunks*), while a massive payload of 772 chunks concluded in 451 seconds (0.58 *s/chunks*). This extreme consistency proves that the system's packetization and serialization protocol operates without introducing processing bottlenecks, buffer delays, or cumulative software overhead as file sizes grow. The fixed transmission rate confirms that the wireless communication link is highly reliable and predictable under fluctuating data loads. This allows the central hub to accurately anticipate reassembly times based strictly on incoming file sizes, validating the use



of low-power LoRa networks for remote, image-based telemetry.

4. Develop an automatic SMS Notification module via a Telegram Bot API to inform owners or administrators of their missing or stray dogs, respectively.

Trial	Duration of Command in Seconds	Time Received in Seconds	Received Message
1	1.851	12	1
2	1.330	7	1
3	1.681	8	1
4	1.145	9	1
5	1.516	7	1
6	1.242	8	1
7	1.731	11	1
8	1.415	7	1
9	1.610	10	1
10	1.198	8	1
11	1.554	9	1
12	1.382	7	1
13	1.674	11	1
14	1.295	8	1
15	1.488	10	1
Average Receiving Time		$\frac{132}{15}$	8.80s

The system demonstrated high operational efficiency, requiring a mean Duration of Command of only 1.474 seconds for the central workstation to process the high-dimensional feature verification match and dispatch the outbound alert request, which proves that the background automation scripts introduce negligible processing overhead. Concurrently, the end-to-end network delivery time—measured from the exact moment of command execution to physical cellular handset reception—yielded a cumulative total of 132 seconds across all iterations, establishing a highly efficient arithmetic mean Time Received of 8.800 seconds ($\frac{132}{15}$). Minor variations across individual trials, which ranged tightly between 7 and 12 seconds, are strictly attributable to standard localized cellular network routing, gateway handshakes, and carrier signaling behaviors. Furthermore, the framework achieved a flawless 100% message delivery success rate across all 15 operational trials, with zero packet drops, gateway timeouts, or API transmission failures encountered during continuous stress testing. Taken together, these data streams validate that the integrated notification module functions as a highly dependable and low-latency utility; by maintaining a combined local processing and cellular transmission loop that consistently averages well under 10 seconds, the system successfully satisfies the real-world operational demands necessary to provide immediate, actionable intelligence to community authorities and dog owners.

5. Implement the system to verify its accuracy and latency in identifying dogs and sending notifications.

The comprehensive integration of the prototype successfully unified the decentralized edge-computing nodes, the low-power wireless transmission layer, the centralized relational database, and the automated alerting gateway into a single, continuous processing loop. The end-to-end processing latency of the system is primarily governed by the physical constraints of the low-power LoRa telemetry link during image packet reassembly, whereas the local edge processing, central database matching, and notification initialization occur almost instantaneously. The cumulative timeline required to progress from initial field detection to final mobile handset notification remains well within acceptable operational boundaries for localized community alerting, proving the viability of a monitoring framework that does not rely on cellular internet at the edge. In live environments, the unified pipeline successfully filters out environmental noise, moving shadows, and non-target animals, ensuring high classification accuracy while preventing the central base station from being flooded with redundant data or false notifications. The automated communication gateway demonstrated exceptional network reliability during continuous field operations, maintaining a flawless message delivery success rate with zero packet drops, script execution failures, or gateway timeouts. The overall implementation confirms that the decentralized hardware components and centralized software layers are integrated.

VI. CONCLUSION

Based on the documented findings, the following conclusions are established:

1. Integrate dedicated power-management circuits, such as localized solar harvesting modules, alongside optimized thermal-throttling enclosure designs to guarantee continuous uptime and protect edge systems deployed outdoors.
2. Design a portable version of the edge device and scale the software architecture from a singular village network to a centralized municipality deployment. Integrating this layout into local city tracking systems will allow direct data coordination with animal welfare units, advancing public health goals under the Philippine Anti-Rabies Act (Republic Act No. 9482).
3. Strengthen the feature extraction program by exploring deeper or more advanced backbone networks (such as modified CSPDarknet or attention-mechanism blocks like CBAM). This algorithmic optimization will allow the system to capture more distinct structural and textural features under challenging conditions, minimizing classification errors and improving overall model confidence.



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